

vanshikamunjal665

Containerization & Devops Lab

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[Repository Link](#)

[Website Link](#)

vanshikamunjal665

CONTAINERIZATION AND DEVOPS LAB

Experiment 1:

Comparison of Virtual Machines (VMs) and Containers using Ubuntu and Nginx

Reference Link:

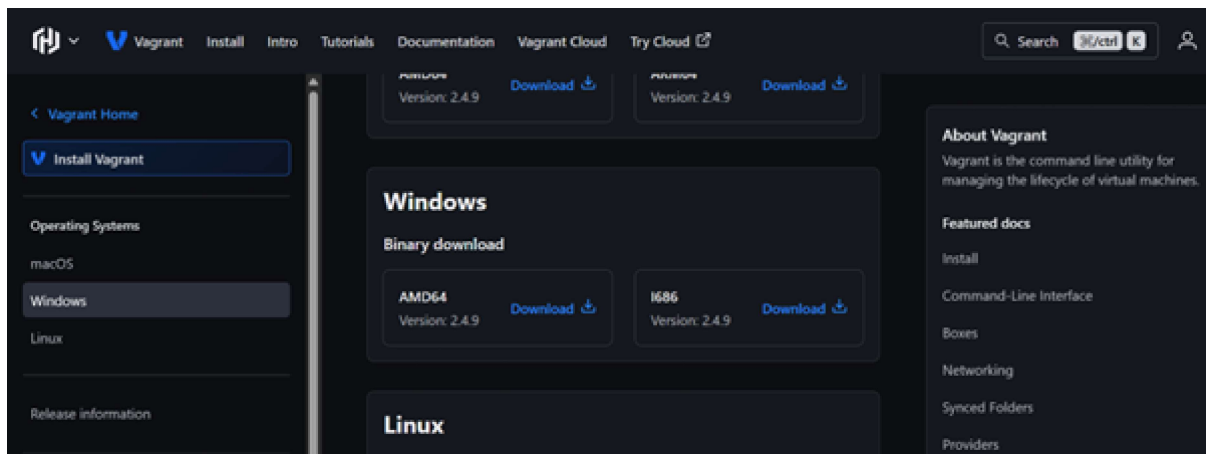
<https://upessocs.github.io/#dir=/Lectures/Containerization%20and%20DevOps/Lab/&file=Experiment%201%20Compare%20VM%20with%20Container.md>

Objective

1. To understand the conceptual and practical differences between Virtual Machines and Containers.
2. To install and configure a Virtual Machine using VirtualBox and Vagrant on Windows.
3. To install and configure Containers using Docker inside WSL.
4. To deploy an Ubuntu-based Nginx web server in both environments.
5. To compare resource utilization, performance, and operational characteristics of Virtual Machines and Containers.

Part A: Virtual Machine (Windows)

1. Install Vagrant for Windows and Verify Installation



```
vagrant@vagrant: ~
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Vanshika Munjal\Desktop\C&D Lab> vagrant --v
Vagrant 2.4.9
```

2. Initialize Vagrant with Ubuntu Box

```
PS C:\Users\Vanshika Munjal\Desktop\C&D Lab> vagrant init hashicorp/bionic64
A 'Vagrantfile' has been placed in this directory. You are now
ready to 'vagrant up' your first virtual environment! Please read
the comments in the Vagrantfile as well as documentation on
'vagrantup.com' for more information on using Vagrant.
```

3. Start the Virtual Machine

```
PS C:\Users\Vanshika Munjal\Desktop\C&D Lab> vagrant up
Bringing machine 'default' up with 'virtualbox' provider...
==> default: Box 'hashicorp/bionic64' could not be found. Attempting to find and install...
default: Box Provider: virtualbox
default: Box Version: >= 0
==> default: Loading metadata for box 'hashicorp/bionic64'
default: URL: https://vagrantcloud.com/api/v2/vagrant/hashicorp/bionic64
==> default: Adding box 'hashicorp/bionic64' (v1.0.282) for provider: virtualbox
default: Downloading: https://vagrantcloud.com/hashicorp/boxes/bionic64/versions/1.0.282/providers/virtualbox/unknown/vagrant.box
default:
==> default: Successfully added box 'hashicorp/bionic64' (v1.0.282) for 'virtualbox'!
==> default: Importing base box 'hashicorp/bionic64'...
==> default: Matching MAC address for NAT networking...
==> default: Checking if box 'hashicorp/bionic64' version '1.0.282' is up to date...
==> default: Setting the name of the VM: CDLab.default_1769224438570_76528
Vagrant is currently configured to create VirtualBox synced folders with
the 'SharedFoldersEnableSymlinksCreate' option enabled. If the Vagrant
guest is not trusted, you may want to disable this option. For more
information on this option, please refer to the VirtualBox manual:

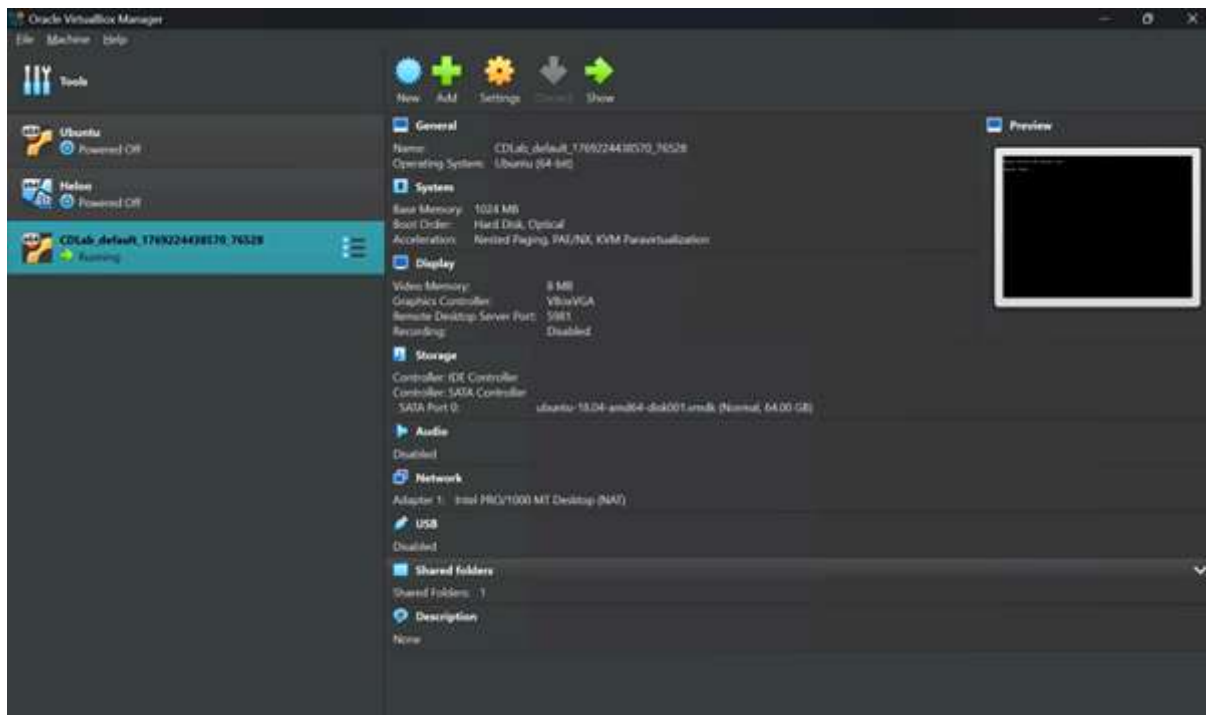
https://www.virtualbox.org/manual/ch04.html#sharedfolders

This option can be disabled globally with an environment variable:

VAGRANT_DISABLE_VBOXSYMLINKCREATE=1

or on a per folder basis within the Vagrantfile:

config.vm.synced_folder '/host/path', '/guest/path', SharedFoldersEnableSymlinksCreate: false
==> default: Clearing any previously set network interfaces...
==> default: Preparing network interfaces based on configuration...
default: Adapter 1: nat
==> default: Forwarding ports...
default: 22 (guest) => 2222 (host) (adapter 1)
==> default: Booting VM...
==> default: Waiting for machine to boot. This may take a few minutes...
default: SSH address: 127.0.0.1:2222
```



4. Access the Virtual Machine

```
PS C:\Users\Vanshika Munjal\Desktop\C&D Lab> vagrant ssh
vagrant@127.0.0.1's password:
Welcome to Ubuntu 18.04.3 LTS (GNU/Linux 4.15.0-58-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

System information as of Sat Jan 24 03:16:24 UTC 2026

System load:  0.2               Processes:    92
Usage of /:   2.5% of 61.80GB   Users logged in:  0
Memory usage: 11%              IP address for eth0: 10.0.2.15
Swap usage:  0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

0 packages can be updated.
0 updates are security updates.
```

5. Install Nginx inside the Virtual Machine


```
vagrant@vagrant:~$ sudo apt update
Get:1 http://security.ubuntu.com/ubuntu bionic-security InRelease [102 kB]
Hit:2 http://archive.ubuntu.com/ubuntu bionic InRelease
Get:3 http://archive.ubuntu.com/ubuntu bionic-updates InRelease [102 kB]
Get:4 http://archive.ubuntu.com/ubuntu bionic-backports InRelease [102 kB]
Get:5 http://security.ubuntu.com/ubuntu bionic-security/main i386 Packages [1,380 kB]
Get:6 http://archive.ubuntu.com/ubuntu bionic-updates/main i386 Packages [1,666 kB]
Get:7 http://security.ubuntu.com/ubuntu bionic-security/main amd64 Packages [2,717 kB]
Get:8 http://archive.ubuntu.com/ubuntu bionic-updates/main amd64 Packages [3,045 kB]
Get:9 http://security.ubuntu.com/ubuntu bionic-security/main Translation-en [468 kB]
Get:10 http://security.ubuntu.com/ubuntu bionic-security/restricted i386 Packages [33.0 kB]
Get:11 http://security.ubuntu.com/ubuntu bionic-security/restricted amd64 Packages [1,317 kB]
Get:12 http://archive.ubuntu.com/ubuntu bionic-updates/main Translation-en [554 kB]
Get:13 http://archive.ubuntu.com/ubuntu bionic-updates/restricted i386 Packages [39.7 kB]
Get:14 http://archive.ubuntu.com/ubuntu bionic-updates/restricted amd64 Packages [1,347 kB]
Get:15 http://security.ubuntu.com/ubuntu bionic-security/restricted Translation-en [182 kB]
Get:16 http://security.ubuntu.com/ubuntu bionic-security/universe i386 Packages [1,078 kB]
Get:17 http://archive.ubuntu.com/ubuntu bionic-updates/restricted Translation-en [187 kB]
Get:18 http://security.ubuntu.com/ubuntu bionic-security/universe amd64 Packages [1,303 kB]
Get:19 http://archive.ubuntu.com/ubuntu bionic-updates/universe amd64 Packages [1,915 kB]
Get:20 http://security.ubuntu.com/ubuntu bionic-security/universe Translation-en [308 kB]
Get:21 http://security.ubuntu.com/ubuntu bionic-security/multiverse i386 Packages [6,008 B]
Get:22 http://security.ubuntu.com/ubuntu bionic-security/multiverse amd64 Packages [19.8 kB]
Get:23 http://security.ubuntu.com/ubuntu bionic-security/multiverse Translation-en [3,928 B]
Get:24 http://archive.ubuntu.com/ubuntu bionic-updates/universe i386 Packages [1,663 kB]
Get:25 http://archive.ubuntu.com/ubuntu bionic-updates/universe Translation-en [421 kB]
```

```
vagrant@vagrant:~$ sudo apt install -y nginx
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  fontconfig-config fonts-dejavu-core libfontconfig1 libgd3 libjpeg8 libjpeg-turbo8 libpng16-0 libnginx-mod-http-geoip
  libnginx-mod-http-image-filter libnginx-mod-http-xslt-filter libnginx-mod-mail libnginx-mod-stream libtiff5 libwebp6 libx11-6
  libx11-data libxcb1 libxpm4 nginx-common nginx-core
Suggested packages:
  libgd-tools fcgiwrap nginx-doc ssl-cert
The following NEW packages will be installed:
  fontconfig-config fonts-dejavu-core libfontconfig1 libgd3 libjpeg8 libjpeg-turbo8 libpng16-0 libnginx-mod-http-geoip
  libnginx-mod-http-image-filter libnginx-mod-http-xslt-filter libnginx-mod-mail libnginx-mod-stream libtiff5 libwebp6 libx11-6
  libx11-data libxcb1 libxpm4 nginx nginx-common nginx-core
0 upgraded, 21 newly installed, 0 to remove and 294 not upgraded.
Need to get 3,195 kB of archives.
After this operation, 11.4 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu bionic-updates/main amd64 libjpeg-turbo8 amd64 1.5.2-0ubuntu5.18.04.6 [111 kB]
Get:2 http://archive.ubuntu.com/ubuntu bionic-updates/main amd64 libx11-6 amd64 1:1.8.4-0ubuntu1 [15.5 kB]
```

```
vagrant@vagrant:~$ sudo systemctl start nginx
```

6. Verify Nginx Installation

```
vagrant@vagrant:~$ curl localhost
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
  body {
    width: 35em;
    margin: 0 auto;
    font-family: Tahoma, Verdana, Arial, sans-serif;
  }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
```

7. VM Observation Commands

```
vagrant@vagrant:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:           985M        67M        440M          4.9M        477M        775M
Swap:          979M          0B        979M
```

```
CPU: 1.3% Tasks: 25, 17 thr; 1 running
Mem: 73.1M/985M Load average: 0.05 0.18 0.09
Sep: 0M/989M Uptime: 00:03:58
```

PID	USER	PR	NI	VIRT	RES	SHR	%CPU	MEM%	TIME+	Command
1009	vagrant	20	0	26980	4232	3400	0.8	0.4	0:00.01	htop
1	root	20	0	155M	8808	6676	0.0	0.9	0:04.05	/sbin/init
404	root	19	-1	80604	13536	12844	0.0	1.3	0:00.19	/lib/systemd/systemd-journald
420	root	20	0	103M	1776	1552	0.0	0.2	0:00.00	/sbin/lvmtd -f
421	root	20	0	46112	4988	3140	0.0	0.5	0:00.55	/lib/systemd/systemd-udevd
455	root	20	0	47600	3552	3152	0.0	0.4	0:00.03	/sbin/rpcbind -f -w
464	systemd-n	20	0	80044	5436	4832	0.0	0.5	0:00.12	/lib/systemd/systemd-networkd
478	systemd-r	20	0	70628	5148	4592	0.0	0.5	0:00.06	/lib/systemd/systemd-resolved
617	syslog	20	0	256M	4372	2600	0.0	0.4	0:00.00	/usr/sbin/rsyslogd -n
618	syslog	20	0	256M	4372	2600	0.0	0.4	0:00.00	/usr/sbin/rsyslogd -n
619	syslog	20	0	256M	4372	2600	0.0	0.4	0:00.00	/usr/sbin/rsyslogd -n
612	syslog	20	0	256M	4372	2600	0.0	0.4	0:00.03	/usr/sbin/rsyslogd -n
615	root	20	0	90540	1604	1472	0.0	0.2	0:00.00	/usr/bin/lsnfs /var/lib/lsnfs/
616	root	20	0	90540	1604	1472	0.0	0.2	0:00.00	/usr/bin/lsnfs /var/lib/lsnfs/
613	root	20	0	90540	1604	1472	0.0	0.2	0:00.00	/usr/bin/lsnfs /var/lib/lsnfs/
614	messagebu	20	0	50000	4060	3768	0.0	0.4	0:00.10	/usr/bin/dbus-daemon --system --address=systemd: --nofork --nopidfile
779	root	20	0	166M	17364	9580	0.0	1.7	0:00.00	/usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-triggers
627	root	20	0	166M	17364	9580	0.0	1.7	0:00.29	/usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-triggers
630	daemon	20	0	28332	2452	2248	0.0	0.2	0:00.00	/usr/sbin/atd -f
631	root	20	0	31320	3248	2968	0.0	0.3	0:00.00	/usr/sbin/cron -f
632	root	20	0	70608	6112	5384	0.0	0.6	0:00.06	/lib/systemd/systemd-logind
650	root	20	0	280M	6804	5952	0.0	0.7	0:00.02	/usr/lib/accounts-service/accounts-daemon
657	root	20	0	285M	6804	5952	0.0	0.7	0:00.00	/usr/lib/accounts-service/accounts-daemon
633	root	20	0	285M	6804	5952	0.0	0.7	0:00.08	/usr/lib/accounts-service/accounts-daemon
674	root	20	0	16180	1968	1832	0.0	0.2	0:00.01	/sbin/agetty -o -p -- \u --noclear tty1 linux
686	root	20	0	284M	7224	6176	0.0	0.7	0:00.00	/usr/lib/policykit-1/polkitd --no-debug
688	root	20	0	284M	7224	6176	0.0	0.7	0:00.00	/usr/lib/policykit-1/polkitd --no-debug
682	root	20	0	284M	7224	6176	0.0	0.7	0:00.11	/usr/lib/policykit-1/polkitd --no-debug
909	root	20	0	241M	2532	2172	0.0	0.3	0:00.00	/usr/sbin/VBoxService --pidfile /var/run/vboxadd-service.sh
950	root	20	0	241M	2532	2172	0.0	0.3	0:00.00	/usr/sbin/VBoxService --pidfile /var/run/vboxadd-service.sh

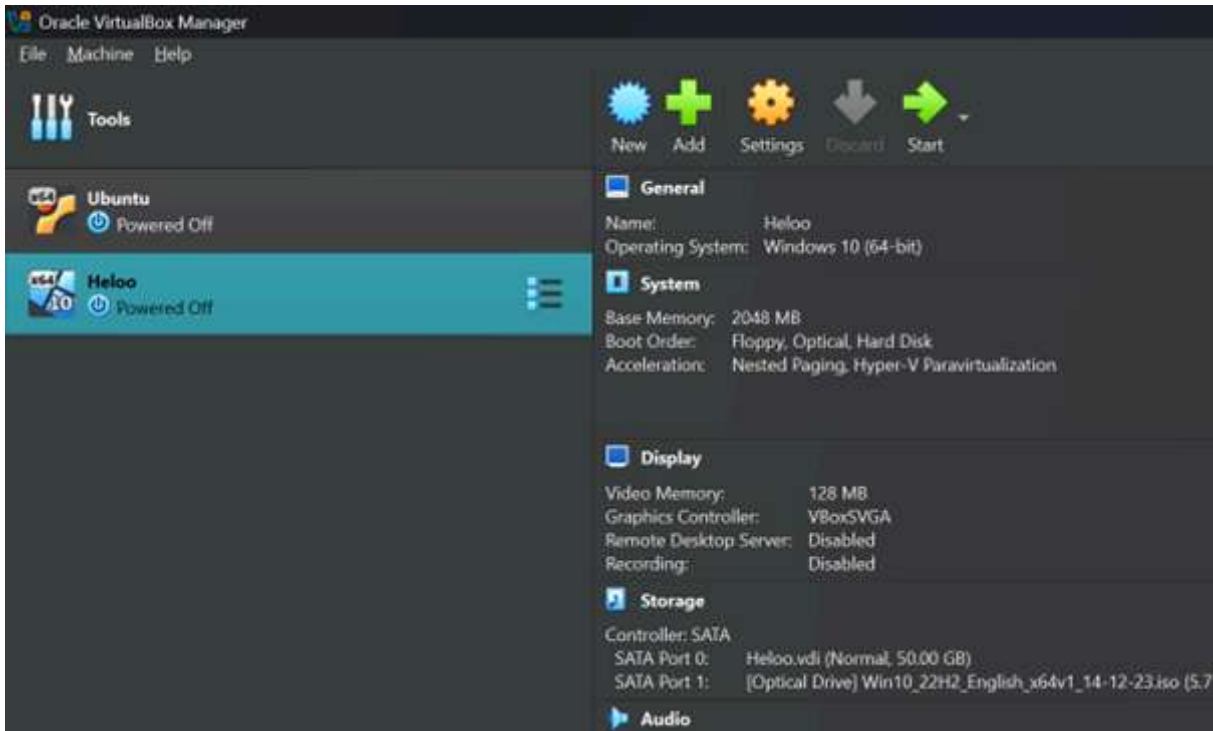
```
vagrant@vagrant:~$ systemd-analyze
Startup finished in 6.925s (kernel) + 32.585s (userspace) = 39.510s
```

8. Exit, Stop, and Remove the Virtual Machine

```
vagrant@vagrant:~$ exit
logout
PS C:\Users\Vanshika Munjal\Desktop\C&D Lab> vagrant halt
==> default: Attempting graceful shutdown of VM...
```



```
PS C:\Users\Vanshika Munjal\Desktop\C&D Lab> vagrant destroy
default: Are you sure you want to destroy the 'default' VM? [y/N] y
==> default: Destroying VM and associated drives...
PS C:\Users\Vanshika Munjal\Desktop\C&D Lab> |
```



Part B: Containers using WSL (Windows)

1. Install Docker Engine inside WSL

```
PS C:\Users\HP> wsl
vanshika_munj1@DESKTOP-1D7G070:/mnt/c/Users/HP$ sudo apt update
[sudo] password for vanshika_munj1:
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1489 kB]
Get:5 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [229 kB]
Get:6 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [9808 B]
Get:8 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [924 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [289 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.3 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [19.7 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [208 B]
Get:13 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [28.0 kB]
Get:14 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [6472 B]
Get:15 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [208 B]
Get:16 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1699 kB]
Get:18 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [318 kB]

vanshika_munj1@DESKTOP-1D7G070:/mnt/c/Users/HP$ sudo apt install -y docker.io
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
docker.io is already the newest version (28.2.2-0ubuntu1~24.04.1).
0 upgraded, 0 newly installed, 0 to remove and 121 not upgraded.
vanshika_munj1@DESKTOP-1D7G070:/mnt/c/Users/HP$ sudo systemctl start docker
vanshika_munj1@DESKTOP-1D7G070:/mnt/c/Users/HP$ sudo usermod -aG docker $USER
```


2. Run Ubuntu Container with Nginx

```

vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/HP$ docker pull ubuntu
Using default tag: latest
permission denied while trying to connect to the Docker daemon socket at unix:///var/run/docker.sock: Post "http://%2Fvar%2Frun%2Fdocker.sock/v1.50/images/create?fromImage=docker.io%2Flibrary%2Fubuntu&tag=latest": dial unix /var/run/docker.sock: connect: permission denied
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/HP$ sudo docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
Digest: sha256:cddba651b3880c3686ecf4e3c4220f026b521fb76978881737d24f208828b2b
Status: Image is up to date for ubuntu:latest
docker.io/library/ubuntu:latest
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/HP$ docker run -d -p 8080:80 --name nginx-container nginx
docker: permission denied while trying to connect to the Docker daemon socket at unix:///var/run/docker.sock: Head "http://%2Fvar%2Frun%2Fdocker.sock/_ping": dial unix /var/run/docker.sock: connect: permission denied

Run 'docker run --help' for more information
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/HP$ sudo docker run -d -p 8080:80 --name nginx-container nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
119d43ee815: Pull complete
780146c8ad64: Pull complete
d989180b8a84: Pull complete
560799c38424: Pull complete
10b68cfefe1: Pull complete
57f0dd1befe2: Pull complete
eaf8753feae0: Pull complete
Digest: sha256:c881927c4077710ac6b1da63b83aal63937fb47457950c267d92f7e4dedf4aac
Status: Downloaded newer image for nginx:latest
3506c46dc18ad0745553bb4aee9090b72fd5c8f838d2699883d5afal4057b693

```

3. Verify Nginx in Container

```

vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/HP$ curl localhost:8080
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>

```

4. Container Observation Commands

```

vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/HP$ sudo docker stats

```

CONTAINER ID	NAME	CPU %	MEM USAGE / LIMIT	MEM %	NET I/O	BLOCK I/O	PIOS
3506c46dc18a	nginx-container	0.08%	10.47MiB / 7.697GiB	0.13%	2.02kB / 1.4kB	0B / 12.3kB	13

```

vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/HP$ free -h

```

	total	used	free	shared	buff/cache	available
Mem:	7.6Gi	573Mi	6.9Gi	3.6Mi	294Mi	7.0Gi
Swap:	2.0Gi	0B	2.0Gi			

```

vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/HP$

```

Result

The experiment demonstrates that containers are significantly more lightweight and resource-efficient compared to virtual machines, while virtual machines provide stronger isolation and full OS-level abstraction.

Conclusion

Virtual Machines are suitable for full OS isolation and legacy workloads, whereas Containers are ideal for microservices, rapid deployment, and efficient resource utilization.

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CONTAINERIZATION AND DEVOPS LAB

Experiment 2:

Docker Installation, Configuration, and Running Images

Reference Link:

<https://share.google/Jqjmq73cZkjqQ4xv6>

Objective

1. Pull docker images.
2. Run containers.
3. Manage container lifecycle

1. Pull Image

```
PS C:\Users\HP> wsl
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ docker -v
Docker version 28.2.2, build 28.2.2-0ubuntu1-24.04.1
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
Digest: sha256:c881927c4077710ac4b1da63b83aal63937fb47457950c267d92f7e4dedf4aec
Status: Image is up to date for nginx:latest
docker.io/library/nginx:latest
```

2. Run container with port mapping

```
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ docker run -d -p 8080:80 nginx
5ce3887ef93033fdcf7641137a8977238f1025fd6f2e628ca471190a406d8c89
```

3. Verify running containers

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
5ce3887ef93033fdcf7641137a8977238f1025fd6f2e628ca471190a406d8c89	nginx	"/docker-entrypoint..."	12 seconds ago	Up 11 seconds	0.0.0.0:8080->80/tcp, [::]:8080->80/tcp	vibrant_blackburn

4. Stop and Remove Container

```

vanshika_munj1@DESKTOP-107G070:/mnt/c/Users/HP$ docker stop 5ce38
5ce38
vanshika_munj1@DESKTOP-107G070:/mnt/c/Users/HP$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS   NAMES
vanshika_munj1@DESKTOP-107G070:/mnt/c/Users/HP$ docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS   NAMES
5ce3857ef938   nginx     "/docker-entrypoint.…"   43 seconds ago   Exited (0) 14 seconds ago   vibrant_blackbur
n
299349015eb0   java-app:1.0   "bash"     20 hours ago   Exited (0) 20 hours ago   stoic_taussig
9b56a1ceb40d   java-app:1.0   "java Hello"   20 hours ago   Exited (0) 20 hours ago   trusting_cannon
5d3a38405fda   java-app:1.0   "bash"     2 days ago   Exited (129) 33 hours ago   compassionate_th
ompson
bf8666bdd98b   java-app:1.0   "java Hello"   2 days ago   Exited (0) 2 days ago   magical_engelbar
t
2f58b4cefd52   myrepo/java-app:1.0   "bash"     3 days ago   Exited (127) 3 days ago   gracious_vaughan
3506c46dc18a   nginx     "/docker-entrypoint.…"   6 days ago   Exited (0) 6 days ago   nginx-container
b33fa91cdb51   ubuntu    "bash"     7 days ago   Exited (255) 2 days ago   javac
eae3b1a2f374   ubuntu    "/bin/bash"   8 days ago   Exited (255) 7 days ago   gifted_yonath
22a45aef84f6   ubuntu    "/bin/bash"   8 days ago   Exited (0) 8 days ago   dazzling_johnson
1519b58e8caa   1b44b5a3e06a   "/hello"     9 days ago   Exited (0) 8 days ago   kind_zhukovsky
vanshika_munj1@DESKTOP-107G070:/mnt/c/Users/HP$ docker rm 5ce38
5ce38
vanshika_munj1@DESKTOP-107G070:/mnt/c/Users/HP$ docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS   NAMES
299349015eb0   java-app:1.0   "bash"     20 hours ago   Exited (0) 20 hours ago   stoic_taussig
9b56a1ceb40d   java-app:1.0   "java Hello"   20 hours ago   Exited (0) 20 hours ago   trusting_cannon
5d3a38405fda   java-app:1.0   "bash"     2 days ago   Exited (129) 33 hours ago   compassionate_th
ompson
bf8666bdd98b   java-app:1.0   "java Hello"   2 days ago   Exited (0) 2 days ago   magical_engelbart
2f58b4cefd52   myrepo/java-app:1.0   "bash"     3 days ago   Exited (127) 3 days ago   gracious_vaughan
3506c46dc18a   nginx     "/docker-entrypoint.…"   6 days ago   Exited (0) 6 days ago   nginx-container
b33fa91cdb51   ubuntu    "bash"     7 days ago   Exited (255) 2 days ago   javac
eae3b1a2f374   ubuntu    "/bin/bash"   8 days ago   Exited (255) 7 days ago   gifted_yonath
22a45aef84f6   ubuntu    "/bin/bash"   8 days ago   Exited (0) 8 days ago   dazzling_johnson
1519b58e8caa   1b44b5a3e06a   "/hello"     9 days ago   Exited (0) 8 days ago   kind_zhukovsky

```

5. Remove Image

```

vanshika_munj1@DESKTOP-107G070:/mnt/c/Users/HP$ docker rmi nginx
Error response from daemon: conflict: unable to remove repository reference "nginx" (must force) - container 3506c46dc18a is using it
s referenced image 4af177a024eb
vanshika_munj1@DESKTOP-107G070:/mnt/c/Users/HP$ sudo docker rmi nginx
[sudo] password for vanshika_munj1:
Error response from daemon: conflict: unable to remove repository reference "nginx" (must force) - container 3506c46dc18a is using it
s referenced image 4af177a024eb
vanshika_munj1@DESKTOP-107G070:/mnt/c/Users/HP$ sudo docker rmi nginx -f
Untagged: nginx:latest
Untagged: nginx@sha256:c881927c4077710ac4b1da63b83aa163937fb47457950c267d92f7e4dedf4aec
Deleted: sha256:4af177a024eb8a1e43f4fb6c66735bb8260115cb5925a64f51673219bd97c144
vanshika_munj1@DESKTOP-107G070:/mnt/c/Users/HP$ |

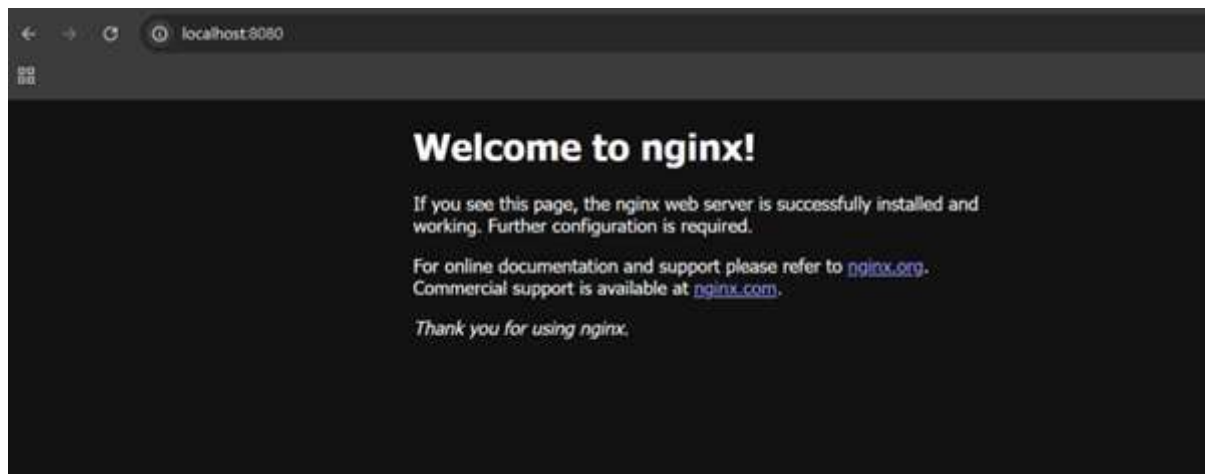
```

6. Verify

```

vanshika_munj1@DESKTOP-107G070:/mnt/c/Users/HP$ docker run -d -p 8080:80 nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
119d43e0c815: Already exists
700146c8ad64: Already exists
d989109b8a84: Already exists
500799c30424: Already exists
10b68cfefee1: Already exists
57f0dd1befe2: Already exists
eaf8753feae8: Already exists
Digest: sha256:c881927c4077710ac4b1da63b83aa163937fb47457950c267d92f7e4dedf4aec
Status: Downloaded newer image for nginx:latest
749fbde8cdd5edd8f83fee07ea6603f261c5f8c29f13940543b32c7e4f0b358c1
vanshika_munj1@DESKTOP-107G070:/mnt/c/Users/HP$ curl -I localhost:8080
HTTP/1.1 200 OK
Server: nginx/1.29.4
Date: Sat, 31 Jan 2026 03:07:09 GMT
Content-Type: text/html
Content-Length: 615
Last-Modified: Tue, 09 Dec 2025 18:28:10 GMT
Connection: keep-alive
ETag: "69385a3a-267"
Accept-Ranges: bytes
vanshika_munj1@DESKTOP-107G070:/mnt/c/Users/HP$ |

```

Result

Docker images were successfully pulled, containers executed, and lifecycle commands performed.

Conclusion

This lab demonstrated virtualization using **Vagrant** + **VirtualBox** and containerization using **Docker**, highlighting clear performance and resource efficiency differences. Containers are better suited for rapid deployment and microservices, while virtual machines provide stronger isolation.

vanshikamunjal665

CONTAINERIZATION AND DEVOPS LAB

Experiment 3:

Deploying Nginx using different base images and comparing image layers

Reference Link:

<https://upessocs.github.io/#dir=/Lectures/Containerization%20and%20DevOps/Theory/Containerization%20and%20DevOps/Lab/&file=Experiment%203%20Nginx.md>

Objective

1. Deploy NGINX using:
 - Official `nginx` image
 - Ubuntu-based image
 - Alpine-based image
2. Understand Docker image layers and size differences
3. Compare performance, security, and use-cases of each approach
4. Explain real-world use of NGINX in containerized system

Part A: Deploy Nginx using official image

0. Starting docker demon first

```

Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\HP> wsl
vanshika_munjaj@DESKTOP-107G070:/mnt/c/Users/HP$ docker pull nginx:latest
Cannot connect to the Docker daemon at unix:///var/run/docker.sock. Is the docker daemon running?
vanshika_munjaj@DESKTOP-107G070:/mnt/c/Users/HP$ sudo apt update && sudo apt upgrade -y
[sudo] password for vanshika_munjaj:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1739 kB]
Get:6 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.2 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:9 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1528 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:15 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7296 B]
Get:16 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [10.5 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
Get:18 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Fetched 4322 kB in 15s (293 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
81 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done

```

```

vanshika_munjaj@DESKTOP-107G070:/mnt/c/Users/HP$ sudo apt install docker.io -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
docker.io is already the newest version (28.2.2-0ubuntu1-24.04.1).
The following packages were automatically installed and are no longer required:
  libllvm19 libwayland-server0
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
vanshika_munjaj@DESKTOP-107G070:/mnt/c/Users/HP$ sudo service socker start
Failed to start socker.service: Unit socker.service not found.
vanshika_munjaj@DESKTOP-107G070:/mnt/c/Users/HP$ sudo service docker start
Job for docker.service failed because the control process exited with error code.
See "systemctl status docker.service" and "journalctl -xeu docker.service" for details.
vanshika_munjaj@DESKTOP-107G070:/mnt/c/Users/HP$ sudo service docker status
* docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: failed (Result: exit-code) since Sat 2026-02-07 03:41:04 UTC; 1s ago
   TriggeredBy: * docker.socket
     Docs: https://docs.docker.com
    Process: 5213 ExecStart=/usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock (code=exited, status=1/FAILURE)
   Main PID: 5213 (code=exited, status=1/FAILURE)
    CPU: 86ms

Feb 07 03:41:04 DESKTOP-107G070 systemd[1]: docker.service: Scheduled restart job, restart counter is at 3.
Feb 07 03:41:04 DESKTOP-107G070 systemd[1]: docker.service: Start request repeated too quickly.
Feb 07 03:41:04 DESKTOP-107G070 systemd[1]: docker.service: Failed with result 'exit-code'.
Feb 07 03:41:04 DESKTOP-107G070 systemd[1]: Failed to start docker.service - Docker Application Container Engine.
vanshika_munjaj@DESKTOP-107G070:/mnt/c/Users/HP$ sudo dockerd
INFO[2026-02-07T03:41:40.523209778Z] Starting up
WARN[2026-02-07T03:41:40.526648630Z] Binding to IP address without --tlsverify is insecure and gives root access on this machine to e
everyone who has access to your network. host="tcp://0.0.0.0:2375"
WARN[2026-02-07T03:41:40.526823200Z] Binding to an IP address, even on localhost, can also give access to scripts run in a browser. B
e safe out there! host="tcp://0.0.0.0:2375"
WARN[2026-02-07T03:41:40.526855029Z] [DEPRECATION NOTICE] In future versions this will be a hard failure preventing the daemon from s
tarting! Learn more at: https://docs.docker.com/go/api-security/ host="tcp://0.0.0.0:2375"
WARN[2026-02-07T03:41:41.527784584Z] Binding to an IP address without --tlsverify is deprecated. Startup is intentionally being slowe

```

```

In future versions this will be a hard failure preventing the daemon from starting! Learn more at: https://docs.docker.com/go/api-security/
INFO[2026-02-07T03:41:57.729157882Z] Docker daemon
commit="28.2.2-0ubuntu1-24.04.1" containerd-snapsh
otter=false storage-driver=overlay2 version=28.2.2
INFO[2026-02-07T03:41:57.729820943Z] Initializing buildkit
WARN[2026-02-07T03:41:57.734506206Z] CDI setup error /var/run/cdi: failed to monitor for changes: no such file or directory
WARN[2026-02-07T03:41:57.734578979Z] CDI setup error /etc/cdi: failed to monitor for changes: no such file or directory
INFO[2026-02-07T03:41:57.769327979Z] Completed buildkit initialization
INFO[2026-02-07T03:41:57.780196827Z] Daemon has completed initialization
INFO[2026-02-07T03:41:57.780363578Z] API listen on [::]:2375
INFO[2026-02-07T03:41:57.780367382Z] API listen on /var/run/docker.sock
*INFO[2026-02-07T03:42:39.663399148Z] Processing signal 'interrupt'
INFO[2026-02-07T03:42:39.668738148Z] stopping event stream following graceful shutdown error="<nil>" module=libcontainerd namespace=
moby
INFO[2026-02-07T03:42:39.672800073Z] Daemon shutdown complete
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HF$ sudo service docker status
* docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: failed (Result: exit-code) since Sat 2026-02-07 03:41:04 UTC; 1min 45s ago
   TriggeredBy: * docker.socket
   Docs: https://docs.docker.com
   Process: 5213 ExecStart=/usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock (code=exited, status=1/FAILURE)
   Main PID: 5213 (code=exited, status=1/FAILURE)
   CPU: 86ms

Feb 07 03:41:04 DESKTOP-107G070 systemd[1]: docker.service: Scheduled restart job, restart counter is at 3.
Feb 07 03:41:04 DESKTOP-107G070 systemd[1]: docker.service: Start request repeated too quickly.
Feb 07 03:41:04 DESKTOP-107G070 systemd[1]: docker.service: Failed with result 'exit-code'.
Feb 07 03:41:04 DESKTOP-107G070 systemd[1]: Failed to start docker.service - Docker Application Container Engine.
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HF$ docker ps
Cannot connect to the Docker daemon at unix:///var/run/docker.sock. Is the docker daemon running?
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HF$ sudo dockerd
INFO[2026-02-07T03:43:28.087795390Z] Starting up
WARN[2026-02-07T03:43:28.088212888Z] Binding to IP address without --tlsverify is insecure and gives root access on this machine to e
veryone who has access to your network. host="tcp://0.0.0.0:2375"
WARN[2026-02-07T03:43:28.088248900Z] Binding to an IP address, even on localhost, can also give access to scripts run in a browser. B
e safe out there! host="tcp://0.0.0.0:2375"

```

```

Starting! Learn more at: https://docs.docker.com/go/api-security/ host="tcp://0.0.0.0:2375"
WARN[2026-02-07T03:43:29.088959757Z] Binding to an IP address without --tlsverify is deprecated. Startup is intentionally being slowe
d down to show this message. host="tcp://0.0.0.0:2375"
WARN[2026-02-07T03:43:29.089057763Z] Please consider generating tls certificates with client validation to prevent exposing unauthent
icated root access to your network. host="tcp://0.0.0.0:2375"
WARN[2026-02-07T03:43:29.089102733Z] You can override this by explicitly specifying '--tls=false' or '--tlsverify=false' host="tcp://
0.0.0.0:2375"
WARN[2026-02-07T03:43:29.089153940Z] Support for listening on TCP without authentication or explicit intent to run without authentica
tion will be removed in the next release. host="tcp://0.0.0.0:2375"
INFO[2026-02-07T03:43:44.111692537Z] OTEL tracing is not configured, using no-op tracer provider
INFO[2026-02-07T03:43:44.170426414Z] Creating a containerd client address=/run/containerd/containerd.sock timeout=1m
0s
INFO[2026-02-07T03:43:44.216639251Z] [graphdriver] using prior storage driver: overlay2
INFO[2026-02-07T03:43:44.272082259Z] Loading containers: start.
WARN[2026-02-07T03:43:45.546932933Z] Error (Unable to complete atomic operation, key modified) deleting object [endpoint_count 02fddf
545c206c819fb362b793d4cb63787e53792449debb0f8b891216e66622], retrying....
WARN[2026-02-07T03:43:45.604454879Z] error locating sandbox id 1104bf0599c4379d2f02ad113de9e517d38bf6876cb0840cd6de173148cc27f: sand
box 1104bf0599c4379d2f02ad113de9e517d38bf6876cb0840cd6de173148cc27f not found
WARN[2026-02-07T03:43:45.604503680Z] error locating sandbox id 8b14cd7d50b209da2ceb4ee073d384e164e772563d11dcde517b3d5af8b38e23: sand
box 8b14cd7d50b209da2ceb4ee073d384e164e772563d11dcde517b3d5af8b38e23 not found
INFO[2026-02-07T03:43:45.605166448Z] Loading containers: done.
WARN[2026-02-07T03:43:45.629099522Z] [DEPRECATION NOTICE]: API is accessible on http://0.0.0.0:2375 without encryption.
Access to the remote API is equivalent to root access on the host. Refer
to the 'Docker daemon attack surface' section in the documentation for
more information: https://docs.docker.com/go/attack-surface/
In future versions this will be a hard failure preventing the daemon from starting! Learn more at: https://docs.docker.com/go/api-security/
INFO[2026-02-07T03:43:45.629173966Z] Docker daemon
commit="28.2.2-0ubuntu1-24.04.1" containerd-snapsh
otter=false storage-driver=overlay2 version=28.2.2
INFO[2026-02-07T03:43:45.629247475Z] Initializing buildkit
WARN[2026-02-07T03:43:45.633516206Z] CDI setup error /etc/cdi: failed to monitor for changes: no such file or directory
WARN[2026-02-07T03:43:45.633562184Z] CDI setup error /var/run/cdi: failed to monitor for changes: no such file or directory
INFO[2026-02-07T03:43:45.675438568Z] Completed buildkit initialization
INFO[2026-02-07T03:43:45.693175527Z] Daemon has completed initialization
INFO[2026-02-07T03:43:45.693374584Z] API listen on [::]:2375
INFO[2026-02-07T03:43:45.693450648Z] API listen on /var/run/docker.sock

```

1. Pull the Image


```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\HP> wsl
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/HP$ docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS        NAMES
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/HP$ docker pull nginx:latest
latest: Pulling from library/nginx
bc8d55a45c0d: Already exists
46bf3a120c8e: Pull complete
4f4efe02d542: Pull complete
7b6cb8ccac7b: Pull complete
773400a233fd: Pull complete
47cd406a84ef: Pull complete
bae5a1799a80: Pull complete
Digest: sha256:341bf0f3ce6c5277d6002cf6e1fb0319fa4252add24ab6a0e262e0056d313208
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
```

2. Run Container

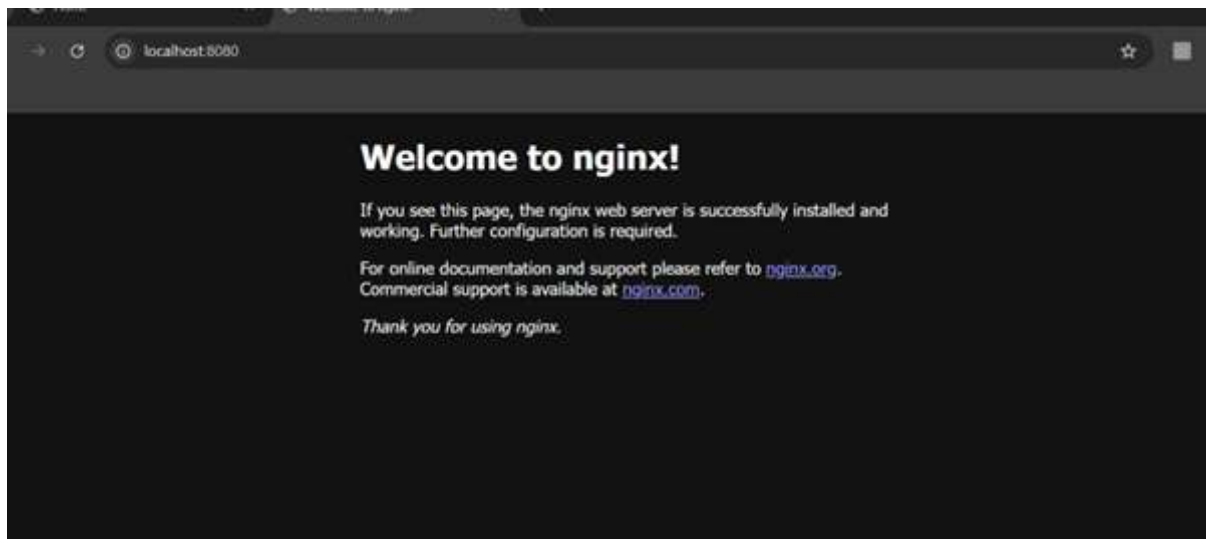
```
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/HP$ docker run -d --name nginx-official -p 8080:80 nginx
f156a23b33187ffefbb6813585d368a1ca45889fa07dbe5f41416304930981a1
```

3. Verify

```
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/HP$ curl http://localhost:8080
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
```

4. Key Observations

```
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/HP$ docker images nginx
REPOSITORY   TAG       IMAGE ID       CREATED        SIZE
nginx        latest    5cdef4ac3335   2 days ago    161MB
nginx        <none>    248d2326f351   4 days ago    161MB
nginx        <none>    4af177a824eb   3 weeks ago    161MB
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/HP$
```

Part B: Custom Nginx using Ubuntu base image

1. Create Dockerfile

File Edit View

```
FROM ubuntu:22.04
```

```
RUN apt-get update && \  
    apt-get install -y nginx && \  
    apt-get clean && \  
    rm -rf /var/lib/apt/lists/*
```

```
EXPOSE 80
```

```
CMD ["nginx", "-g", "daemon off;"]
```

2. Change Directory

```
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/HP$ cd "/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp3 file"  
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp3 file$ ls  
Dockerfile.txt  
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp3 file$ mv Dockerfile.txt Dockerfile  
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp3 file$ ls  
Dockerfile
```

3. Build Image

```

vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp3 file$ docker build -t nginx-ubuntu .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  2.048kB
Step 1/4 : FROM ubuntu:22.04
--> 65c77cbc27c2
Step 2/4 : RUN apt-get update &&          apt-get install -y nginx &&          apt-get clean &&          rm -rf /var/lib/apt/lists/*
--> Running in dce1271aca8b
Get:1 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Get:2 http://archive.ubuntu.com/ubuntu jammy InRelease [270 kB]
Get:3 http://security.ubuntu.com/ubuntu jammy-security/multiverse amd64 Packages [62.6 kB]
Get:4 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages [3677 kB]
Get:5 http://archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:6 http://archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:7 http://security.ubuntu.com/ubuntu jammy-security/restricted amd64 Packages [6388 kB]
Get:8 http://archive.ubuntu.com/ubuntu jammy/main amd64 Packages [1792 kB]
Get:9 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 Packages [1297 kB]
Get:10 http://archive.ubuntu.com/ubuntu jammy/multiverse amd64 Packages [266 kB]
Get:11 http://archive.ubuntu.com/ubuntu jammy/restricted amd64 Packages [164 kB]
Get:12 http://archive.ubuntu.com/ubuntu jammy/universe amd64 Packages [17.5 MB]
Get:13 http://archive.ubuntu.com/ubuntu jammy-updates/restricted amd64 Packages [6607 kB]
Get:14 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 Packages [1688 kB]
Get:15 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [4014 kB]
Get:16 http://archive.ubuntu.com/ubuntu jammy-updates/multiverse amd64 Packages [70.9 kB]
Get:17 http://archive.ubuntu.com/ubuntu jammy-backports/main amd64 Packages [83.9 kB]
Get:18 http://archive.ubuntu.com/ubuntu jammy-backports/universe amd64 Packages [37.2 kB]
Fetched 44.2 MB in 7s (5996 kB/s)
Reading package lists...
Reading package lists...
Building dependency tree...
Reading state information...
The following additional packages will be installed:
  fontconfig-config fonts-dejavu-core iproute2 libatm1 libbpf0 libbrotli1

```

4. Run Container

```

vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp3 file$ docker run -d --name nginx-ubuntu -p 8081:80
nginx-ubuntu
c27efca3fd1d98f92abd69ac688cbfb2f9a9bf5de7c72bf396a5ebbf4ce9d028

```

5. Observations

```

vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp3 file$ docker images nginx-ubuntu
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
nginx-ubuntu  latest   b901f358ac03   22 seconds ago  134MB
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp3 file$

```

Part C: Custom Nginx using Alpine base image

1. Create Dockerfile

```
FROM alpine:latest

RUN apk add --no-cache nginx

EXPOSE 80

CMD ["nginx", "-g", "daemon off;"]
```

2. Build Image

```
vanshika_munjial@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&O Lab/exp3 file$ docker build -t nginx-alpine .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
  Install the buildx component to build images with BuildKit:
  https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  2.048kB
Step 1/4 : FROM alpine:latest
latest: Pulling from library/alpine
589002ba0eae: Pulling fs layer
589002ba0eae: Verifying Checksum
589002ba0eae: Download complete
589002ba0eae: Pull complete
Digest: sha256:25109184c71bdad752c8312a8623239606a9a2071e8825f20acb8f2198c3f659
Status: Downloaded newer image for alpine:latest
--> a40c03cbb81c
Step 2/4 : RUN apk add --no-cache nginx
--> Running in f8b070498032
(1/2) Installing pcre2 (10.47-r0)
(2/2) Installing nginx (1.28.2-r0)
  Executing nginx-1.28.2-r0.pre-install
  Executing nginx-1.28.2-r0.post-install
Executing busybox-1.37.0-r30.trigger
OK: 9.9 MiB in 18 packages
--> Removed intermediate container f8b070498032
--> f3454fa55eb6
Step 3/4 : EXPOSE 80
--> Running in 6639e63731dc
--> Removed intermediate container 6639e63731dc
--> a2fefbe398d5
Step 4/4 : CMD ["nginx", "-g", "daemon off;"]
--> Running in e11bc740eef8
--> Removed intermediate container e11bc740eef8
--> cf1b942d9cfd
Successfully built cf1b942d9cfd
Successfully tagged nginx-alpine:latest
```

3. Run Container

```
vanshika_munjial@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&O Lab/exp3 file$ docker run -d --name nginx-alpine -p 8082:80
nginx-alpine
8d03257a8109212b57a6cee27e8bfacbdb2071ff321d5b20fbae17aee40eaf682
```

4. Observations

```
vanshika_munjial@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&O Lab/exp3 file$ docker images nginx-alpine
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
nginx-alpine   latest    cf1b942d9cfd   42 seconds ago  10.4MB
vanshika_munjial@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&O Lab/exp3 file$ |
```

Result

The experiment

Conclusion

Virtual Machines

vanshikamunjal665

CONTAINERIZATION AND DEVOPS LAB

Experiment 4:

Docker Essentials [🔗](#)

Reference Link:

<https://upessocs.github.io/#dir=/Lectures/Containerization%20and%20DevOps/Lab/&file=Experiment%204%20Docker%20Essentials%20Dockerfile%20dockerignore%20tagging%20and%20publishing.md>

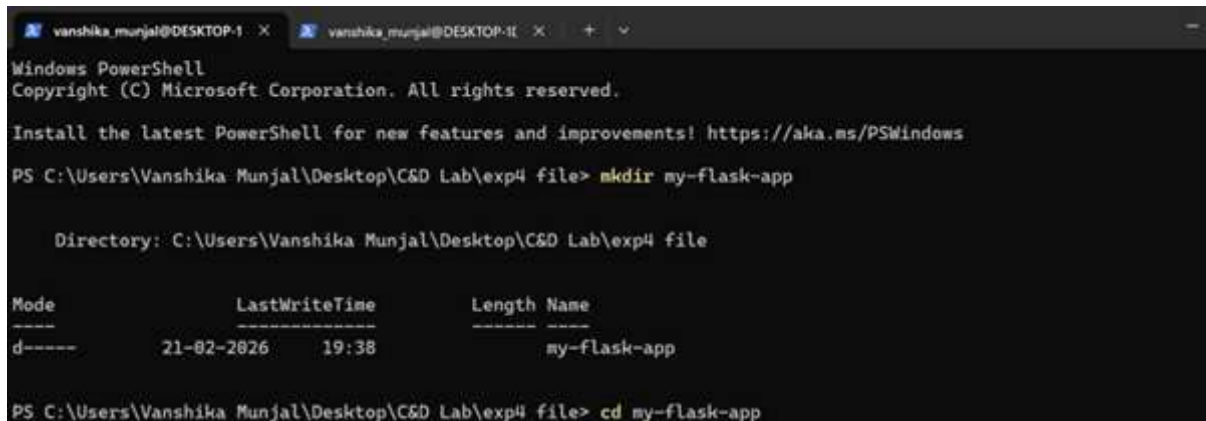
Objective

1. Dockerfile
2. .dockerignore
3. Tagging
4. Publishing

Part 1: Containerization Applications with Dockerfile

1. Create a Simple Application

python flask app



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Vanshika Munjal\Desktop\C&D Lab\exp4 file> mkdir my-flask-app

Directory: C:\Users\Vanshika Munjal\Desktop\C&D Lab\exp4 file

Mode                LastWriteTime         Length Name
----                -
d-----         21-02-2026    19:38             my-flask-app

PS C:\Users\Vanshika Munjal\Desktop\C&D Lab\exp4 file> cd my-flask-app
```

app.py



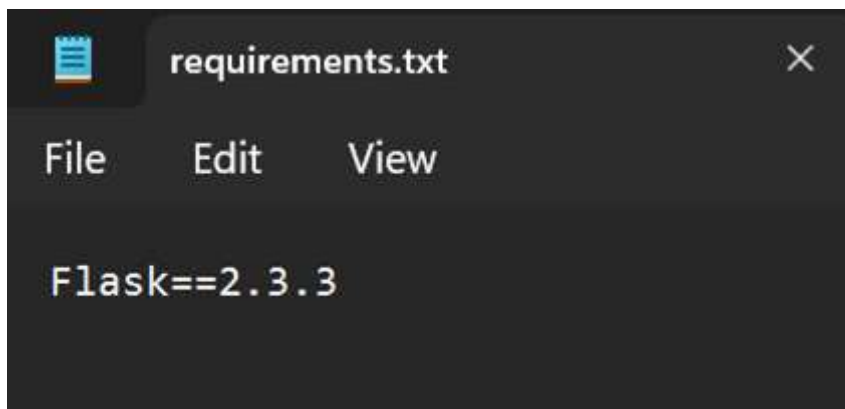
```
app.py - C:/Users/Vanshika Munjal/Desktop/C&D Lab/exp4 file/my-flask-app/app.py (3.12.7)
File Edit Format Run Options Window Help
from flask import Flask
app = Flask(__name__)

@app.route('/')
def hello():
    return "Hello from Docker!"

@app.route('/health')
def health():
    return "OK"

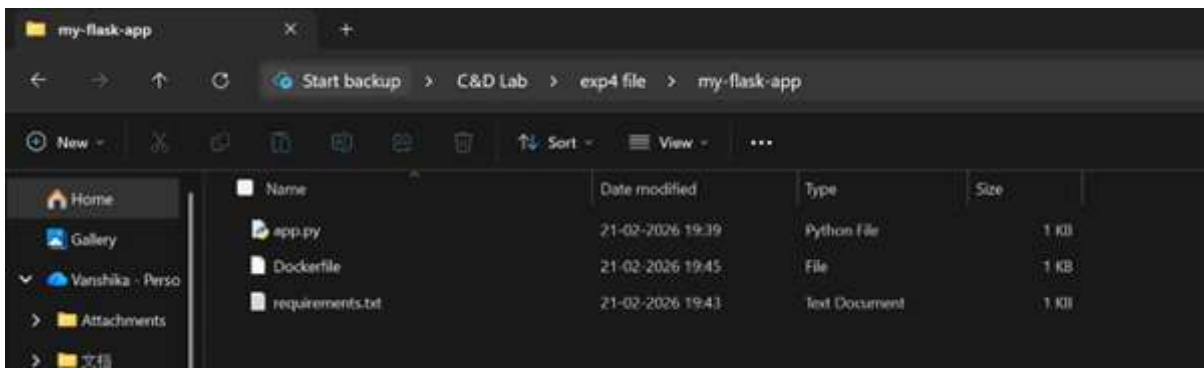
if __name__ == '__main__':
    app.run(host='0.0.0.0', port=5000)
```

requirements.txt



```
requirements.txt
File Edit View
Flask==2.3.3
```

folders



2. Create Dockerfile

```

# Use Python base image
FROM python:3.9-slim

# Set working directory
WORKDIR /app

# Copy requirements file
COPY requirements.txt .

# Install dependencies
RUN pip install --no-cache-dir -r requirements.txt

# Copy application code
COPY app.py .

# Expose port
EXPOSE 5000

# Run the application
CMD ["python", "app.py"]

```

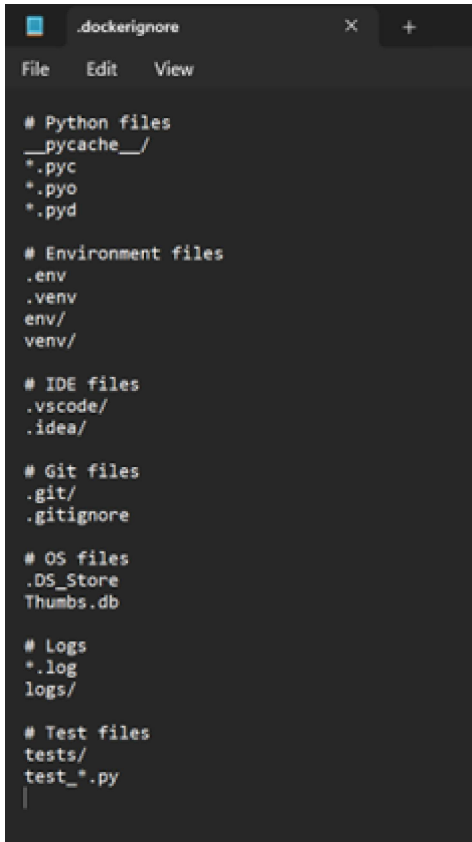
```

PS C:\Users\HP> mkdir my-flask-app
vanishika_munjal@DESKTOP-107G070: /mnt/c/Users/HP$ cd "/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp4 file"
vanishika_munjal@DESKTOP-107G070: /mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp4 file$ mv Dockerfile.txt Dockerfile
mv: cannot stat 'Dockerfile.txt': No such file or directory
vanishika_munjal@DESKTOP-107G070: /mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp4 file$ cd my-flask-app
vanishika_munjal@DESKTOP-107G070: /mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp4 file/my-flask-app$ mv Dockerfile.txt Dockerfile
vanishika_munjal@DESKTOP-107G070: /mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp4 file/my-flask-app$

```

Part 2: Using .dockerignore

1. Create .dockerignore File



```
.dockerignore
File Edit View

# Python files
__pycache__/
*.pyc
*.pyo
*.pyd

# Environment files
.env
.venv
env/
venv/

# IDE files
.vscode/
.idea/

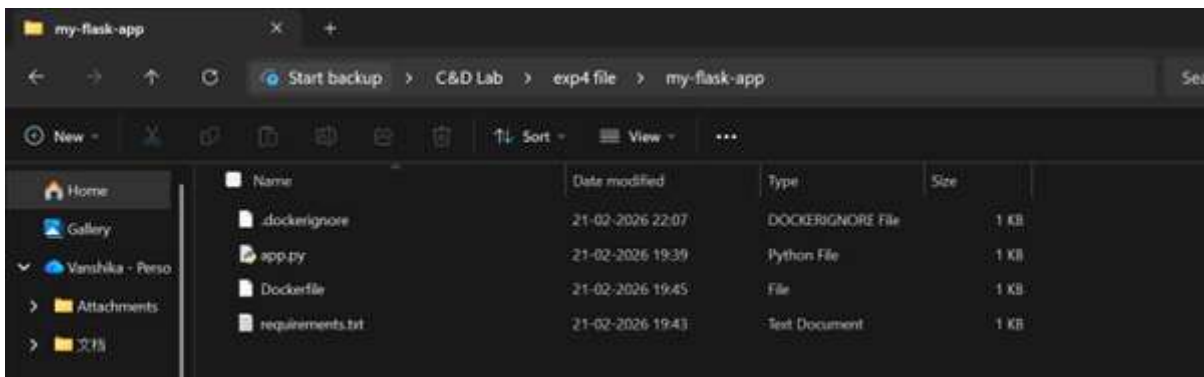
# Git files
.git/
.gitignore

# OS files
.DS_Store
Thumbs.db

# Logs
*.log
logs/

# Test files
tests/
test_*.py
```

folders



Why .dockerignore is important?

1. Prevents unnecessary files from being copied
2. Reduces image size
3. Improves build speed

4. Increases security

Part 3: Building Docker Images

1. Build image from Dockerfile

```

vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp4 file/my-flask-app$ docker build -t my-flask-app .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  5.12kB
Step 1/7 : FROM python:3.9-slim
3.9-slim: Pulling from library/python
38513bd72563: Pulling fs layer
b3ec39b36ae8: Pulling fs layer
fc7443084902: Pulling fs layer
ea56f685404a: Pulling fs layer
ea56f685404a: Waiting
b3ec39b36ae8: Verifying Checksum
b3ec39b36ae8: Download complete
ea56f685404a: Verifying Checksum
ea56f685404a: Download complete
fc7443084902: Verifying Checksum
fc7443084902: Download complete
38513bd72563: Verifying Checksum
38513bd72563: Download complete
38513bd72563: Pull complete
b3ec39b36ae8: Pull complete
fc7443084902: Pull complete
ea56f685404a: Pull complete
Digest: sha256:2d97f6910b16bd338d3060f261f53f1d4965f755599aabbacdale13cf1731b1b
Status: Downloaded newer image for python:3.9-slim
--> 085da638e1b8
Step 2/7 : WORKDIR /app
--> Running in f2e155e115a6
--> Removed intermediate container f2e155e115a6
--> a31aaec2cb3
Step 3/7 : COPY requirements.txt .
--> b7c356dac109
Step 4/7 : RUN pip install --no-cache-dir -r requirements.txt

```

2. Check build images

```

vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp4 file/my-flask-app$ docker images

```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
my-flask-app	latest	89bbe1326a7	5 seconds ago	132MB
c-app	1.0	937d4a9fe50a	10 days ago	900kB
<none>	<none>	0e937fe85245	11 days ago	334MB
nginx-alpine	latest	cf1b942d9cfd	2 weeks ago	10.4MB
nginx-ubuntu	latest	b901f358ac03	2 weeks ago	134MB
python-app	1.0	0a18d6cfd91c	2 weeks ago	1.21GB
mysql	latest	f66b7a288113	2 weeks ago	922MB
nginx	latest	5cdef4ac3335	2 weeks ago	161MB
python	latest	9a7c5808942d	2 weeks ago	1.12GB
nginx	<none>	248d2326f351	2 weeks ago	161MB
javaapp	2.0	1964998c1ba7	3 weeks ago	773MB
vanshikamunjal2005/javaapp	2.0	1964998c1ba7	3 weeks ago	773MB
java-app	1.0	270358796c98	3 weeks ago	773MB
alpine	latest	a40c03cbb81c	3 weeks ago	8.44MB
myrepo/java-app	1.0	4f3c89e386c2	3 weeks ago	922MB
ubuntu	latest	493218ed0f40	5 weeks ago	78.1MB
nginx	<none>	4af177a024eb	5 weeks ago	161MB
ubuntu	22.04	65c77cbc27c2	6 weeks ago	77.9MB
python	3.9-slim	085da638e1b8	3 months ago	122MB
busybox	latest	af3f0f48a24e	17 months ago	4.43MB

3. Tag with version number (alternative)


```

vanshika_munjal@DESKTOP-107G0770:/mnt/c/Users/Vanshika_Munjal/Desktop/C&D Lab/exp4 file/my-flask-app$ docker build -t my-flask-app:1.0
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon   5.12kB
Step 1/7 : FROM python:3.9-slim
--> 085da638e1b8
Step 2/7 : WORKDIR /app
--> Using cache
--> a31aaec2cb3
Step 3/7 : COPY requirements.txt .
--> Using cache
--> b7c356dac109
Step 4/7 : RUN pip install --no-cache-dir -r requirements.txt
--> Using cache
--> cd1f92a50c4a
Step 5/7 : COPY app.py .
--> Using cache
--> a52b14ceae60
Step 6/7 : EXPOSE 5000
--> Using cache
--> 2a4cf1081b76
Step 7/7 : CMD ["python", "app.py"]
--> Using cache
--> 89bbe1326a7
Successfully built 89bbe1326a7
Successfully tagged my-flask-app:1.0

```

4. Show image history

```

vanshika_munjal@DESKTOP-107G0770:/mnt/c/Users/Vanshika_Munjal/Desktop/C&D Lab/exp4 file/my-flask-app$ docker history my-flask-app

```

IMAGE	CREATED	CREATED BY	SIZE	COMMENT
89bbe1326a7	About a minute ago	/bin/sh -c #(nop) CMD ["python" "app.py"]	0B	
2a4cf1081b76	About a minute ago	/bin/sh -c #(nop) EXPOSE 5000	0B	
a52b14ceae60	About a minute ago	/bin/sh -c #(nop) COPY file:49edba258cd3ee6c_	241B	
cd1f92a50c4a	About a minute ago	/bin/sh -c pip install --no-cache-dir -r req_	10.5MB	
b7c356dac109	About a minute ago	/bin/sh -c #(nop) COPY file:b83229e65f7256a2_	12B	
a31aaec2cb3	About a minute ago	/bin/sh -c #(nop) WORKDIR /app	0B	
085da638e1b8	3 months ago	CMD ["python3"]	0B	buildkit.dockerfile.v0
<missing>	3 months ago	RUN /bin/sh -c set -eux; for src in idle3 p_	36B	buildkit.dockerfile.v0
<missing>	3 months ago	RUN /bin/sh -c set -eux; savedAptMark=\$(a_	39.5MB	buildkit.dockerfile.v0
<missing>	3 months ago	ENV PYTHON_SHA256=00e07d7c0f2f0cc002432d1ee8_	0B	buildkit.dockerfile.v0
<missing>	3 months ago	ENV PYTHON_VERSION=3.9.25	0B	buildkit.dockerfile.v0
<missing>	3 months ago	ENV GPG_KEY=E3FF2839C048B25C084DEBE9B26995E3_	0B	buildkit.dockerfile.v0
<missing>	3 months ago	RUN /bin/sh -c set -eux; apt-get update; a_	3.81MB	buildkit.dockerfile.v0
<missing>	3 months ago	ENV LANG=C.UTF-8	0B	buildkit.dockerfile.v0
<missing>	3 months ago	ENV PATH=/usr/local/bin:/usr/local/sbin:/usr_	0B	buildkit.dockerfile.v0
<missing>	4 months ago	# debian.sh --arch 'amd64' out/ 'trixie' 'el_	78.6MB	debuerreotype 0.16

5. Inspect image details

```

vanshika_munjal@DESKTOP-107G0770:/mnt/c/Users/Vanshika_Munjal/Desktop/C&D Lab/exp4 file/my-flask-app$ docker inspect my-flask-app
[
  {
    "Id": "sha256:89bbe1326a7823a9241a4efd154c92884ec4a56428e65d4d617860e80322a70",
    "RepoTags": [
      "my-flask-app:1.0",
      "my-flask-app:latest"
    ],
    "RepoDigests": [],
    "Parent": "sha256:2a4cf1081b76cc8b09bccaaff7df892b26f1064d03813e719943e7ed512ddd3",
    "Comment": "",
    "Created": "2026-02-21T16:46:02.951589607Z",
    "DockerVersion": "28.2.2",
    "Author": "",
    "Config": {
      "ExposedPorts": {
        "5000/tcp": {}
      },
      "Env": [
        "PATH=/usr/local/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin",
        "LANG=C.UTF-8",
        "GPG_KEY=E3FF2839C048B25C084DEBE9B26995E310250568",
        "PYTHON_VERSION=3.9.25",
        "PYTHON_SHA256=00e07d7c0f2f0cc002432d1ee84d2a40dae40a99303e3f97701c10966c91834"
      ],
      "Cmd": [
        "python",
        "app.py"
      ],
      "WorkingDir": "/app"
    }
  }
]

```

Part 4: Running Containers

1. Run container with port mapping

```
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C60 Lab/exp4 file/my-flask-app$ docker run -d -p 5000:5000 --name flask-container my-flask-app
814851ff943bdaf92848c2df6ac8e49cc27e58c7cc4f3588d52d8d976c423b7a
```

2. Test the application

```
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C60 Lab/exp4 file/my-flask-app$ curl http://localhost:5000
Hello from Docker!vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C60 Lab/exp4 file/my-flask-app$ docker ps
```



3. View running containers

```
Hello from Docker!vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C60 Lab/exp4 file/my-flask-app$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
814851ff943b   my-flask-app   "python app.py"         3 minutes ago Up 3 minutes   0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp   flask-container
```

4. View container logs

```
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C60 Lab/exp4 file/my-flask-app$ docker logs flask-container
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.18.0.2:5000
Press CTRL+C to quit
172.18.0.1 - - [21/Feb/2026 17:39:51] "GET / HTTP/1.1" 200 -
172.18.0.1 - - [21/Feb/2026 17:39:58] "GET / HTTP/1.1" 200 -
172.18.0.1 - - [21/Feb/2026 17:39:58] "GET /favicon.ico HTTP/1.1" 404 -
```

5. Stop container

```
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C60 Lab/exp4 file/my-flask-app$ docker stop flask-container
flask-container
```

6. Start stopped container

```
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C60 Lab/exp4 file/my-flask-app$ docker start flask-container
flask-container
```

7. Remove container

```
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp4 file/my-flask-app$ docker rm flask-container
Error response from daemon: cannot remove container "flask-container": container is running: stop the container before removing or force remove
```

8. Remove container forcefully

```
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp4 file/my-flask-app$ docker rm -f flask-container
flask-container
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp4 file/my-flask-app$ |
```

Part 5: Multi-stage Builds

Why Multi-stage Builds?

1. Smaller final image size
2. Better security (remove build tools)
3. Separate build and runtime environments

1. Simple multi-stage Dockerfile

```

Dockerfile.multistage
File Edit View

# STAGE 1: Builder stage
FROM python:3.9-slim AS builder

WORKDIR /app

# Copy requirements
COPY requirements.txt .

# Install dependencies in virtual environment
RUN python -m venv /opt/venv
ENV PATH="/opt/venv/bin:$PATH"
RUN pip install --no-cache-dir -r requirements.txt

# STAGE 2: Runtime stage
FROM python:3.9-slim

WORKDIR /app

# Copy virtual environment from builder
COPY --from=builder /opt/venv /opt/venv
ENV PATH="/opt/venv/bin:$PATH"

# Copy application code
COPY app.py .

# Create non-root user
RUN useradd -m -u 1000 appuser
USER appuser

# Expose port
EXPOSE 5000

# Run application
CMD ["python", "app.py"]

```

2. Build regular image

```

vanshika_munjal@DESKTOP-107G070: /mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp4 file/oy-flask-app$ docker build -t flask-regular .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  6.656kB
Step 1/7 : FROM python:3.9-slim
--> 885da638e1b8
Step 2/7 : WORKDIR /app
--> Using cache
--> a31aaec2cb3
Step 3/7 : COPY requirements.txt .
--> Using cache
--> b7c356dac109
Step 4/7 : RUN pip install --no-cache-dir -r requirements.txt
--> Using cache
--> cd1f92a50c4a
Step 5/7 : COPY app.py .
--> Using cache
--> a52b14ceae60
Step 6/7 : EXPOSE 5000
--> Using cache
--> 2a4ef1081b76
Step 7/7 : CMD ["python", "app.py"]
--> Using cache
--> 89bbe1326a7
Successfully built 89bbe1326a7
Successfully tagged flask-regular:latest

```

3. Build multi-stage image

```

vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp4 file/my-flask-app$ docker build -f Dockerfile.multistage -t flask-multistage .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon 6.656kB
Step 1/15 : FROM python:3.9-slim AS builder
--> 085da638e1b8
Step 2/15 : WORKDIR /app
--> Using cache
--> a31aaac2cb3
Step 3/15 : COPY requirements.txt .
--> Using cache
--> b7c356dac109
Step 4/15 : RUN python -m venv /opt/venv
--> Running in cfs91160ad2a
--> Removed intermediate container cfs91160ad2a
--> a61d6b1ae3e4
Step 5/15 : ENV PATH="/opt/venv/bin:$PATH"
--> Running in b49f269dc806
--> Removed intermediate container b49f269dc806
--> 7d141012f1a3
Step 6/15 : RUN pip install --no-cache-dir -r requirements.txt

```

4. Compare sizes

```

vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp4 file/my-flask-app$ docker images | grep flask-
flask-multistage   latest      4c293534cd2c   13 seconds ago   145MB
flask-regular      latest     89bbe1326a7    About an hour ago 132MB
my-flask-app       1.0       89bbe1326a7    About an hour ago 132MB
my-flask-app       latest     89bbe1326a7    About an hour ago 132MB

```

Part 6: Publishing to Docker Hub

1. Login to Docker Hub

```

vanshika_munjal@DESKTOP-107G070: ~
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\HP> wsl
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ cd "/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp4 file"
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp4 file$ docker login
Authenticating with existing credentials... [Username: vanshikamunjal2005]

Info → To login with a different account, run 'docker logout' followed by 'docker login'

Login Succeeded

```

2. Tag image for Docker Hub

```

vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp4 file$ docker tag my-flask-app:latest vanshikamunjal2005/my-flask-app:1.0

```

3. Push to Docker Hub


```

vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp4 file$ docker push vanshikamunjal2005/my-flask-app:1.0
The push refers to repository [docker.io/vanshikamunjal2005/my-flask-app]
191fd3598a56: Pushed
a8cb1b257f61: Pushed
b9549719be5e: Pushed
83d410503ed2: Pushed
c8f6b54339a8: Mounted from library/python
298992e09a63: Mounted from library/python
4f237755fbae: Mounted from library/python
d7c97cb6f1fe: Mounted from library/python
1.0: digest: sha256:139601673e13bee6109c43a2075246f0f92943e2e4cdee2081a4d01f80d3f443 size: 1990

```

4. Pull from Docker Hub

```

vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp4 file$ docker pull vanshikamunjal2005/my-flask-app:latest
Error response from daemon: manifest for vanshikamunjal2005/my-flask-app:latest not found: manifest unknown: manifest unknown
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp4 file$ docker pull vanshikamunjal2005/my-flask-app:1.0
1.0: Pulling from vanshikamunjal2005/my-flask-app
Digest: sha256:139601673e13bee6109c43a2075246f0f92943e2e4cdee2081a4d01f80d3f443
Status: Image is up to date for vanshikamunjal2005/my-flask-app:1.0
docker.io/vanshikamunjal2005/my-flask-app:1.0

```

5. Run the pulled image

```

vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp4 file$ docker run -d -p 5000:5000 vanshikamunjal2005/my-flask-app:1.0
99a6721be1bbc61090bbd406893a4d8d83df0869af34d6b4c9c445aa319c67e0
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp4 file$ |

```

Part 7: Node.js Example

1. Node.js Application

```

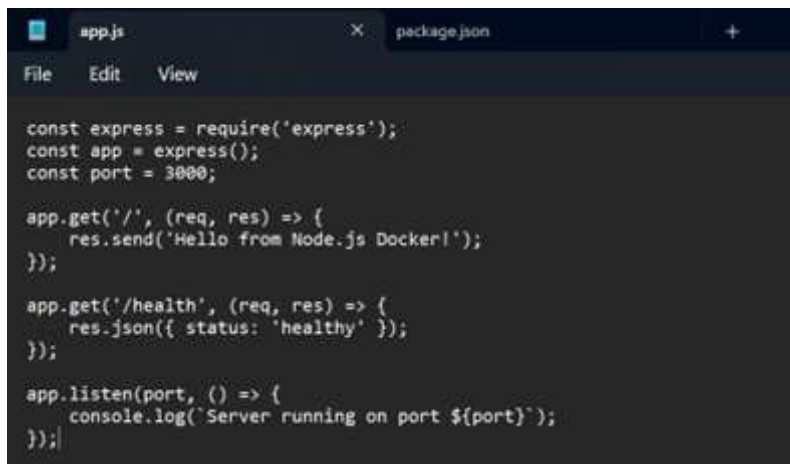
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\HP> wsl
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ cd "/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp4 file"
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp4 file$ mkdir my-node-app
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp4 file$ cd my-node-app
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp4 file/my-node-app$ mv Dockerfile.txt Dockerfile
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp4 file/my-node-app$

```

app.js



```
const express = require('express');
const app = express();
const port = 3000;

app.get('/', (req, res) => {
  res.send('Hello from Node.js Docker!');
});

app.get('/health', (req, res) => {
  res.json({ status: 'healthy' });
});

app.listen(port, () => {
  console.log(`Server running on port ${port}`);
});
```

package.json



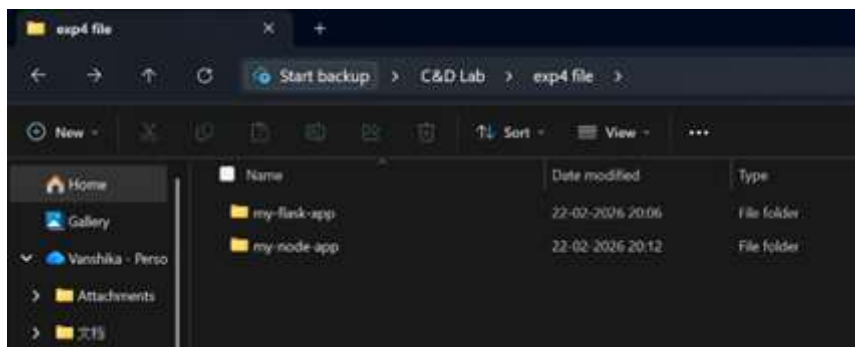
```
{
  "name": "node-docker-app",
  "version": "1.0.0",
  "main": "app.js",
  "dependencies": {
    "express": "^4.18.2"
  }
}
```

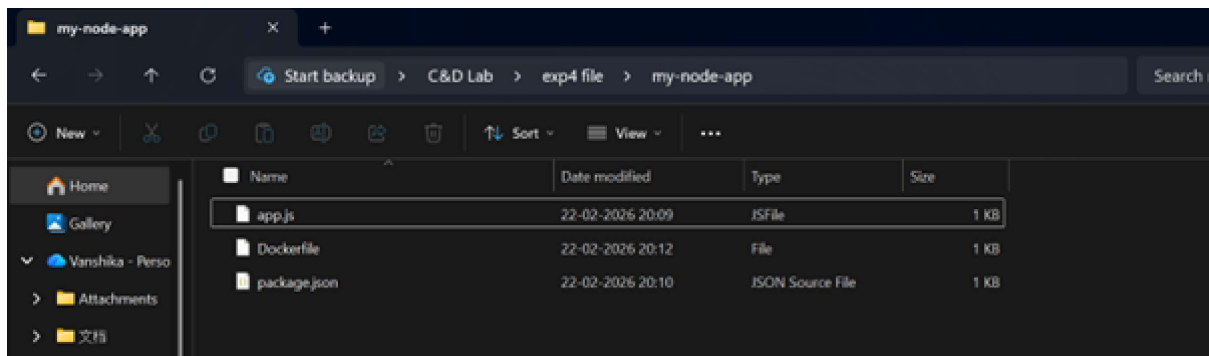
Dockerfile



```
FROM node:18-alpine
WORKDIR /app
COPY package.json ./
RUN npm install --only=production
COPY app.js .
EXPOSE 3000
CMD ["node", "app.js"]
```

folders





2. Build Image

```
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp4 file/my-node-app$ docker build -t my-node-app .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  4.096kB
Step 1/7 : FROM node:18-alpine
18-alpine: Pulling from library/node
f18232174bc9: Pulling fs layer
dd71dde834b5: Pulling fs layer
1e5a4c89cee5: Pulling fs layer
25ff2da83641: Pulling fs layer
25ff2da83641: Waiting
1e5a4c89cee5: Verifying Checksum
1e5a4c89cee5: Download complete
f18232174bc9: Download complete
f18232174bc9: Pull complete
25ff2da83641: Download complete
dd71dde834b5: Download complete
dd71dde834b5: Pull complete
1e5a4c89cee5: Pull complete
25ff2da83641: Pull complete
```

3. Run container

```
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp4 file/my-node-app$ docker run -d -p 3000:3000 --name
node-container my-node-app
ee68f892c2ce28153dfc5ec4062b6d841d123cd238ede33f74404dfbbc449464
```

4. Test

```
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp4 file/my-node-app$ curl http://localhost:3000
Hello from Node.js Docker!vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp4 file/my-node-app$ |
```



vanshikamunjal665

CONTAINERIZATION AND DEVOPS LAB

Experiment 5:

Docker- Volumes, Environment Variables, Monitoring & Networks

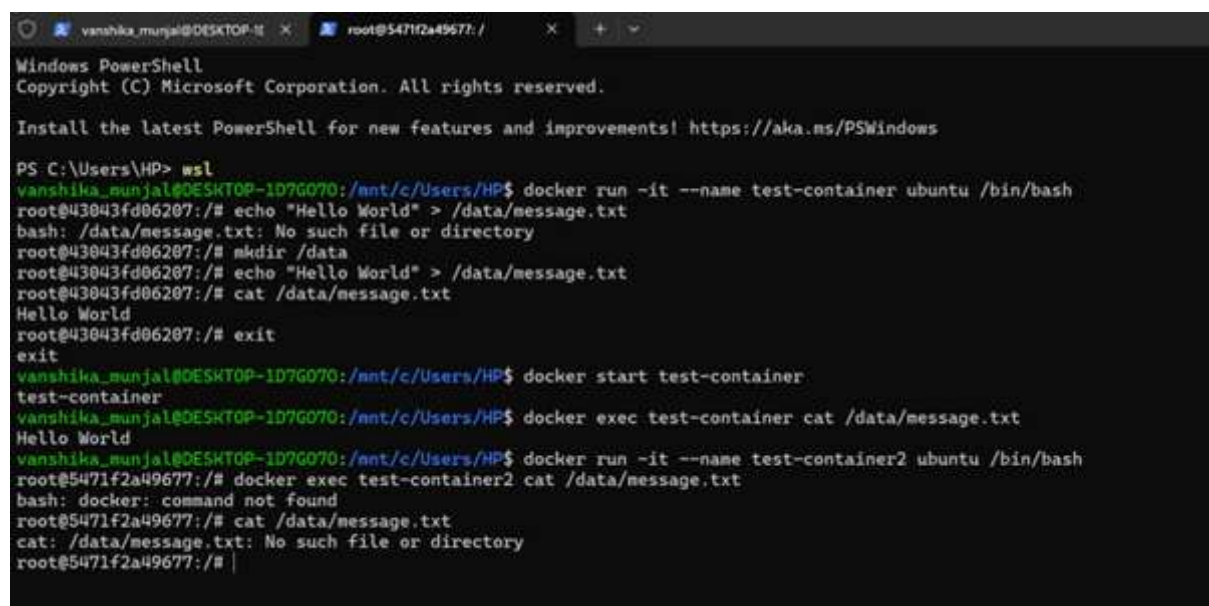
Reference Link:

<https://upessocs.github.io/#dir=/Lectures/Containerization%20and%20DevOps/Lab/&file=Experiment%2005%20Docker%20Volumes,%20Environment%20Variables,%20Monitoring,%20Networks%20Final.md>

Part 1: Docker Volumes- Persistent Data Storage

Lab 1: Understanding Data Persistence

The Problem: Container Data is Ephemeral



```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\HP> wsl
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/HP$ docker run -it --name test-container ubuntu /bin/bash
root@43043fd06207:/# echo "Hello World" > /data/message.txt
bash: /data/message.txt: No such file or directory
root@43043fd06207:/# mkdir /data
root@43043fd06207:/# echo "Hello World" > /data/message.txt
root@43043fd06207:/# cat /data/message.txt
Hello World
root@43043fd06207:/# exit
exit
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/HP$ docker start test-container
test-container
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/HP$ docker exec test-container cat /data/message.txt
Hello World
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/HP$ docker run -it --name test-container2 ubuntu /bin/bash
root@5471f2a49677:/# docker exec test-container2 cat /data/message.txt
bash: docker: command not found
root@5471f2a49677:/# cat /data/message.txt
cat: /data/message.txt: No such file or directory
root@5471f2a49677:/#
```

Solution: Volumes

Lab 2: Volume Types

1. Anonymous Volumes

```
vanshika_munjial@DESKTOP-107G070:/mnt/c/Users/HP$ docker run -d -v /app/data --name web1 nginx
8fb48ad27a577eda2d901770b8b9812bb5bd361e7940fee609db824762de5609
vanshika_munjial@DESKTOP-107G070:/mnt/c/Users/HP$ docker volume ls
DRIVER      VOLUME NAME
local       1b98d65bb3d767f925a58930476de00d88d28b860f1482b8baca23f82506c384
local       5d2e578b14128a809f370a10f002e14d67d6d660d7789719064c2af5b96f5d95
local       7ca4b803667911817f228440136bab6419fdc5750f7729c025fb39d4d9b52b57
local       11a9fe249bdf6a609686d75709d056a86bd9fc309463dbc643ba35089b49176
local       14cb18358c4dcfa09657cf0a68447f708a9da4ae8089507f406calb1c26983b8
local       25feb-class_mysql_data
local       25feb-class_wp_content
local       26feb-class_mysql_data
local       26feb-class_wp_content
local       37b3f46b3c81e6b3e340672661a7355a425a79ca409fbd432bbab4290c90dba1
local       49af95d171db743ae8a15940af4e2a4553d97f7fac2e37cf5c94d217ffbf2a2
local       63d456d51a0cb6f49071702398658f92c22f819641ac0da5333ec291a5e28ee7
local       ae41b0e6b4e33ef5938627e214c938a307600f75d1a47bf268e3a41a412bd0d6
local       afc2a5a7de2eaa163fe9e26d5cafff53c4ffc2af38a82985b728294c715a5254
local       b9dc2ffabd12388d7010267d22c9f9c4ddc9ab91a4836405c22e2f7f0a617218
local       b01530fb0ae69c9eb0a1a8f009957d3c3d1be33564abe8ef7776e336975f1cf9
local       cdab12f55ab0175bd1bcaebe7dc71075d205bfc290d8c7e4e1533aa41339d89e
local       ce92eb3a7a0ce60031a3aabe44c6f1405e36ac2391444b7c5c61f3a2ed3c329c
local       containerization-project-1_postgres_data
local       de8ff57a54f4941d6fa842eb2b9a9dbf43fbf73818b10051d12ac75d9d465181
local       ef44bb4654fcc01c53ad5d51363bf77b8453e9ef8035ae81409264d4878a11bd
local       fb058b5137c74e4a6cab010bc33b66d7f5502847ecf0cc40e6732d3bb305f64c
local       k3d-mycluster-images
local       myvolume
local       portainer_data
local       test
vanshika_munjial@DESKTOP-107G070:/mnt/c/Users/HP$ docker inspect web1 | grep -A 5 Mounts
"Mounts": [
  {
    "Type": "volume",
    "Name": "afc2a5a7de2eaa163fe9e26d5cafff53c4ffc2af38a82985b728294c715a5254",
    "Source": "/var/lib/docker/volumes/afc2a5a7de2eaa163fe9e26d5cafff53c4ffc2af38a82985b728294c715a5254/_data",
    "Destination": "/app/data",
```

2. Named Volumes

```
vanshika_munjial@DESKTOP-107G070:/mnt/c/Users/HP$ docker volume create mydata
mydata
vanshika_munjial@DESKTOP-107G070:/mnt/c/Users/HP$ docker run -d -v mydata:/app/data --name web2 nginx
3357050f706c0409b24af4b836cfcd1a789e1fd62dc2638f8e78a9eb360d9387
vanshika_munjial@DESKTOP-107G070:/mnt/c/Users/HP$ docker volume ls
DRIVER      VOLUME NAME
local       1b98d65bb3d767f925a58930476de00d88d28b860f1482b8baca23f82506c384
local       5d2e578b14128a809f370a10f002e14d67d6d660d7789719064c2af5b96f5d95
local       7ca4b803667911817f228440136bab6419fdc5750f7729c025fb39d4d9b52b57
local       11a9fe249bdf6a609686d75709d056a86bd9fc309463dbc643ba35089b49176
local       14cb18358c4dcfa09657cf0a68447f708a9da4ae8089507f406calb1c26983b8
local       25feb-class_mysql_data
local       25feb-class_wp_content
local       26feb-class_mysql_data
local       26feb-class_wp_content
local       37b3f46b3c81e6b3e340672661a7355a425a79ca409fbd432bbab4290c90dba1
local       49af95d171db743ae8a15940af4e2a4553d97f7fac2e37cf5c94d217ffbf2a2
local       63d456d51a0cb6f49071702398658f92c22f819641ac0da5333ec291a5e28ee7
local       ae41b0e6b4e33ef5938627e214c938a307600f75d1a47bf268e3a41a412bd0d6
local       afc2a5a7de2eaa163fe9e26d5cafff53c4ffc2af38a82985b728294c715a5254
local       b9dc2ffabd12388d7010267d22c9f9c4ddc9ab91a4836405c22e2f7f0a617218
local       b01530fb0ae69c9eb0a1a8f009957d3c3d1be33564abe8ef7776e336975f1cf9
local       cdab12f55ab0175bd1bcaebe7dc71075d205bfc290d8c7e4e1533aa41339d89e
local       ce92eb3a7a0ce60031a3aabe44c6f1405e36ac2391444b7c5c61f3a2ed3c329c
local       containerization-project-1_postgres_data
local       de8ff57a54f4941d6fa842eb2b9a9dbf43fbf73818b10051d12ac75d9d465181
local       ef44bb4654fcc01c53ad5d51363bf77b8453e9ef8035ae81409264d4878a11bd
local       fb058b5137c74e4a6cab010bc33b66d7f5502847ecf0cc40e6732d3bb305f64c
local       k3d-mycluster-images
local       mydata
local       myvolume
local       portainer_data
local       test
```

```
vanshika_munjial@DESKTOP-107G070:/mnt/c/Users/HP$ docker volume inspect mydata
[
  {
    "CreatedAt": "2026-03-27T08:54:02Z",
    "Driver": "local",
    "Labels": null,
    "Mountpoint": "/var/lib/docker/volumes/mydata/_data",
    "Name": "mydata",
    "Options": null,
    "Scope": "local"
  }
]
```


1. Bind Mounts

```
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ mkdir ~/myapp-data
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ docker run -d -v ~/myapp-data:/app/data --name web3 nginx
6274e0d3f87cc459c35cbbd8b32a56267calcb374ee7614bcb4bcbf2e05109341
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ echo "From Host" > ~/myapp-data/host-file.txt
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ docker exec web3 cat /app/data/host-file.txt
From Host
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ |
```

Lab 3: Practical Volume Examples

Example 1: Database with Persistent Storage

```
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ docker run -d \
--name mysql-db \
-v mysql-data:/var/lib/mysql \
-e MYSQL_ROOT_PASSWORD=secret \
mysql:8.0
Unable to find image 'mysql:8.0' locally
8.0: Pulling from library/mysql
d4c7048d1cf1: Pull complete
6375ec02106f: Pull complete
771773a1fe7a: Pull complete
c3b8cfb10b41: Pull complete
1378718ee973: Pull complete
5915b0e5580c: Pull complete
60c3210ab1f6: Pull complete
440208b3f29f: Pull complete
10ee8ef3b9fc: Pull complete
cd8097e7c26b: Pull complete
0940ecf5daec: Pull complete
Digest: sha256:64756cc92f707eb504496d774353990bcb0f6999ddf598b6ad188f2da66bd000
Status: Downloaded newer image for mysql:8.0
5bba7e8f8e8311b2c8372da5da14704844fef6109a122b559fbf8c20094dc84a
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ docker stop mysql-db
mysql-db
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ docker rm mysql-db
mysql-db
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ docker run -d \
--name new-mysql \
-v mysql-data:/var/lib/mysql \
-e MYSQL_ROOT_PASSWORD=secret \
mysql:8.0
924fbc1f458fc92f0e9ee8f716a47d532b99c72b8a858fd73c0e18fde9300152
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ |
```

Example 2: Web App with Configuration Files

```
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ nano ~/nginx-config/nginx.conf
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ docker run -d \
--name nginx-custom \
-p 8080:80 \
-v ~/nginx-config/nginx.conf:/etc/nginx/conf.d/default.conf \
nginx
docker: Error response from daemon: Conflict. The container name "/nginx-custom" is already in use by container "a0e6c2a717f0cdbad64a
ceb32368695af6118b508886cb09b7d1d008b2a75d72". You have to remove (or rename) that container to be able to reuse that name.

Run 'docker run --help' for more information
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ docker rm -f nginx-custom
nginx-custom
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ docker run -d --name nginx-custom -p 8080:80 -v ~/nginx-config/nginx.conf:/etc
/nginx/conf.d/default.conf nginx
d70058ff092c50f5c004b31cedde7ea27e5abe5c02b90ac15280306938d7d8e
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ curl http://localhost:8080
Hello from mounted config!vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ |
```

```
GNU nano 7.2 New Buffer *
server {
    listen 80;
    server_name localhost;
    location / {
        return 200 "Hello from mounted config!";
    }
}
```

Lab 4: Volume Management Commands

```

vanshika_munjal@DESKTOP-107G070: /mnt/c/Users/HP$ curl http://localhost:8080
Hello from mounted config!vanshika_munjal@DESKTOP-107G070: /mnt/c/Users/HP$ docker volume ls
DRIVER    VOLUME NAME
local     1b98d65bb3d767f925a58930476de00d88d28b860f1482b8baca23f82506c384
local     5d2e578b14128a809f370a10f082e14d67d6d660d7789719064c2af5b96f5d95
local     7ca4b083667911817f228440136bab6419fdc5750f7729c025fb39d4d9b52b57
local     11a9fe249bdf6a609686d75709d056a86bd9fc309463dbc643ba35089b49176
local     14cb18358c4dcfa09657cf0a68447f708a9da4ae8089507f406calb1c26983b8
local     25feb-class_mysql_data
local     25feb-class_wp_content
local     26feb-class_mysql_data
local     26feb-class_wp_content
local     37b3f46b3c81e6b3e340672661a7355a425a79ca409fbd432bbab4290c90dba1
local     49af95d171db743ae8a15940af4e2a4553d97f7fac2e37cf5c94d217ffbf2a2
local     63d456d51a0cb6f9071702390658f92c22f819641ac0da533ec291a5e28ee7
local     ae41b0e6b4e33ef5938627e214c938a307600f75d1a47bf268e3a4a1412bd0d6
local     afc2a5a7de2aaf163fe9e26d5caff53c4ffc2af38a82985b728294c715a5254
local     b9dc2ffabd12388d7010267d22c9f9c4ddc9ab91a4836405c22e2f7f0a617218
local     b0153f0b0ae69c9eb0a1a8f089957d3c3d1be33564abe8ef7776e336975f1cf9
local     cdab12f55ab0175bd1bcaebe7dc71075d205bfc290d8c7e4e1533aa41339d89e
local     ce92eb3a7a0ce60031a3aabe44c6f1405e36ac2391444b7c5c61f3a2ed3c329c
local     containerization-project-1_postgres_data
local     de08ff5a54f4941d6fa842eb2b9a9dbf43fbf72818b10651d12ac75d9d465181
local     ef44b4b654fcc01c53ad5d51363b7f7b8453e9ef8035ae81409264d4870a11bd
local     fb058b5137c74e4a6cab010bc33b66d7f5502847ecf0cc40e6732d3bb305f64c
local     k3d-mycluster-images
local     mydata
local     mysql-data
local     myvolume
local     portainer_data
local     test
vanshika_munjal@DESKTOP-107G070: /mnt/c/Users/HP$ docker volume create app-volume
app-volume
vanshika_munjal@DESKTOP-107G070: /mnt/c/Users/HP$ docker volume inspect app-volume
[
  {
    "CreatedAt": "2026-03-27T09:14:47Z",

```

```

vanshika_munjal@DESKTOP-107G070: /mnt/c/Users/HP$ docker volume inspect app-volume
[
  {
    "CreatedAt": "2026-03-27T09:14:47Z",
    "Driver": "local",
    "Labels": null,
    "Mountpoint": "/var/lib/docker/volumes/app-volume/_data",
    "Name": "app-volume",
    "Options": null,
    "Scope": "local"
  }
]
vanshika_munjal@DESKTOP-107G070: /mnt/c/Users/HP$ docker volume prune
WARNING! This will remove anonymous local volumes not used by at least one container.
Are you sure you want to continue? [y/N] y
Deleted Volumes:
b9dc2ffabd12388d7010267d22c9f9c4ddc9ab91a4836405c22e2f7f0a617218
1b98d65bb3d767f925a58930476de00d88d28b860f1482b8baca23f82506c384
ef44b4b654fcc01c53ad5d51363b7f7b8453e9ef8035ae81409264d4870a11bd
37b3f46b3c81e6b3e340672661a7355a425a79ca409fbd432bbab4290c90dba1
ce92eb3a7a0ce60031a3aabe44c6f1405e36ac2391444b7c5c61f3a2ed3c329c
cdab12f55ab0175bd1bcaebe7dc71075d205bfc290d8c7e4e1533aa41339d89e

Total reclaimed space: 246.1MB
vanshika_munjal@DESKTOP-107G070: /mnt/c/Users/HP$ docker volume rm test
Error response from daemon: remove test: volume is in use ~ [e7c11219b87715918de6e38a20ba7831f69691fc8763898a4cc714a3d9225753]
vanshika_munjal@DESKTOP-107G070: /mnt/c/Users/HP$ docker ps -a
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS
d70858ff092c   nginx     "/docker-entrypoint..." 2 minutes ago   Up 2 minutes   0.0.0.0:80
80->80/tcp, [::]:8080->80/tcp
924fbelf458f   mysql:8.0 "/docker-entrypoint.s..." 15 minutes ago   Up 15 minutes   3306/tcp,
33060/tcp
6274e0d3f87c   nginx     "/docker-entrypoint..." 20 minutes ago   Up 20 minutes   80/tcp
3357050f706c   nginx     "/docker-entrypoint..." 22 minutes ago   Up 22 minutes   80/tcp
web2

```

```

vanshika_munjal@DESKTOP-107G070: /mnt/c/Users/HP$ docker rm -f test2
test2
vanshika_munjal@DESKTOP-107G070: /mnt/c/Users/HP$ docker volume rm test
test
vanshika_munjal@DESKTOP-107G070: /mnt/c/Users/HP$ docker cp local-file.txt container-name:/path/in/volume
lsat /mnt/c/Users/HP/local-file.txt: no such file or directory
vanshika_munjal@DESKTOP-107G070: /mnt/c/Users/HP$ echo "Hello Docker Volume" > local-file.txt
vanshika_munjal@DESKTOP-107G070: /mnt/c/Users/HP$ docker cp local-file.txt nginx-custom:/tmp/
Successfully copied 2.05kB to nginx-custom:/tmp/
vanshika_munjal@DESKTOP-107G070: /mnt/c/Users/HP$ docker exec nginx-custom cat /tmp/local-file.txt
Hello Docker Volume
vanshika_munjal@DESKTOP-107G070: /mnt/c/Users/HP$ |

```

Part 2: Environment Variables

Lab 1: Setting Environment Variables

Method 1: Using -e flag

```
vanshika_munjial@DESKTOP-1D7G070:/mnt/c/Users/HP$ docker run -d \
--name app2 \
-e DATABASE_URL="postgres://user:pass@db:5432/mydb" \
-e DEBUG="true" \
-p 3001:3000 \
my-node-app
39e661272145b0dcd459e7133f1238d19ec8464994a8ae4f05bbbc52f4710b73
```

```
vanshika_munjial@DESKTOP-1D7G070:/mnt/c/Users/HP$ docker run -d \
--name app3 \
-e VAR1=value1 \
-e VAR2=value2 \
-e VAR3=value3 \
my-node-app
1769b49aca5db4780d432a3e20dbf5b4bc49a6cf18a09d9c5d800b8a19dc2980
vanshika_munjial@DESKTOP-1D7G070:/mnt/c/Users/HP$
```



Hello from Node.js Docker!

Method 2: Using --env-file

```
vanshika_munjial@DESKTOP-1D7G070:/mnt/c/Users/HP$ echo "DATABASE_HOST=localhost" > .env
vanshika_munjial@DESKTOP-1D7G070:/mnt/c/Users/HP$ echo "DATABASE_PORT=5432" >> .env
vanshika_munjial@DESKTOP-1D7G070:/mnt/c/Users/HP$ echo "API_KEY=secret123" >> .env
```

```
vanshika_munjial@DESKTOP-1D7G070:/mnt/c/Users/HP$ docker run -d \
--env-file .env \
--name app4 \
-p 3002:3000 \
my-node-app
bbb9a97b17722f2e5dea89c665d649da10ce61ea0a688ebf97071f221b7d5f11
```

Method 3: In Dockerfile

Set default environment variables

ENV NODE_ENV=production

ENV PORT=3000

ENV APP_VERSION=1.0.0

Part 3: Docker Monitoring

Lab 1: Basic Monitoring Commands

CONTAINER ID	NAME	CPU %	MEM USAGE / LIMIT	MEM %	NET I/O	BLOCK I/O	PIDS
bbb9a97b1772	app4	0.00%	15.65MiB / 7.607GiB	0.20%	1.01kB / 126B	0B / 0B	7
1769b49aca5d	app3	0.00%	15.64MiB / 7.607GiB	0.20%	1.13kB / 126B	0B / 0B	7
39e661272145	app2	0.00%	18.94MiB / 7.607GiB	0.24%	3.84kB / 1.99kB	6MB / 0B	7
d70058ff092c	nginx-custom	0.00%	10.42MiB / 7.607GiB	0.13%	2.17kB / 670B	356kB / 4.1kB	13
924fbelf458f	new-mysql	1.18%	354.2MiB / 7.607GiB	4.55%	1.85kB / 126B	65.5kB / 15.4MB	37
6274e0d3f87c	web3	0.00%	10.37MiB / 7.607GiB	0.13%	2.1kB / 126B	49.2kB / 12.3kB	13
3357050f706c	web2	0.00%	10.38MiB / 7.607GiB	0.13%	2.22kB / 126B	0B / 12.3kB	13
8fb48ad27a57	web1	0.00%	13.79MiB / 7.607GiB	0.18%	2.35kB / 126B	14.3MB / 12.3kB	13
43043fd06207	test-container	0.00%	924KiB / 7.607GiB	0.01%	2.6kB / 126B	0B / 0B	1
ea81f68bde87	portainer	0.00%	15.08MiB / 7.607GiB	0.19%	3.38kB / 126B	66.3MB / 139kB	11
924e1452e147	k3d-mycluster-serverlb	0.01%	27.18MiB / 7.607GiB	0.35%	1.78kB / 126B	23MB / 16.4kB	23
0fa96e9ee63a	k3d-mycluster-server-0	15.26%	722.5MiB / 7.607GiB	9.28%	68kB / 33.2kB	486MB / 114MB	608
f41e72cb4ef4	backend-api	0.00%	18.28MiB / 7.607GiB	0.23%	2.88kB / 1.25kB	496kB / 0B	11
7883abb665c2	postgres_db	0.00%	29.22MiB / 7.607GiB	0.38%	3.47kB / 2.01kB	32.7MB / 303kB	6

CONTAINER ID	NAME	CPU %	MEM USAGE / LIMIT	MEM %	NET I/O	BLOCK I/O	PIDS
61eaa2be19b7	container1	0.00%	0B / 0B	0.00%	0B / 0B	0B / 0B	0
e5c0d79603bf	container2	0.00%	0B / 0B	0.00%	0B / 0B	0B / 0B	0

NAME	CPU %	MEM USAGE / LIMIT	NET I/O
app4	0.00%	15.66MiB / 7.607GiB	1.01kB / 126B
app3	0.00%	15.64MiB / 7.607GiB	1.13kB / 126B
app2	0.00%	18.94MiB / 7.607GiB	3.84kB / 1.99kB
nginx-custom	0.00%	10.42MiB / 7.607GiB	2.17kB / 670B
new-mysql	1.02%	354.2MiB / 7.607GiB	1.85kB / 126B
web3	0.00%	10.37MiB / 7.607GiB	2.1kB / 126B
web2	0.00%	10.38MiB / 7.607GiB	2.22kB / 126B
web1	0.00%	13.79MiB / 7.607GiB	2.35kB / 126B
test-container	0.00%	924KiB / 7.607GiB	2.6kB / 126B
portainer	0.00%	15.08MiB / 7.607GiB	3.38kB / 126B
k3d-mycluster-serverlb	0.01%	27.18MiB / 7.607GiB	1.78kB / 126B
k3d-mycluster-server-0	13.12%	722.1MiB / 7.607GiB	68kB / 33.2kB
backend_api	0.00%	18.28MiB / 7.607GiB	2.88kB / 1.25kB
postgres_db	0.01%	29.22MiB / 7.607GiB	3.47kB / 2.01kB

```
vanshika_munjial@DESKTOP-1D7G070:/mnt/c/Users/HP$ docker stats --no-stream
```

CONTAINER ID	NAME	CPU %	MEM USAGE / LIMIT	MEM %	NET I/O	BLOCK I/O	PIDS
bbb9a97b1772	app4	0.00%	15.66MiB / 7.607GiB	0.20%	1.08kB / 126B	0B / 0B	7
1769b49aca5d	app3	0.00%	15.64MiB / 7.607GiB	0.20%	1.2kB / 126B	0B / 0B	7
39e661272145	app2	0.00%	18.94MiB / 7.607GiB	0.24%	3.91kB / 1.99kB	6MB / 0B	7
d70058ff092c	nginx-custom	0.00%	10.42MiB / 7.607GiB	0.13%	2.17kB / 670B	356kB / 4.1kB	13
924fbel1f458f	new-mysql	1.52%	354.2MiB / 7.607GiB	4.55%	1.85kB / 126B	65.5kB / 15.4MB	37
6274e0d3f87c	web3	0.00%	10.37MiB / 7.607GiB	0.13%	2.1kB / 126B	49.2kB / 12.3kB	13
3357050f706c	web2	0.00%	10.38MiB / 7.607GiB	0.13%	2.22kB / 126B	0B / 12.3kB	13
8fb48ad27a57	web1	0.00%	13.79MiB / 7.607GiB	0.18%	2.35kB / 126B	14.3MB / 12.3kB	13
43043fd06207	test-container	0.00%	924KiB / 7.607GiB	0.01%	2.6kB / 126B	0B / 0B	1
ea81f68bde87	portainer	0.00%	15.08MiB / 7.607GiB	0.19%	3.38kB / 126B	66.3MB / 139kB	11
924e1452e147	k3d-mycluster-serverlb	0.01%	27.18MiB / 7.607GiB	0.35%	1.78kB / 126B	23MB / 16.4kB	23
0fa96e9ee63a	k3d-mycluster-server-0	16.09%	720.3MiB / 7.607GiB	9.25%	68kB / 33.2kB	486MB / 117MB	608
f41e72cb4ef4	backend_api	0.00%	18.28MiB / 7.607GiB	0.23%	2.88kB / 1.25kB	496kB / 0B	11
7883abb665c2	postgres_db	0.01%	29.23MiB / 7.607GiB	0.38%	3.47kB / 2.01kB	32.7MB / 303kB	6

```
vanshika_munjial@DESKTOP-1D7G070:/mnt/c/Users/HP$
```

```
Container: app4 | CPU: 0.00% | Memory: 0.20%
Container: app3 | CPU: 0.00% | Memory: 0.20%
Container: app2 | CPU: 0.00% | Memory: 0.24%
Container: nginx-custom | CPU: 0.00% | Memory: 0.13%
Container: new-mysql | CPU: 1.45% | Memory: 4.55%
Container: web3 | CPU: 0.00% | Memory: 0.13%
Container: web2 | CPU: 0.00% | Memory: 0.13%
Container: web1 | CPU: 0.00% | Memory: 0.18%
Container: test-container | CPU: 0.00% | Memory: 0.01%
Container: portainer | CPU: 0.00% | Memory: 0.19%
Container: k3d-mycluster-serverlb | CPU: 0.00% | Memory: 0.35%
Container: k3d-mycluster-server-0 | CPU: 20.54% | Memory: 9.27%
Container: backend_api | CPU: 0.00% | Memory: 0.23%
Container: postgres_db | CPU: 0.02% | Memory: 0.38%
```

```

vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ docker stats --format json --no-stream
[{"BlockIO": "0B / 0B", "CPUPerc": "0.00%", "Container": "bbb9a97b1772", "ID": "bbb9a97b1772", "MemPerc": "0.20%", "MemUsage": "15.66MiB / 7.607GiB", "Name": "app4", "NetIO": "1.08kB / 126B", "PIDs": "7"}, {"BlockIO": "0B / 0B", "CPUPerc": "0.00%", "Container": "1769b49aca5d", "ID": "1769b49aca5d", "MemPerc": "0.20%", "MemUsage": "15.64MiB / 7.607GiB", "Name": "app3", "NetIO": "1.2kB / 126B", "PIDs": "7"}, {"BlockIO": "6MB / 0B", "CPUPerc": "0.00%", "Container": "39e661272145", "ID": "39e661272145", "MemPerc": "0.24%", "MemUsage": "18.94MiB / 7.607GiB", "Name": "app2", "NetIO": "3.91kB / 1.99kB", "PIDs": "7"}, {"BlockIO": "356kB / 4.1kB", "CPUPerc": "0.00%", "Container": "d70058ff092c", "ID": "d70058ff092c", "MemPerc": "0.13%", "MemUsage": "10.42MiB / 7.607GiB", "Name": "nginx-custom", "NetIO": "2.17kB / 670B", "PIDs": "13"}, {"BlockIO": "65.5kB / 15.4MB", "CPUPerc": "1.05%", "Container": "924fbef458f", "ID": "924fbef458f", "MemPerc": "4.55%", "MemUsage": "354.2MiB / 7.607GiB", "Name": "new-mysql", "NetIO": "1.85kB / 126B", "PIDs": "37"}, {"BlockIO": "49.2kB / 12.3kB", "CPUPerc": "0.00%", "Container": "6274e0d3f87c", "ID": "6274e0d3f87c", "MemPerc": "0.13%", "MemUsage": "10.37MiB / 7.607GiB", "Name": "web3", "NetIO": "2.1kB / 126B", "PIDs": "13"}, {"BlockIO": "0B / 12.3kB", "CPUPerc": "0.00%", "Container": "3357050f706c", "ID": "3357050f706c", "MemPerc": "0.13%", "MemUsage": "10.38MiB / 7.607GiB", "Name": "web2", "NetIO": "2.22kB / 126B", "PIDs": "13"}, {"BlockIO": "14.3MB / 12.3kB", "CPUPerc": "0.00%", "Container": "8fb48ad27a57", "ID": "8fb48ad27a57", "MemPerc": "0.18%", "MemUsage": "13.79MiB / 7.607GiB", "Name": "web1", "NetIO": "2.35kB / 126B", "PIDs": "13"}, {"BlockIO": "0B / 0B", "CPUPerc": "0.00%", "Container": "43043fd06207", "ID": "43043fd06207", "MemPerc": "0.01%", "MemUsage": "924KiB / 7.607GiB", "Name": "test-container", "NetIO": "2.6kB / 126B", "PIDs": "11"}, {"BlockIO": "66.3MB / 139kB", "CPUPerc": "0.13%", "Container": "ea81f68bde87", "ID": "ea81f68bde87", "MemPerc": "0.19%", "MemUsage": "15.08MiB / 7.607GiB", "Name": "portainer", "NetIO": "3.38kB / 126B", "PIDs": "11"}, {"BlockIO": "23MB / 16.4kB", "CPUPerc": "0.01%", "Container": "924e1452e147", "ID": "924e1452e147", "MemPerc": "0.35%", "MemUsage": "27.18MiB / 7.607GiB", "Name": "k3d-mycluster-serverlb", "NetIO": "1.78kB / 126B", "PIDs": "23"}, {"BlockIO": "486MB / 117MB", "CPUPerc": "16.23%", "Container": "0fa96e9ee63a", "ID": "0fa96e9ee63a", "MemPerc": "9.26%", "MemUsage": "721.6MiB / 7.607GiB", "Name": "k3d-mycluster-server-0", "NetIO": "68kB / 33.2kB", "PIDs": "608"}, {"BlockIO": "496kB / 0B", "CPUPerc": "0.00%", "Container": "f41e72cb4ef4", "ID": "f41e72cb4ef4", "MemPerc": "0.23%", "MemUsage": "18.28MiB / 7.607GiB", "Name": "backend_api", "NetIO": "2.88kB / 1.25kB", "PIDs": "11"}, {"BlockIO": "32.7MB / 303kB", "CPUPerc": "0.00%", "Container": "7883abb665c2", "ID": "7883abb665c2", "MemPerc": "0.38%", "MemUsage": "29.22MiB / 7.607GiB", "Name": "postgres_db", "NetIO": "3.47kB / 2.01kB", "PIDs": "6"}]
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$

```

```

vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ docker stats --no-stream --no-trunc
CONTAINER ID        NAME               CPU %               MEM USAGE / LIMIT   MEM %
NET I/O           BLOCK I/O         PIDS
bbb9a97b1772f2e5dea89c665d649da10ce61ea8a688ebf97071f221b7d5f11  app4               0.00%               15.66MiB / 7.607GiB  0.20%
1.08kB / 126B      0B / 0B            7
1769b49aca5db4780d432a3e20dbf5b4bc49a6cf18a09d0c5d800b8a19dc2980  app3               0.00%               15.64MiB / 7.607GiB  0.20%
1.2kB / 126B      0B / 0B            7
39e661272145b0dcd459e7133f1238d19ec8064994a8ae4f05bbbc52f4710b73  app2               0.00%               18.94MiB / 7.607GiB  0.24%
3.91kB / 1.99kB   6MB / 0B           7
d70058ff092c50f5c004b31ceddee7ea27e5abe5c02b90ac15280306938d78e   nginx-custom      0.00%               10.42MiB / 7.607GiB  0.13%
2.17kB / 670B    356kB / 4.1kB      13
924fbef458f92f0e9ee8f716a47d532b99c72b8a58fd73c0e18fd9300152     new-mysql         1.05%               354.2MiB / 7.607GiB  4.55%
1.85kB / 126B    65.5kB / 15.4MB    37
6274e0d3f87cc459c35cbbd8b32a56267ca1cb374ee7614bcb4bfc2e05109341  web3               0.00%               10.37MiB / 7.607GiB  0.13%
2.1kB / 126B     49.2kB / 12.3kB    13
3357050f706c0409b24af4b836cfdclaf789e1fd62dc2638f8e78a9eb360d9387  web2               0.00%               10.38MiB / 7.607GiB  0.13%
2.22kB / 126B    0B / 12.3kB        13
8fb48ad27a577eda2d801770b8b9812bb5bd361e7940fee669db824762de5609  web1               0.00%               13.79MiB / 7.607GiB  0.18%
2.35kB / 126B    14.3MB / 12.3kB    13
43043fd0620720fdb2f0c05809c2bcb87526276742dd4e73a34a7628e189b79f  test-container    0.00%               924KiB / 7.607GiB   0.01%
2.6kB / 126B     0B / 0B             1
ea81f68bde87dc911181d1906efe4d1b4c795c16f4fee4624a23d172d7ec6e7   portainer         0.06%               15.08MiB / 7.607GiB  0.19%
3.38kB / 126B    66.3MB / 139kB     11
924e1452e147e4353c695f0d5ea8f7a275fb5aafebc2847b42f56a3557afae84   k3d-mycluster-serverlb  0.00%               27.18MiB / 7.607GiB  0.35%
1.78kB / 126B    23MB / 16.4kB      23
0fa96e9ee63af593538644484cafe0c8de6f6a5627a1e790a47e978ce63c75     k3d-mycluster-server-0  13.84%              721.2MiB / 7.607GiB  9.26%
68kB / 33.2kB   486MB / 117MB      608
f41e72cb4ef4d5968beabdad12813e5d71b64f787bdb2a7581b9d6922c838859  backend_api       0.00%               18.28MiB / 7.607GiB  0.23%
2.88kB / 1.25kB  496kB / 0B         11
7883abb665c23d55601c32c765dabf3a34f2d41caf120e2f851af4684d568a98   postgres_db       0.00%               29.23MiB / 7.607GiB  0.38%
3.47kB / 2.01kB  32.7MB / 303kB     6

```

Lab 2: docker top- Process Monitoring

```

vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ docker top 43043fd06207
UID                PID                PPID              C                STIME             TTY              TIME
root               4923               4899              0                08:43             pts/0            00:00:00
/bin/bash
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ docker top 43043fd06207 -ef
UID                PID                PPID              C                STIME             TTY              TIME
root               4923               4899              0                08:43             pts/0            00:00:00
/bin/bash
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ ps aux | grep docker
root      569  0.0  0.0  14312  6992 pts/0    S+   08:39   0:00 sudo dockerd
root      573  0.0  0.0  14312  2532 pts/2    Ss   08:39   0:00 sudo dockerd
root      574  1.4  1.3 5874616 104276 pts/2    Sl+  08:39   1:21 dockerd
root      1312  0.0  0.0   1056   640 ?        Ss   08:39   0:00 /sbin/docker-init -- /bin/k3d-entrypoint.sh server --tls-san 0.0.0
.0 --tls-san k3d-mycluster-serverlb
root      1316  0.0  0.0   1056   640 ?        Ss   08:39   0:00 /sbin/docker-init -- /bin/sh -c nginx-proxy
root      1419  0.0  0.0  1745100 3968 pts/2    Sl+  08:39   0:00 /usr/bin/docker-proxy -proto tcp -host-ip 0.0.0.0 -host-port 8000
-container-ip 172.18.0.2 -container-port 8000 -use-listen-fd
root      1427  0.0  0.0  1818832 4096 pts/2    Sl+  08:39   0:00 /usr/bin/docker-proxy -proto tcp -host-ip :: -host-port 8000 -cont
ainer-ip 172.18.0.2 -container-port 8000 -use-listen-fd
root      1434  0.0  0.0  1671368 3840 pts/2    Sl+  08:39   0:00 /usr/bin/docker-proxy -proto tcp -host-ip 0.0.0.0 -host-port 9443
-container-ip 172.18.0.2 -container-port 9443 -use-listen-fd
root      1443  0.0  0.0  1671368 3840 pts/2    Sl+  08:39   0:00 /usr/bin/docker-proxy -proto tcp -host-ip :: -host-port 9443 -cont
ainer-ip 172.18.0.2 -container-port 9443 -use-listen-fd
root      1535  0.0  0.0  1892820 3968 pts/2    Sl+  08:39   0:00 /usr/bin/docker-proxy -proto tcp -host-ip 0.0.0.0 -host-port 39691
-container-ip 172.24.0.3 -container-port 6443 -use-listen-fd
root      1835  0.0  0.0  1745100 4096 pts/2    Sl+  08:39   0:00 /usr/bin/docker-proxy -proto tcp -host-ip 0.0.0.0 -host-port 3000
-container-ip 172.23.0.2 -container-port 3000 -use-listen-fd
root      1842  0.0  0.0  1671368 4096 pts/2    Sl+  08:39   0:00 /usr/bin/docker-proxy -proto tcp -host-ip :: -host-port 3000 -cont
ainer-ip 172.23.0.2 -container-port 3000 -use-listen-fd
root      7793  0.0  0.0  1671368 3968 pts/2    Sl+  09:11   0:00 /usr/bin/docker-proxy -proto tcp -host-ip 0.0.0.0 -host-port 8080
-container-ip 172.18.0.8 -container-port 80 -use-listen-fd
root      7799  0.0  0.0  1745100 3968 pts/2    Sl+  09:11   0:00 /usr/bin/docker-proxy -proto tcp -host-ip :: -host-port 8000 -cont
ainer-ip 172.18.0.8 -container-port 80 -use-listen-fd
root      10208  0.0  0.0  1671368 4096 pts/2    Sl+  09:47   0:00 /usr/bin/docker-proxy -proto tcp -host-ip 0.0.0.0 -host-port 3001

```


Lab 3: docker logs- Application Logs

```
vanshika_munjial@DESKTOP-107G070:/mnt/c/Users/HP$ docker logs 8fb48ad27a57
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/03/27 08:52:28 [notice] 1#1: using the "epoll" event method
2026/03/27 08:52:28 [notice] 1#1: nginx/1.29.5
2026/03/27 08:52:28 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/03/27 08:52:28 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
2026/03/27 08:52:28 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/03/27 08:52:28 [notice] 1#1: start worker processes
2026/03/27 08:52:28 [notice] 1#1: start worker process 29
2026/03/27 08:52:28 [notice] 1#1: start worker process 30
2026/03/27 08:52:28 [notice] 1#1: start worker process 31
2026/03/27 08:52:28 [notice] 1#1: start worker process 32
2026/03/27 08:52:28 [notice] 1#1: start worker process 33
2026/03/27 08:52:28 [notice] 1#1: start worker process 34
2026/03/27 08:52:28 [notice] 1#1: start worker process 35
2026/03/27 08:52:28 [notice] 1#1: start worker process 36
2026/03/27 08:52:28 [notice] 1#1: start worker process 37
2026/03/27 08:52:28 [notice] 1#1: start worker process 38
2026/03/27 08:52:28 [notice] 1#1: start worker process 39
2026/03/27 08:52:28 [notice] 1#1: start worker process 40
vanshika_munjial@DESKTOP-107G070:/mnt/c/Users/HP$ docker logs -f 8fb48ad27a57
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
```

```
vanshika_munjial@DESKTOP-107G070:/mnt/c/Users/HP$ docker logs --tail 100 8fb48ad27a57
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/03/27 08:52:28 [notice] 1#1: using the "epoll" event method
2026/03/27 08:52:28 [notice] 1#1: nginx/1.29.5
2026/03/27 08:52:28 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/03/27 08:52:28 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
2026/03/27 08:52:28 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/03/27 08:52:28 [notice] 1#1: start worker processes
2026/03/27 08:52:28 [notice] 1#1: start worker process 29
2026/03/27 08:52:28 [notice] 1#1: start worker process 30
2026/03/27 08:52:28 [notice] 1#1: start worker process 31
2026/03/27 08:52:28 [notice] 1#1: start worker process 32
2026/03/27 08:52:28 [notice] 1#1: start worker process 33
2026/03/27 08:52:28 [notice] 1#1: start worker process 34
2026/03/27 08:52:28 [notice] 1#1: start worker process 35
2026/03/27 08:52:28 [notice] 1#1: start worker process 36
2026/03/27 08:52:28 [notice] 1#1: start worker process 37
2026/03/27 08:52:28 [notice] 1#1: start worker process 38
2026/03/27 08:52:28 [notice] 1#1: start worker process 39
2026/03/27 08:52:28 [notice] 1#1: start worker process 40
```

```

vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ docker logs -t 8fb48ad27a57
2026-03-27T08:52:28.336172993Z /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
2026-03-27T08:52:28.336209888Z /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
2026-03-27T08:52:28.339646764Z /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
2026-03-27T08:52:28.352608579Z 10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
2026-03-27T08:52:28.372494140Z 10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
2026-03-27T08:52:28.372620352Z /docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
2026-03-27T08:52:28.373063298Z /docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
2026-03-27T08:52:28.377969097Z /docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
2026-03-27T08:52:28.388120410Z /docker-entrypoint.sh: Configuration complete; ready for start up
2026-03-27T08:52:28.408027395Z 2026/03/27 08:52:28 [notice] 1#1: using the "epoll" event method
2026-03-27T08:52:28.408118291Z 2026/03/27 08:52:28 [notice] 1#1: nginx/1.29.5
2026-03-27T08:52:28.408121087Z 2026/03/27 08:52:28 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026-03-27T08:52:28.408122814Z 2026/03/27 08:52:28 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
2026-03-27T08:52:28.408124514Z 2026/03/27 08:52:28 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026-03-27T08:52:28.408365695Z 2026/03/27 08:52:28 [notice] 1#1: start worker processes
2026-03-27T08:52:28.408600130Z 2026/03/27 08:52:28 [notice] 1#1: start worker process 29
2026-03-27T08:52:28.408887205Z 2026/03/27 08:52:28 [notice] 1#1: start worker process 30
2026-03-27T08:52:28.409393175Z 2026/03/27 08:52:28 [notice] 1#1: start worker process 31
2026-03-27T08:52:28.409785172Z 2026/03/27 08:52:28 [notice] 1#1: start worker process 32
2026-03-27T08:52:28.410030404Z 2026/03/27 08:52:28 [notice] 1#1: start worker process 33
2026-03-27T08:52:28.410504408Z 2026/03/27 08:52:28 [notice] 1#1: start worker process 34
2026-03-27T08:52:28.410838648Z 2026/03/27 08:52:28 [notice] 1#1: start worker process 35
2026-03-27T08:52:28.411398480Z 2026/03/27 08:52:28 [notice] 1#1: start worker process 36
2026-03-27T08:52:28.41185500Z 2026/03/27 08:52:28 [notice] 1#1: start worker process 37
2026-03-27T08:52:28.412219638Z 2026/03/27 08:52:28 [notice] 1#1: start worker process 38
2026-03-27T08:52:28.41264315Z 2026/03/27 08:52:28 [notice] 1#1: start worker process 39
2026-03-27T08:52:28.413006771Z 2026/03/27 08:52:28 [notice] 1#1: start worker process 40
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ docker logs --since 2024-01-15 8fb48ad27a57
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh

```

```

vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ docker logs -f --tail 50 -t 8fb48ad27a57
2026-03-27T08:52:28.336172993Z /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
2026-03-27T08:52:28.336209888Z /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
2026-03-27T08:52:28.339646764Z /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
2026-03-27T08:52:28.352608579Z 10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
2026-03-27T08:52:28.372494140Z 10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
2026-03-27T08:52:28.372620352Z /docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
2026-03-27T08:52:28.373063298Z /docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
2026-03-27T08:52:28.377969097Z /docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
2026-03-27T08:52:28.388120410Z /docker-entrypoint.sh: Configuration complete; ready for start up
2026-03-27T08:52:28.408027395Z 2026/03/27 08:52:28 [notice] 1#1: using the "epoll" event method
2026-03-27T08:52:28.408118291Z 2026/03/27 08:52:28 [notice] 1#1: nginx/1.29.5
2026-03-27T08:52:28.408121087Z 2026/03/27 08:52:28 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026-03-27T08:52:28.408122814Z 2026/03/27 08:52:28 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
2026-03-27T08:52:28.408124514Z 2026/03/27 08:52:28 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026-03-27T08:52:28.408365695Z 2026/03/27 08:52:28 [notice] 1#1: start worker processes
2026-03-27T08:52:28.408600130Z 2026/03/27 08:52:28 [notice] 1#1: start worker process 29
2026-03-27T08:52:28.408887205Z 2026/03/27 08:52:28 [notice] 1#1: start worker process 30
2026-03-27T08:52:28.409393175Z 2026/03/27 08:52:28 [notice] 1#1: start worker process 31
2026-03-27T08:52:28.409785172Z 2026/03/27 08:52:28 [notice] 1#1: start worker process 32
2026-03-27T08:52:28.410030404Z 2026/03/27 08:52:28 [notice] 1#1: start worker process 33
2026-03-27T08:52:28.410504408Z 2026/03/27 08:52:28 [notice] 1#1: start worker process 34
2026-03-27T08:52:28.410838648Z 2026/03/27 08:52:28 [notice] 1#1: start worker process 35
2026-03-27T08:52:28.411398480Z 2026/03/27 08:52:28 [notice] 1#1: start worker process 36
2026-03-27T08:52:28.41185500Z 2026/03/27 08:52:28 [notice] 1#1: start worker process 37
2026-03-27T08:52:28.412219638Z 2026/03/27 08:52:28 [notice] 1#1: start worker process 38
2026-03-27T08:52:28.41264315Z 2026/03/27 08:52:28 [notice] 1#1: start worker process 39
2026-03-27T08:52:28.413006771Z 2026/03/27 08:52:28 [notice] 1#1: start worker process 40

```

Lab 4: Container Inspection

```

*vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ docker inspect 924fbef458f
[
  {
    "Id": "924fbef458fc92f0e9ee8f716a47d532b99c72b8a858fd73c0e18fde9300152",
    "Created": "2026-03-27T09:00:29.5972961Z",
    "Path": "/docker-entrypoint.sh",
    "Args": [
      "mysql"
    ],
    "State": {
      "Status": "running",
      "Running": true,
      "Paused": false,
      "Restarting": false,
      "OOMKilled": false,
      "Dead": false,
      "Pid": 6730,
      "ExitCode": 0,
      "Error": "",
      "StartedAt": "2026-03-27T09:00:29.716439594Z",
      "FinishedAt": "0001-01-01T00:00:00Z"
    },
    "Image": "sha256:a123df39bc7c56bb6e40f8e5ef90469cc191dd7afcbf6e0d618c61a12e225dde",
    "ResolvConfPath": "/var/lib/docker/containers/924fbef458fc92f0e9ee8f716a47d532b99c72b8a858fd73c0e18fde9300152/resolv.conf",
    "HostnamePath": "/var/lib/docker/containers/924fbef458fc92f0e9ee8f716a47d532b99c72b8a858fd73c0e18fde9300152/hostname",
    "HostsPath": "/var/lib/docker/containers/924fbef458fc92f0e9ee8f716a47d532b99c72b8a858fd73c0e18fde9300152/hosts",
    "LogPath": "/var/lib/docker/containers/924fbef458fc92f0e9ee8f716a47d532b99c72b8a858fd73c0e18fde9300152/924fbef458fc92f0e9ee8f716a47d532b99c72b8a858fd73c0e18fde9300152-tsn.log"
  }
]

```



```

vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ docker inspect --format='{{.State.Status}}' 924fbelf458f
running
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ docker inspect --format='{{.NetworkSettings.IPAddress}}' 924fbelf458f
172.18.0.7
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ docker inspect --format='{{.Config.Env}}' 924fbelf458f
[MYSQL_ROOT_PASSWORD=secret PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin GOSU_VERSION=1.19 MYSQL_MAJOR=8.0 MYSQL
_VERSION=8.0.45-1.el9 MYSQL_SHELL_VERSION=8.0.45-1.el9]
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ docker inspect --format='{{.HostConfig.Memory}}' 924fbelf458f
0
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ docker inspect --format='{{.HostConfig.NanoCpus}}' 924fbelf458f
0
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ |

```

Lab 5: Practical Monitoring Script

```

GNU nano 7.2 monitor.sh
#!/bin/bash
# monitor.sh - Simple Docker monitoring

echo "=== Docker Monitoring Dashboard ==="
echo "Time: $(date)"
echo

echo "1. Running Containers:"
docker ps --format "table {{.Names}}\t{{.Status}}\t{{.Ports}}"
echo

echo "2. Resource Usage:"
docker stats --no-stream --format "table {{.Name}}\t{{.CPUPerc}}\t{{.MemUsage}}\t{{.NetIO}}\t{{.BlockIO}}"
echo

echo "3. Recent Events:"
docker events --since '5m' --until '0s' --format '{{.Time}} {{.Type}} {{.Action}}' | tail -5
echo

echo "4. System Info:"
docker system df

```

```

vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ nano monitor.sh
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ chmod +x monitor.sh
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ ./monitor.sh
=== Docker Monitoring Dashboard ===
Time: Fri Mar 27 10:25:33 UTC 2026

1. Running Containers:
NAME                STATUS              PORTS
app4                Up 33 minutes      0.0.0.0:3002->3000/tcp, [::]:3002->3000/tcp
app3                Up 36 minutes      3000/tcp
app2                Up 37 minutes      0.0.0.0:3001->3000/tcp, [::]:3001->3000/tcp
nginx-custom        Up About an hour   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
new-mysql            Up About an hour   3306/tcp, 33060/tcp
web3                Up About an hour   80/tcp
web2                Up 2 hours         80/tcp
web1                Up 2 hours         80/tcp
test-container      Up 2 hours
portainer           Up 2 hours         0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp, 0.0.0.0:9443->9443/tcp, [::]:9443->9443/tcp,
9080/tcp
k3d-mycluster-serverlb Up 2 hours         80/tcp, 0.0.0.0:39691->6443/tcp
k3d-mycluster-server-0 Up 2 hours
backend_api         Up 2 hours         0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp
postgres_db         Up 2 hours         5432/tcp

2. Resource Usage:
NAME                CPU %             MEM USAGE / LIMIT   NET I/O           BLOCK I/O
app4                0.00%            15.67MiB / 7.607GiB  1.15kB / 126B     0B / 0B
app3                0.02%            15.65MiB / 7.607GiB  1.27kB / 126B     0B / 0B
app2                0.00%            18.96MiB / 7.607GiB  3.91kB / 1.99kB    6MB / 0B
nginx-custom        0.00%            10.42MiB / 7.607GiB  2.24kB / 670B     356kB / 4.1kB
new-mysql           1.06%            354.2MiB / 7.607GiB  1.85kB / 126B     65.5kB / 15.4MB
web3                0.00%            10.37MiB / 7.607GiB  2.1kB / 126B      49.2kB / 12.3kB
web2                0.00%            10.38MiB / 7.607GiB  2.22kB / 126B     0B / 12.3kB
web1                0.00%            13.79MiB / 7.607GiB  2.35kB / 126B     14.3MB / 12.3kB
test-container      0.00%            924KiB / 7.607GiB    2.6kB / 126B      0B / 0B
portainer           0.00%            15.09MiB / 7.607GiB  3.38kB / 126B     66.3MB / 139kB
k3d-mycluster-serverlb 0.01%            27.18MiB / 7.607GiB  1.78kB / 126B     23MB / 16.4kB

```

Part 4: Docker Networks

Lab 1: Understanding Docker Network Types

```

vanshika_munjial@DESKTOP-107G070:/mnt/c/Users/HP$ docker network ls
NETWORK ID          NAME                                   DRIVER  SCOPE
c240c021bc32        25feb-class_wordpress-network        bridge  local
ffe9bee69d4d        26feb-class_wordpress-network        bridge  local
a0d4824dc341        bridge                                bridge  local
6145b902375a        containerization-project-1_default   bridge  local
6bfcffe93f47        docker_gwbridge                       bridge  local
21b3242599f7        host                                  host     local
pmvdlnm6v9xq        ingress                               overlay  swarm
531b76a88b56        k3d-mycluster                        bridge  local
ec10f63ed559        lan_net                               macvlan  local
4682c419e924        my_bridge                             bridge  local
301f2539971a        my_ipvlan                             ipvlan   local
mlq0hxxk8rbqm       my_overlay                            overlay  swarm
7464a52939aa        none                                  null     local

```

Lab 2: Network Types Explained

1. Bridge Network

```

vanshika_munjial@DESKTOP-107G070:/mnt/c/Users/HP$ docker network create my-network
49c767ae24e0f4fef11d1589c97edb88f16fe5e3b6db2b567693d578b44c1bf7
vanshika_munjial@DESKTOP-107G070:/mnt/c/Users/HP$ docker network inspect my-network
[
  {
    "Name": "my-network",
    "Id": "49c767ae24e0f4fef11d1589c97edb88f16fe5e3b6db2b567693d578b44c1bf7",
    "Created": "2026-03-27T10:29:31.07815551Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": {},
      "Config": [
        {
          "Subnet": "172.25.0.0/16",
          "Gateway": "172.25.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Containers": {},
    "Options": {},
    "Labels": {}
  }
]

```

```

vanshika_munjial@DESKTOP-107G070:/mnt/c/Users/HP$ docker run -d --name web1a --network my-network nginx
db333ad836633dbdaa8f962919ff8c8db46fb73e322a06dfcac8b5bf38472e376
vanshika_munjial@DESKTOP-107G070:/mnt/c/Users/HP$ docker run -d --name web2a --network my-network nginx
584bf940eccc88e6736d76c6223dd331a36f0c6d35b77ed86d2d12f413c829e5
vanshika_munjial@DESKTOP-107G070:/mnt/c/Users/HP$ docker exec web1a curl http://web2a
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
  0     0     0     0     0     0      0      0  --:--:-- --:--:-- --:--:--     0<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
100   615   100   615     0     0  78105     0  --:--:-- --:--:-- --:--:--  87857

```

1. Host Network

```

vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/HP$ docker run -d --name host-app --network host nginx
d8b208d656beaab718416572ec8b80926e716f58b48b3e856be8847791ef77cf
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/HP$ curl http://localhost
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>

```

2. None Network

```

vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/HP$ docker run -d --name isolated-app --network none alpine sleep 3600
fe852c8528fbf52a05e4a009a6e9989601bade0211822ca188ff096b963fbd21
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/HP$ docker exec isolated-app ifconfig
lo:
Link encap:Local Loopback
inet addr:127.0.0.1 Mask:255.0.0.0
inet6 addr: ::1/128 Scope:Host
UP LOOPBACK RUNNING MTU:65536 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

```

3. Overlay Network (swarm)

```

vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/HP$ docker network create --driver overlay my-overlay
g3cjdmbhrsmr59qw51bkke15z
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/HP$ |

```

Lab 3: Network Management Commands

```

vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/HP$ docker network create \
--driver bridge \
--subnet 192.168.100.0/24 \
--gateway 192.168.100.1 \
my-subnet
2192ec849b8aa44b6eb56b17239a7b017c6c8f5a7964d8f350f57844e17152e9
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/HP$ docker network connect app-network ubuntu
Error response from daemon: No such container: ubuntu
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/HP$ docker network connect app-network 924fbelf458f
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/HP$ docker network disconnect app-network 924fbelf458f
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/HP$ docker network rm app-network
app-network
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/HP$ docker network prune
WARNING! This will remove all custom networks not used by at least one container.
Are you sure you want to continue? [y/N] y
Deleted Networks:
my-overlay
my_overlay
my_ipvlan
my_bridge
25feb-class_wordpress-network
lan_net
26feb-class_wordpress-network
my-subnet

```

Lab 4: Multi-container Application Program


```

vanshika_munj665@DESKTOP-107G070: /mnt/c/Users/HP$ docker network create app-network
7643b49574c5b922996c108dcb98418c536cb047096c3e78cd8e1f137b73b58f
vanshika_munj665@DESKTOP-107G070: /mnt/c/Users/HP$ docker run -d \
--name postgres-db \
--network app-network \
-e POSTGRES_PASSWORD=secret \
-v pgdata:/var/lib/postgresql/data \
postgres:15
Unable to find image 'postgres:15' locally
15: Pulling from library/postgres
ec781dee3f47: Pull complete
ce2467f2f21d: Pull complete
d4ced18622af: Pull complete
af73ff344a10: Pull complete
07cec992154a: Pull complete
fd0dcffac314: Pull complete
bda354b903ce: Pull complete
fcd98f4944fb: Pull complete
45e04ac2d7d2: Pull complete
fla2676f0017: Pull complete
eab3f9a27af8: Pull complete
d2c8c030ed3b: Pull complete
c2e686ef6906: Pull complete
b2c40507d19c: Pull complete
Digest: sha256:c635fa3e3b7421a659d34abdf6d492f679cbe8149e261a501237b55c5a94212
Status: Downloaded newer image for postgres:15
4c8db85030f3e2690354c80f62f7e021c0280ac232ca5fda0cf45004c3c0b76c

```

```

vanshika_munj665@DESKTOP-107G070: /mnt/c/Users/HP$ docker run -d \
--name web-app \
--network app-network \
-p 8081:3000 \
-e DATABASE_URL="postgres://postgres:secret@postgres-db:5432/mydb" \
-e DATABASE_HOST="postgres-db" \
my-node-app
docker: Error response from daemon: Conflict. The container name "/web-app" is already in use by container "cfb7877522754e740eb7ba49c
a16fd706427548759f04f6dba868c2a27721761". You have to remove (or rename) that container to be able to reuse that name.

Run 'docker run --help' for more information
vanshika_munj665@DESKTOP-107G070: /mnt/c/Users/HP$ docker rm -f web-app
web-app
vanshika_munj665@DESKTOP-107G070: /mnt/c/Users/HP$ docker run -d \
--name web-app \
--network app-network \
-p 8081:3000 \
-e DATABASE_URL="postgres://postgres:secret@postgres-db:5432/mydb" \
-e DATABASE_HOST="postgres-db" \
my-node-app
d6328ed3fe7291f5ec9e78abaf4a47a66b35ee3d01b042ef0d4d32a72290749
vanshika_munj665@DESKTOP-107G070: /mnt/c/Users/HP$

```



Hello from Node.js Docker!

Lab 5: Network Inspection & Debugging

```

vanshika_munj665@DESKTOP-107G070: /mnt/c/Users/HP$ docker network inspect bridge
[
  {
    "Name": "bridge",
    "Id": "a0d4824dc3413a83b2941f39f062e5032785476e609573473a2cc6247da94883",
    "Created": "2026-03-27T08:42:26.409788716Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": null,
      "Config": [
        {
          "Subnet": "172.18.0.0/16",
          "Gateway": "172.18.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    }
  }
]

```

```
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ docker inspect --format='{{(range .NetworkSettings.Networks)}}{{.IPAddress}}{{(end)}}' 584bf940eccc
172.15.0.3
```

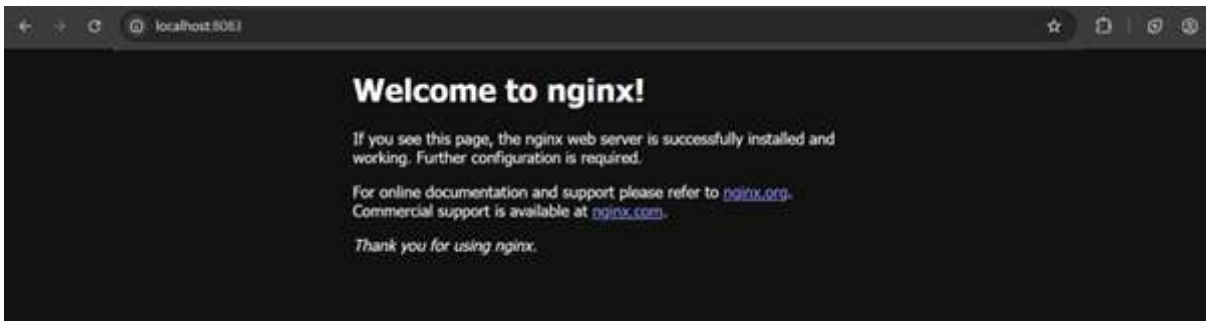
```
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ docker exec web-app ping -c 3 postgres-db
PING postgres-db (172.19.0.2): 56 data bytes
64 bytes from 172.19.0.2: seq=0 ttl=64 time=1.125 ms
64 bytes from 172.19.0.2: seq=1 ttl=64 time=0.171 ms
64 bytes from 172.19.0.2: seq=2 ttl=64 time=0.151 ms

--- postgres-db ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.151/0.482/1.125 ms
```

```
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ docker exec web-app getent hosts postgres-db
172.19.0.2      postgres-db  postgres-db
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ docker port 584bf940eccc
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$
```

Lab 6: Port Publishing vs Exposing

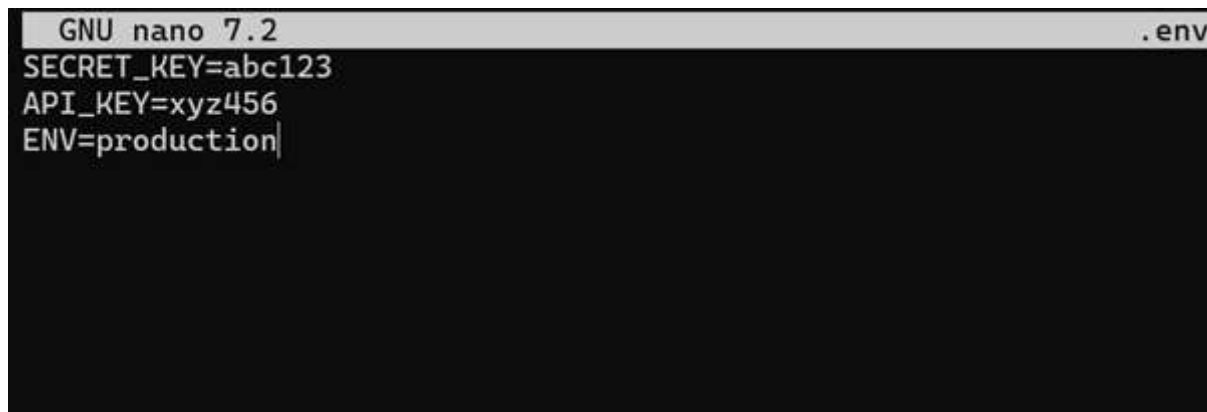
```
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ docker run -d -p 8083:80 --name app7 nginx
441d92590ddd82948cae3be5e8a7d9594f1135891fad3d8125d5792639fce9c
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$
```



Part 5: Complete Real-world Example

Implementation:

```
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ docker network create myapp-network
22cfb0a54a9a722f898765d7b6079b05507a3db696880b02bec313b0af819a160
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ docker run -d \
--name postgres \
--network myapp-network \
-e POSTGRES_PASSWORD=mysecretpassword \
-e POSTGRES_DB=mydatabase \
-v postgres-data:/var/lib/postgresql/data \
postgres:15
5db1ee994cd93690d34bda7ba702990fa3f22d772a546738797b74d8277e0808
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ docker run -d \
--name redis \
--network myapp-network \
-v redis-data:/data \
redis:7-alpine
Unable to find image 'redis:7-alpine' locally
7-alpine: Pulling from library/redis
bc1da058f299: Pull complete
09a5a0c32a23: Pull complete
bd53938b1271: Pull complete
9fe3744a2eac: Pull complete
7ad54b3c4cef: Pull complete
2a97533d89ca: Pull complete
4f4fb700ef54: Pull complete
1c2c8d1ee428: Pull complete
Digest: sha256:8b81dd37ff027bec4e516d41acfbe9fe2460070dc6d4a4570a2ac5b9d59df065
Status: Downloaded newer image for redis:7-alpine
b44969d94ffbc1556b96d9b4b6bb3650488bb7939fce0b8005478459761db6d
```



```
GNU nano 7.2 .env
SECRET_KEY=abc123
API_KEY=xyz456
ENV=production|
```

Key Takeaways

1. Volumes persist data beyond container lifecycle
2. Environment variables configure containers dynamically
3. Monitoring commands help debug and optimize containers
4. Networks enable secure container communication
5. Always use named volumes for production data
6. Custom networks provide better isolation and DNS
7. Monitor resource usage to prevent issues
8. Use .env files for sensitive configuration

This experiment covers essential Docker features for building, configuring, and managing production-ready containerized applications.

vanshikamunjal665

CONTAINERIZATION AND DEVOPS LAB

Experiment 6A:

Comparison of Docker Run and Docker Compose

Reference Link:

<https://upessocs.github.io/#dir=/Lectures/Containerization%20and%20DevOps/Lab/&file=Experiment%2006%20Docker%20Compose%20vs%20Docker%20Run,%20and%20MultiContainer%20Application%20-%20Wordpress+MySQL.md>

Part A: Theory

1. Objective

To understand the relationship between `docker run` and Docker Compose, and to compare their configuration syntax and use cases.

2. Background Theory

2.1 Docker Run (Imperative Approach)

The `docker run` command is used to create and start a container from an image. It requires explicit flags for:

- Port mapping (`-p`)
- Volume mounting (`-v`)
- Environment variables (`-e`)
- Network configuration (`--network`)
- Restart policies (`--restart`)
- Resource limits (`--memory` , `--cpus`)
- Container name (`--name`)

This approach is **imperative**, meaning you specify step-by-step instructions.

Example:

```
docker run -d \  
  --name my-nginx \  
  -p 8080:80 \  
  -v ./html:/usr/share/nginx/html \  
  -e NGINX_HOST=localhost \  
  --restart unless-stopped \  
nginx:alpine
```

2.2 Docker Compose (Declarative Approach)

Docker Compose uses a YAML file (`docker-compose.yml`) to define services, networks, and volumes in a structured format.

Instead of multiple `docker run` commands, a single command is used:

```
docker compose up -d
```

Compose is **declarative**, meaning you define the desired state of the application.

Equivalent Compose File:

```
version: '3.8'  
  
services:  
  nginx:  
    image: nginx:alpine  
    container_name: my-nginx  
    ports:  
      - "8080:80"  
    volumes:  
      - ./html:/usr/share/nginx/html  
    environment:  
      NGINX_HOST: localhost  
    restart: unless-stopped
```

3. Mapping: Docker Run vs Docker Compose

Docker Run Flag	Docker Compose Equivalent
-p 8080:80	ports:
-v host:container	volumes:
-e KEY=value	environment:
--name	container_name:
--network	networks:
--restart	restart:
--memory	deploy.resources.limits.memory
--cpus	deploy.resources.limits.cpus
-d	docker compose up -d

4. Advantages of Docker Compose

1. Simplifies multi-container applications
2. Provides reproducibility
3. Version controllable configuration
4. Unified lifecycle management
5. Supports service scaling

Example:

```
docker compose up --scale web=3
```

Part B: Practical Task

Task 1: Single Container Comparison

Step 1: Run Nginx using Docker Run

```

GNU nano 7.2                                html/index.html
<h1>Hello Vanshika 🚀</h1>
<p>Your Nginx Docker is working!</p>

```

```

vanshika_munjai@DESKTOP-107G07D:/mnt/c/Users/HP$ nano html/index.html
vanshika_munjai@DESKTOP-107G07D:/mnt/c/Users/HP$ docker run -d \
--name lab-nginx \
-p 8081:80 \
-v $(pwd)/html:/usr/share/nginx/html \
nginx:alpine
6efe77f3e6ab03a4f587ef5365f09a1e38dcb3baa08a601dd6a5a9ea35419ab3
vanshika_munjai@DESKTOP-107G07D:/mnt/c/Users/HP$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
6efe77f3e6ab   nginx:alpine                        "/docker-entrypoint.…"   2 seconds ago Up 2 seconds  0.0.0.0:8081->80/tcp, [:
:]8081->80/tcp   lab-nginx
ea81f68bde87   portainer/portainer-ce:latest      "/portainer"            13 days ago   Up 3 minutes  0.0.0.0:8000->8000/tcp, [:
:]8000->8000/tcp, 0.0.0.0:9443->9443/tcp, [::]:9443->9443/tcp, 9000/tcp   portainer
924e1452e147   ghcr.io/k3d-io/k3d-proxy:5.8.3     "/bin/sh -c nginx-pr…"   2 weeks ago   Up 3 minutes  80/tcp, 0.0.0.0:39691->6
443/tcp         k3d-mycluster-serverlb
0fa96e9ee63a   rancher/k3s:v1.31.5-k3s1          "/bin/k3d-entrypoint.…"   2 weeks ago   Up 3 minutes  k3d-mycluster-server-0
f41e72cb4ef4   containerization-project-1_backend  "docker-entrypoint.s…"   2 weeks ago   Up 3 minutes  0.0.0.0:3000->3000/tcp, [:
:]3000->3000/tcp   backend_api
7883abb665c2   containerization-project-1_database "docker-entrypoint.s…"   2 weeks ago   Up 3 minutes  5432/tcp        postgres_db
vanshika_munjai@DESKTOP-107G07D:/mnt/c/Users/HP$ |

```

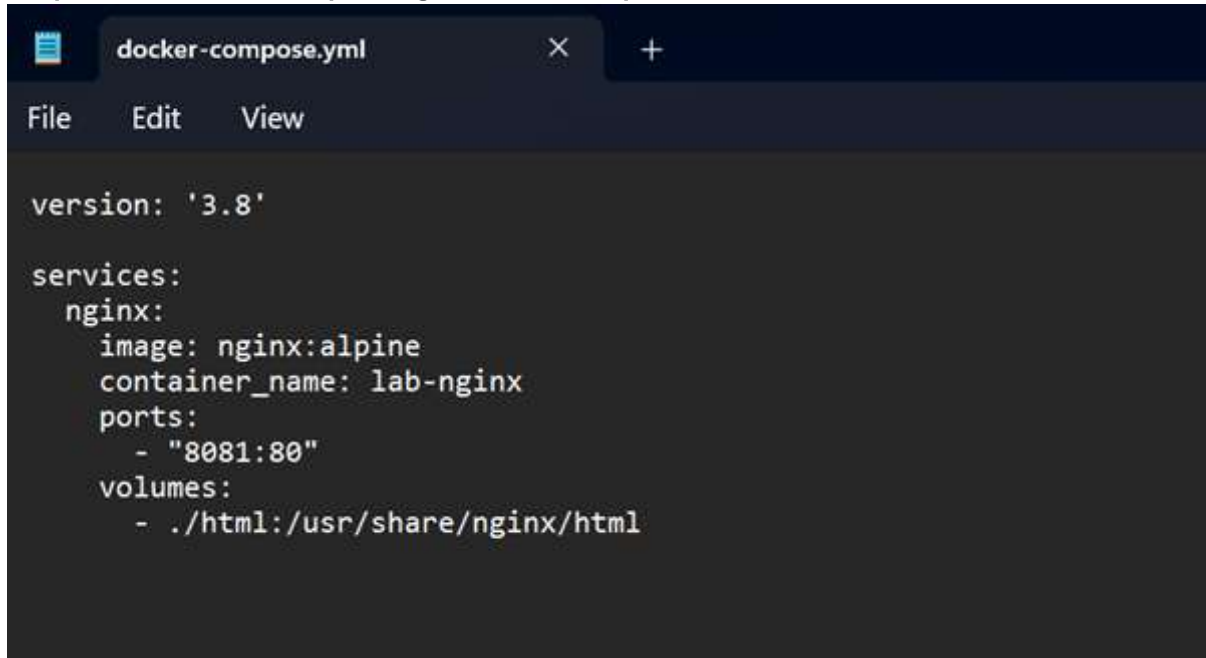


```

vanshika_munjai@DESKTOP-107G07D:/mnt/c/Users/HP$ docker stop lab-nginx
lab-nginx
vanshika_munjai@DESKTOP-107G07D:/mnt/c/Users/HP$ docker rm lab-nginx
lab-nginx
vanshika_munjai@DESKTOP-107G07D:/mnt/c/Users/HP$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
ea81f68bde87   portainer/portainer-ce:latest      "/portainer"            13 days ago   Up 5 minutes  0.0.0.0:8000->8000/tcp, [:
:]8000->8000/tcp, 0.0.0.0:9443->9443/tcp, [::]:9443->9443/tcp, 9000/tcp   portainer
924e1452e147   ghcr.io/k3d-io/k3d-proxy:5.8.3     "/bin/sh -c nginx-pr…"   2 weeks ago   Up 5 minutes  80/tcp, 0.0.0.0:39691->644
3/tcp         k3d-mycluster-serverlb
0fa96e9ee63a   rancher/k3s:v1.31.5-k3s1          "/bin/k3d-entrypoint.…"   2 weeks ago   Up 5 minutes  k3d-mycluster-server-0
f41e72cb4ef4   containerization-project-1_backend  "docker-entrypoint.s…"   2 weeks ago   Up 5 minutes  0.0.0.0:3000->3000/tcp, [:
:]3000->3000/tcp   backend_api
7883abb665c2   containerization-project-1_database "docker-entrypoint.s…"   2 weeks ago   Up 5 minutes  5432/tcp        postgres_db
vanshika_munjai@DESKTOP-107G07D:/mnt/c/Users/HP$ |

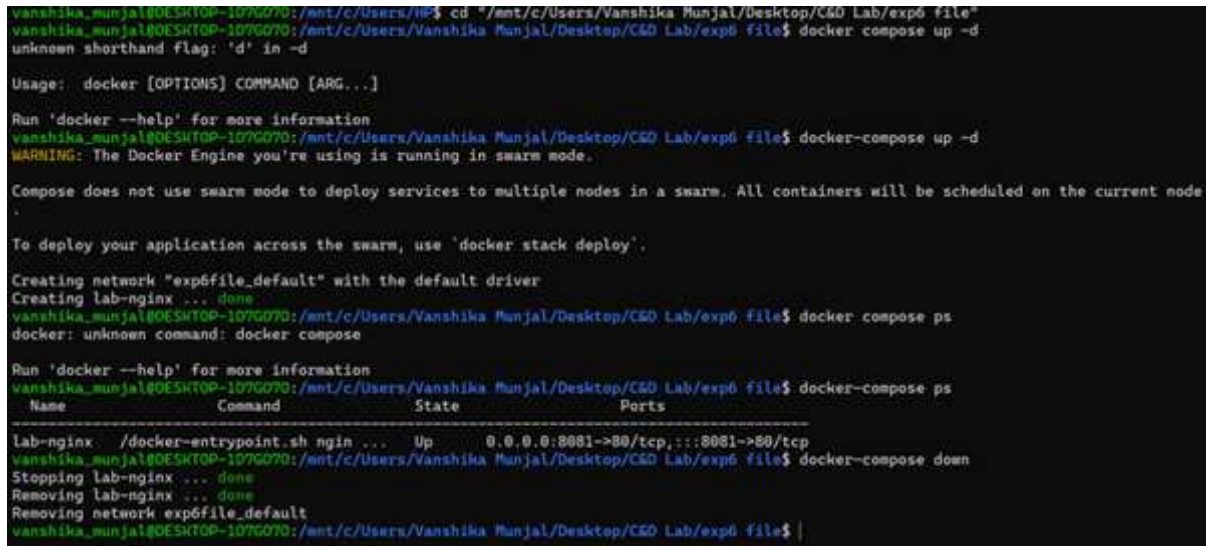
```

Step 2: Run same setup using Docker Compose



```
version: '3.8'

services:
  nginx:
    image: nginx:alpine
    container_name: lab-nginx
    ports:
      - "8081:80"
    volumes:
      - ./html:/usr/share/nginx/html
```



```
vanshika_munjal@BOLSHTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp6 file$
vanshika_munjal@BOLSHTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp6 file$ docker compose up -d
unknown shorthand flag: 'd' in -d

Usage: docker [OPTIONS] COMMAND [ARG...]

Run 'docker --help' for more information
vanshika_munjal@BOLSHTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp6 file$ docker-compose up -d
WARNING: The Docker Engine you're using is running in swarm mode.

Compose does not use swarm mode to deploy services to multiple nodes in a swarm. All containers will be scheduled on the current node.

To deploy your application across the swarm, use 'docker stack deploy'.

Creating network "exp6file_default" with the default driver
Creating lab-nginx ... done
vanshika_munjal@BOLSHTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp6 file$ docker compose ps
docker: unknown command: docker compose

Run 'docker --help' for more information
vanshika_munjal@BOLSHTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp6 file$ docker-compose ps
   Name                  Command                                State      Ports
-----
lab-nginx    /docker-entrypoint.sh nginx ...    Up         0.0.0.0:8081->80/tcp,:::8081->80/tcp
vanshika_munjal@BOLSHTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp6 file$ docker-compose down
Stopping lab-nginx ... done
Removing lab-nginx ... done
Removing network exp6file_default
vanshika_munjal@BOLSHTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp6 file$ |
```

Task 2: Multi-container Application

Objective:

Deploy WordPress with MySQL using:

1. Docker Run (manual way)
2. Docker Compose (structured way)

I. Using Docker Run

1. Create Network

```
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp6 file$ docker network create wp-net
aa03ec1108af123869353d2bcadb937f9be4efd33f0257fa98433b284c51d6fb
```

2. Run MySQL

```
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp6 file$ docker run -d \
--name mysql \
--network wp-net \
-e MYSQL_ROOT_PASSWORD=secret \
-e MYSQL_DATABASE=wordpress \
mysql:5.7
docker: Error response from daemon: Conflict. The container name "/mysql" is already in use by container "aee845296e55ea8732cb15d2b06ca96001710634a07449bed2394f1e1deb61fc". You have to remove (or rename) that container to be able to reuse that name.

Run 'docker run --help' for more information
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp6 file$ docker rm -f mysql
mysql
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp6 file$ docker run -d --name mysql --network wp-net -e MYSQL_ROOT_PASSWORD=secret -e MYSQL_DATABASE=wordpress mysql:5.7
91c80bc197e19118826021ea368ea4daaa53d8dd25bd8294a5994031b070a6cf
```

3. Run WordPress

```
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp6 file$ docker run -d \
--name wordpress \
--network wp-net \
-p 8882:88 \
-e WORDPRESS_DB_HOST=mysql \
-e WORDPRESS_DB_USER=root \
-e WORDPRESS_DB_PASSWORD=secret \
-e WORDPRESS_DB_NAME=wordpress \
wordpress:latest
2f738bd2f38a9546b77538da7e469de1597f2d7044c976747d1233c6da38abf8
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp6 file$
```

4. Test

Welcome

Welcome to the famous five-minute WordPress installation process! Just fill in the information below and you'll be on your way to using the most extendable and powerful personal publishing platform in the world.

Information needed

Please provide the following information. Do not worry, you can always change these settings later.

Site Title:

Username:

Username can have only alphanumeric characters, spaces, underscores, hyphens, periods, and the @ symbol.

Password: [Hide](#)

Strong

Important: You will need this password to log in. Please store it in a secure location.

Your Email:

Double-check your email address before continuing.

Search engine: ☐ Discourage search engines from indexing this site

II. Using Docker Compose

1. Create docker-compose.yml

```
docker-compose.yml
File Edit View

version: '3.8'

services:
  mysql:
    image: mysql:5.7
    environment:
      MYSQL_ROOT_PASSWORD: secret
      MYSQL_DATABASE: wordpress
    volumes:
      - mysql_data:/var/lib/mysql

  wordpress:
    image: wordpress:latest
    ports:
      - "8082:80"
    environment:
      WORDPRESS_DB_HOST: mysql
      WORDPRESS_DB_USER: root
      WORDPRESS_DB_PASSWORD: secret
      WORDPRESS_DB_NAME: wordpress
    depends_on:
      - mysql

volumes:
  mysql_data:
```

2. Run

```
vanshika_munjal@DESKTOP-107G090:/mnt/c/Users/Vanshika_Munjal/Desktop/C&D Lab/exp6 file$ docker-compose up -d
WARNING: The Docker Engine you're using is running in swarm mode.

Compose does not use swarm mode to deploy services to multiple nodes in a swarm. All containers will be scheduled on the current node
.

To deploy your application across the swarm, use 'docker stack deploy'.

Creating network "exp6file_default" with the default driver
Creating volume "exp6file_mysql_data" with default driver
Creating exp6file_mysql_1 ... done
Creating exp6file_wordpress_1 ... done
```

localhost:8082/wp-admin/install.php?step=1

Welcome

Welcome to the famous five-minute WordPress installation process! Just fill in the information below and you'll be on your way to using the most extendable and powerful personal publishing platform in the world.

Information needed

Please provide the following information. Do not worry, you can always change these settings later.

Site Title

Username

Username can have only alphanumeric characters, spaces, underscores, hyphens, periods, and the @ symbol.

Password

Strong

Important: You will need this password to log in. Please store it in a secure location.

Your Email

Double-check your email address before continuing.

Search engine ☐ Discourage search engines from indexing this site

1. Stop

```

vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp6 file$ docker-compose down -v
Stopping exp6file_wordpress_1 ... done
Stopping exp6file_mysql_1 ... done
Removing exp6file_wordpress_1 ... done
Removing exp6file_mysql_1 ... done
Removing network exp6file_default
Removing volume exp6file_mysql_data
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp6 file$

```

Part C: Conversion and Build-based Tasks

Task 3: Convert Docker Run to Docker Compose

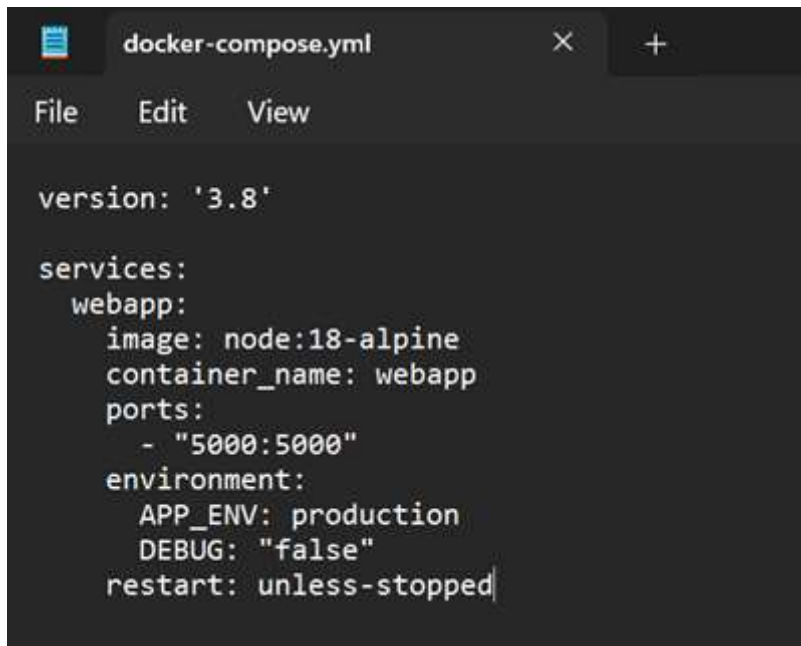
Problem 1: Basic Web Application

Given Docker Run Command:

```

docker run -d \
  --name webapp \
  -p 5000:5000 \
  -e APP_ENV=production \
  -e DEBUG=false \
  --restart unless-stopped \
  node:18-alpine

```



```

version: '3.8'

services:
  webapp:
    image: node:18-alpine
    container_name: webapp
    ports:
      - "5000:5000"
    environment:
      APP_ENV: production
      DEBUG: "false"
    restart: unless-stopped

```

```

vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp6 file$ docker-compose up -d
WARNING: The Docker Engine you're using is running in swarm mode.

Compose does not use swarm mode to deploy services to multiple nodes in a swarm. All containers will be scheduled on the current node
.

To deploy your application across the swarm, use 'docker stack deploy'.

Creating network "exp6file_default" with the default driver
Creating webapp ... done
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp6 file$ docker-compose ps

```

Name	Command	State	Ports
webapp	docker-entrypoint.sh node	Restarting	

Problem 2: Volume + Network Configuration

Given Docker Run Commands:

```
docker network create app-net
```

```
docker run -d \
  --name postgres-db \
  --network app-net \
  -e POSTGRES_USER=admin \
  -e POSTGRES_PASSWORD=secret \
  -v pgdata:/var/lib/postgresql/data \
  postgres:15
```

```
docker run -d \
  --name backend \
  --network app-net \
  -p 8000:8000 \
  -e DB_HOST=postgres-db \
  -e DB_USER=admin \
  -e DB_PASS=secret \
  python:3.11-slim
```

```
docker-compose.yml
File Edit View

version: '3.8'

services:
  postgres-db:
    image: postgres:15
    container_name: postgres-db
    environment:
      POSTGRES_USER: admin
      POSTGRES_PASSWORD: secret
    volumes:
      - pgdata:/var/lib/postgresql/data
    networks:
      - app-net

  backend:
    image: python:3.11-slim
    container_name: backend
    ports:
      - "8001:8000"
    environment:
      DB_HOST: postgres-db
      DB_USER: admin
      DB_PASS: secret
    depends_on:
      - postgres-db
    networks:
      - app-net

volumes:
  pgdata:

networks:
  app-net:
```

```

vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp6 file$ docker-compose up -d
WARNING: The Docker Engine you're using is running in swarm mode.

Compose does not use swarm mode to deploy services to multiple nodes in a swarm. All containers will be scheduled on the current node.

To deploy your application across the swarm, use 'docker stack deploy'.

Creating network "exp6file_app-net" with the default driver
Creating volume "exp6file_pgdata" with default driver
Creating postgres-db ... done
Creating backend ... done
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp6 file$ docker-compose ps

```

Name	Command	State	Ports
backend	python3	Exit 0	
postgres-db	docker-entrypoint.sh postgres	Up	5432/tcp

```

vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp6 file$ docker-compose down
Stopping postgres-db ... done
Removing backend ... done
Removing postgres-db ... done
Removing network exp6file_app-net
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp6 file$ docker-compose down -v --remove-orphans
Removing network exp6file_app-net
WARNING: Network exp6file_app-net not found.
Removing volume exp6file_pgdata
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp6 file$

```

Task 4: Resource Limits Conversion

Given Docker Run Command:

```

docker run -d \
  --name limited-app \
  -p 9000:9000 \
  --memory="256m" \
  --cpus="0.5" \
  --restart always \
  nginx:alpine

```

```

docker-compose.yml
File Edit View

version: '3.8'

services:
  limited-app:
    image: nginx:alpine
    container_name: limited-app
    ports:
      - "9000:9000"
    restart: always

    deploy:
      resources:
        limits:
          memory: 256m
          cpus: "0.5"

```

```

vanshika_munjal@DESKTOP-1D7G07D:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp6 file$ docker-compose up -d
WARNING: The Docker Engine you're using is running in swarm mode.

Compose does not use swarm mode to deploy services to multiple nodes in a swarm. All containers will be scheduled on the current node
.

To deploy your application across the swarm, use 'docker stack deploy'.

Creating limited-app ... done
vanshika_munjal@DESKTOP-1D7G07D:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp6 file$ docker-compose ps
   Name                       Command                                State      Ports
-----
limited-app /docker-entrypoint.sh nginx ... Up        80/tcp, 0.0.0.0:9000->9000/tcp, :::9000->9000/tcp
vanshika_munjal@DESKTOP-1D7G07D:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp6 file$ docker-compose down
Stopping limited-app ... done
Removing limited-app ... done
Removing network exp6file_default
vanshika_munjal@DESKTOP-1D7G07D:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp6 file$ |

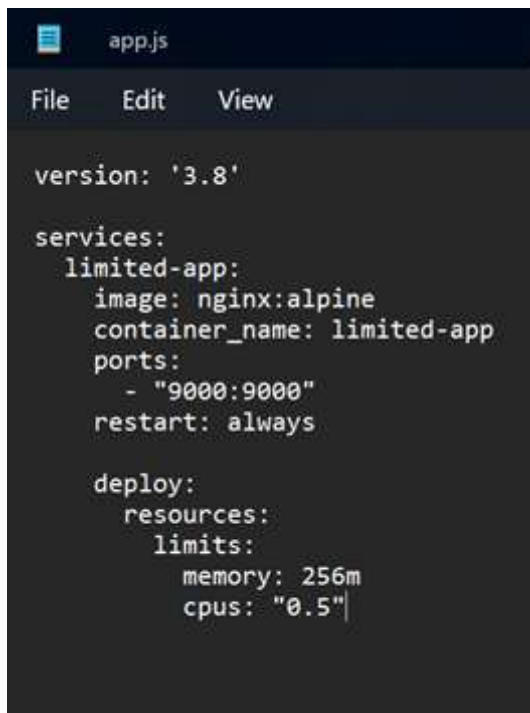
```

Note:

When deploy works: The deploy section works only in Docker Swarm mode and is ignored in normal docker-compose. Difference between Compose and Swarm: Docker Compose is used for local development, while Docker Swarm is used for production and supports features like resource limits and scaling.

Part D: Using Dockerfile instead of Standard Image

Task 5: Replace Standard Image with Dockerfile (Node App)



```

app.js
File Edit View

version: '3.8'

services:
  limited-app:
    image: nginx:alpine
    container_name: limited-app
    ports:
      - "9000:9000"
    restart: always

  deploy:
    resources:
      limits:
        memory: 256m
        cpus: "0.5"

```



```
Dockerfile
File Edit View

FROM node:18-alpine

WORKDIR /app

COPY app.js .

EXPOSE 3000

CMD ["node", "app.js"]
```

```
docker-compose.yml
File Edit View

version: '3.8'

services:
  nodeapp:
    build:
      context: .
      dockerfile: Dockerfile
    container_name: custom-node-app
    ports:
      - "3000:3000"zS
```

1. Build and Run

```
vanshika_munjal@DESKTOP-107G0790:/mnt/c/Users/Vanshika_Munjal/Desktop/C&D Lab/exp6_file$ docker-compose up --build -d
WARNING: The Docker Engine you're using is running in swarm mode.

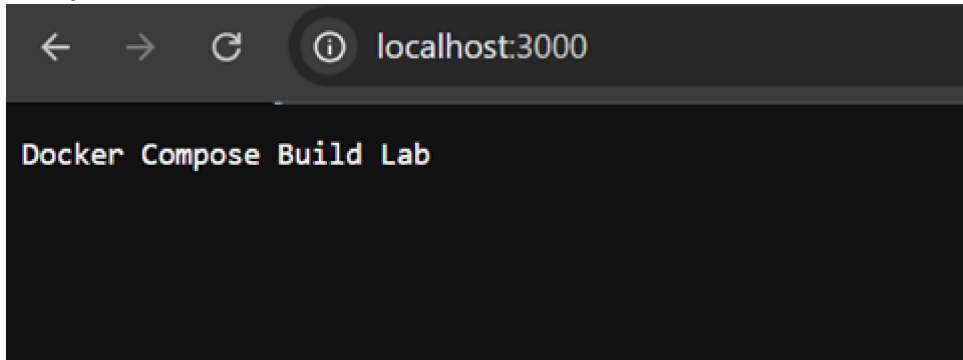
Compose does not use swarm mode to deploy services to multiple nodes in a swarm. All containers will be scheduled on the current node
.

To deploy your application across the swarm, use 'docker stack deploy'.

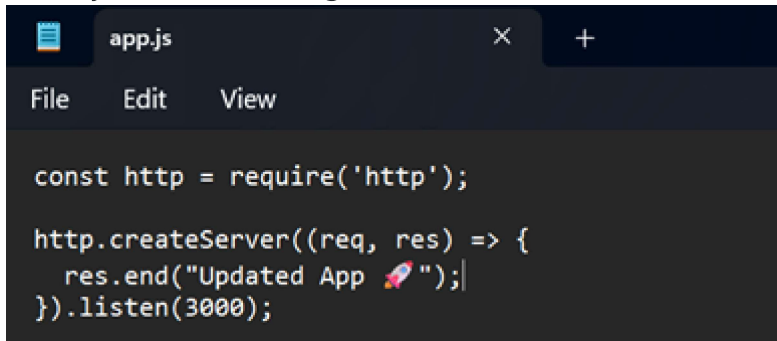
Building nodeapp
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon 4.608kB
Step 1/5 : FROM node:18-alpine
--> ee77c6cd7c18
Step 2/5 : WORKDIR /app
--> Using cache
--> 5cb75b734fd6
Step 3/5 : COPY app.js .
--> Using cache
--> a28a899a822f
Step 4/5 : EXPOSE 3000
--> Using cache
--> 9b81c53f6887
Step 5/5 : CMD ["node", "app.js"]
--> Using cache
--> 0627328a367d
Successfully built 0627328a367d
Successfully tagged exp6file_nodeapp:latest
Starting custom-node-app ... done
```

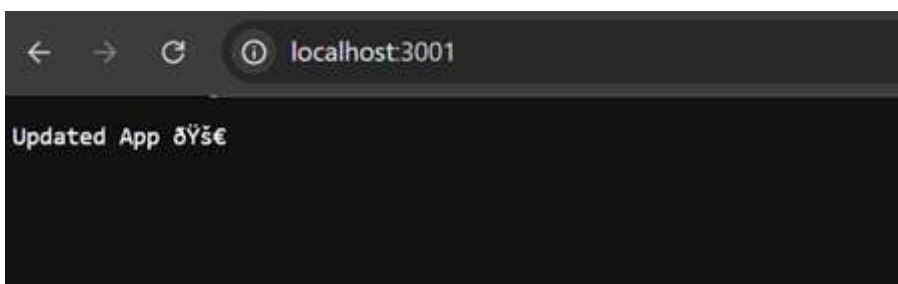
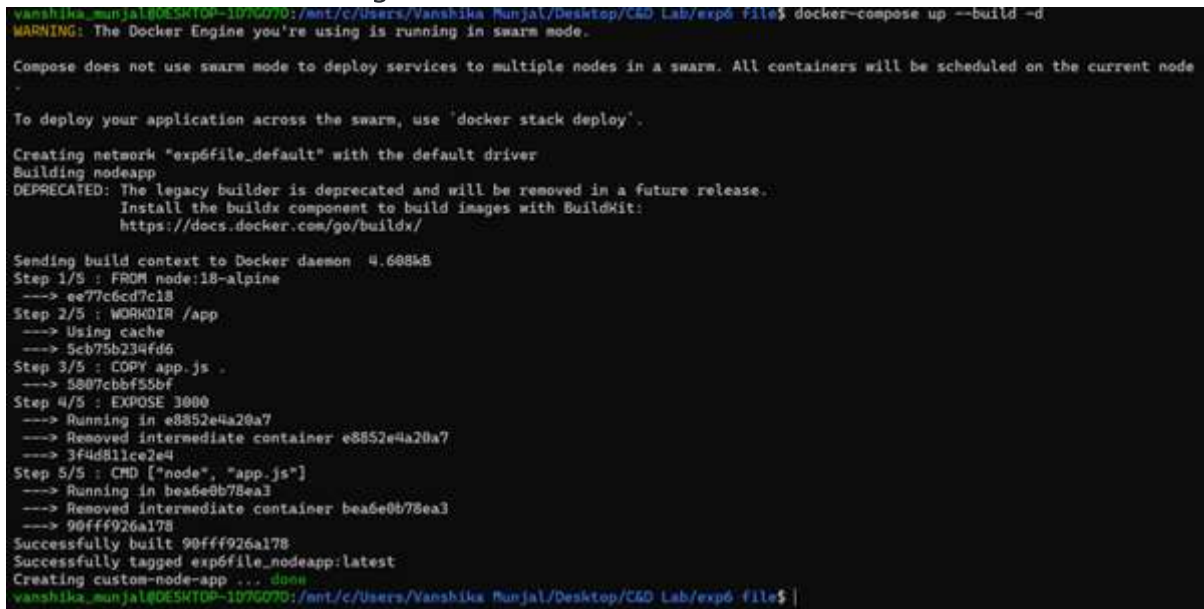
2. Verify in Browser



3. Modify app.js message



4. Rebuild and observe changes



1. Difference between Image and Build

Image:

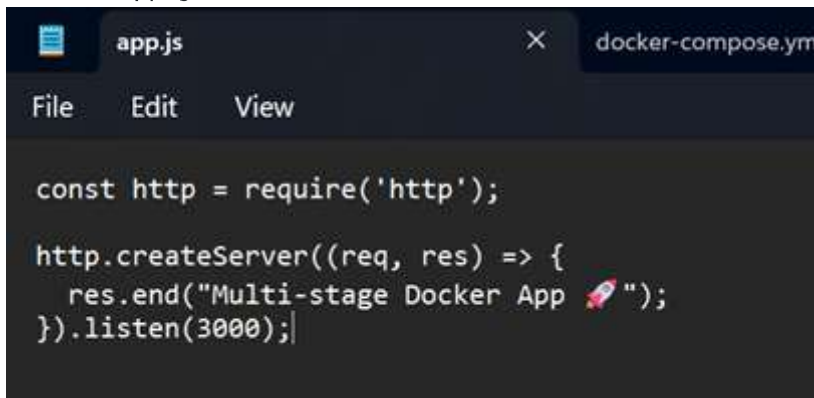
- Used to pull a pre-built image from Docker Hub (or registry)
- No need for Dockerfile
- Faster to run

Build:

- Used to create a custom image using a Dockerfile
- Requires source code + Dockerfile
- Allows customization

Task 6: Multistage Dockerfile with Compose

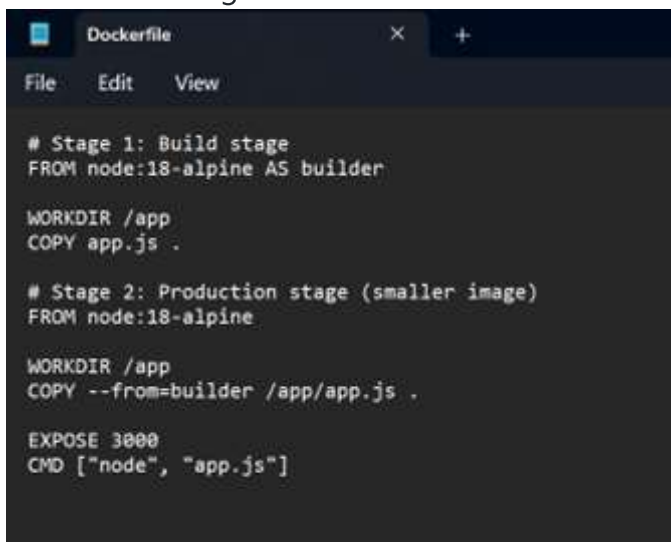
1. Create app.js

A screenshot of a code editor with two tabs: 'app.js' and 'docker-compose.yml'. The 'app.js' tab is active, showing a JavaScript file with the following code:

```
const http = require('http');

http.createServer((req, res) => {
  res.end("Multi-stage Docker App 🚀");
}).listen(3000);
```

2. Create multistage Dockerfile

A screenshot of a code editor with two tabs: 'Dockerfile' and '+'. The 'Dockerfile' tab is active, showing a multistage Dockerfile with the following content:

```
# Stage 1: Build stage
FROM node:18-alpine AS builder

WORKDIR /app
COPY app.js .

# Stage 2: Production stage (smaller image)
FROM node:18-alpine

WORKDIR /app
COPY --from=builder /app/app.js .

EXPOSE 3000
CMD ["node", "app.js"]
```

3. Create docker-compose.yml

```

docker-compose.yml
File Edit View

version: '3.8'

services:
  app:
    build: .
    container_name: multi-stage-app
    ports:
      - "3002:3000"

    environment:
      NODE_ENV: production

    volumes:
      - ./app

```

4. Build and run

```

vanshika_munjal@DESKTOP-1074070:/mnt/c/Users/Vanshika_Munjal/Desktop/C&D Lab/exp6 file$ docker-compose up --build -d
WARNING: The Docker Engine you're using is running in swarm mode.

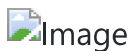
Compose does not use swarm mode to deploy services to multiple nodes in a swarm. All containers will be scheduled on the current node.

To deploy your application across the swarm, use 'docker stack deploy'.

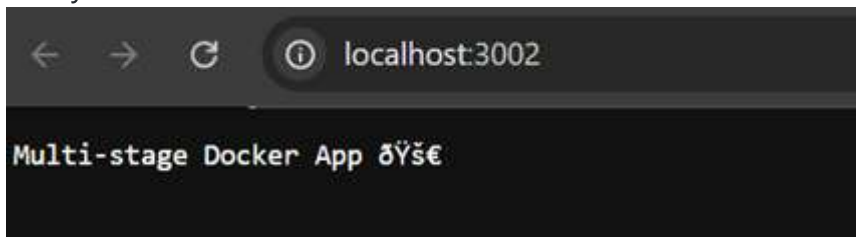
WARNING: Found orphan containers (custom-node-app) for this project. If you removed or renamed this service in your compose file, you
can run this command with the --remove-orphans flag to clean it up.
Building app
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  4.608kB
Step 1/8 : FROM node:18-alpine AS builder
--> ee77c6cd7c18
Step 2/8 : WORKDIR /app
--> Using cache
--> 5cb75b234fd6
Step 3/8 : COPY app.js .
--> 0dd4c17f0e23
Step 4/8 : FROM node:18-alpine
--> ee77c6cd7c18
Step 5/8 : WORKDIR /app
--> Using cache
--> 5cb75b234fd6
Step 6/8 : COPY --from=builder /app/app.js .
--> 23ea5f7df8d9
Step 7/8 : EXPOSE 3000
--> Running in 7e75dee97f91
--> Removed intermediate container 7e75dee97f91
--> c9880d390119
Step 8/8 : CMD ["node", "app.js"]
--> Running in 01eff98af809

```



1. Verify in browser



2. Compare image size

```
vanshika_munjal@DESKTOP-107G070: /mnt/c/Users/Vanshika_Munjal/Desktop/C&D Lab/exp6 file$ docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
exp6file_app        latest             4072755f57b6       About a minute ago 127MB
<none>              <none>            0dd4c17f0e23       About a minute ago 127MB
exp6file_nodeapp    latest             90fff926a178       13 minutes ago     127MB
```

The multi-stage Docker build was implemented successfully. However, the final image size is approximately the same as the base image (127MB). This is because the application is very simple and does not include additional dependencies or build tools.

In real-world applications, multi-stage builds significantly reduce image size by removing unnecessary build dependencies and keeping only the runtime environment.

Experiment 6B: Multi-container Application using Docker Compose (WordPress + Database)

Objective

To deploy a multi-container application using Docker Compose, consisting of:

- WordPress (frontend + PHP)
- MySQL database (backend)

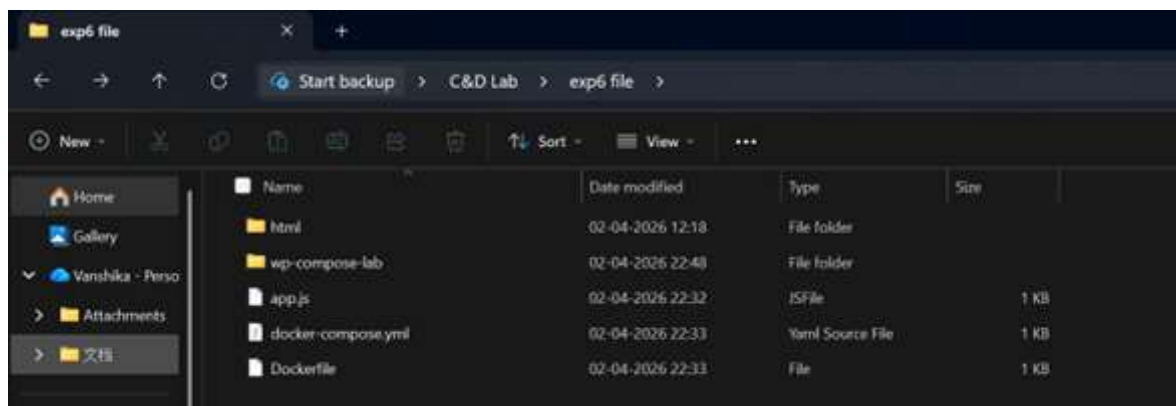
Also:

- Understand container networking & volumes
- Learn how to scale services
- Compare with Docker Swarm for production deployment

Steps

1. Create project directory

```
vanshika_munjal@DESKTOP-107G070: /mnt/c/Users/Vanshika_Munjal/Desktop/C&D Lab/exp6 file$ mkdir wp-compose-lab
vanshika_munjal@DESKTOP-107G070: /mnt/c/Users/Vanshika_Munjal/Desktop/C&D Lab/exp6 file$ cd wp-compose-lab
```

1. Create docker-compose.yml

```

version: '3.9'

services:
  db:
    image: mysql:5.7
    container_name: wordpress_db
    restart: always
    environment:
      MYSQL_ROOT_PASSWORD: rootpass
      MYSQL_DATABASE: wordpress
      MYSQL_USER: wpuser
      MYSQL_PASSWORD: wppass
    volumes:
      - db_data:/var/lib/mysql

  wordpress:
    image: wordpress:latest
    container_name: wordpress_app
    depends_on:
      - db
    ports:
      - "8080:80"
    restart: always
    environment:
      WORDPRESS_DB_HOST: db:3306
      WORDPRESS_DB_USER: wpuser
      WORDPRESS_DB_PASSWORD: wppass
      WORDPRESS_DB_NAME: wordpress
    volumes:
      - wp_data:/var/www/html

volumes:
  db_data:
  wp_data:
  
```

2. Start Application

```

vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp6 file/wp-compose-lab$ docker-compose up -d
WARNING: The Docker Engine you're using is running in swarm mode.

Compose does not use swarm mode to deploy services to multiple nodes in a swarm. All containers will be scheduled on the current node.

To deploy your application across the swarm, use 'docker stack deploy'.

Creating network "wp-compose-lab_default" with the default driver
Creating volume "wp-compose-lab_db_data" with default driver
Creating volume "wp-compose-lab_wp_data" with default driver
Creating wordpress_db ... done
Creating wordpress_app ... done
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp6 file/wp-compose-lab$
  
```

3. Verify in browser



4. Verify containers

```
vanshika_munj665@SKTOP-107G070: /mnt/c/Users/Vanshika_Munj665/Desktop/C&D Lab/exp6 file/wp-compose-lab$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
dc1db51e1c3c	wordpress:latest	"docker-entrypoint.s..."	3 minutes ago	Up 3 minutes	0.0.0.0:8080->80/tcp,
[::]:8080->80/tcp					
0b467bf7372d	mysql:5.7	"docker-entrypoint.s..."	3 minutes ago	Up 3 minutes	3306/tcp, 33060/tcp
480d688cb751	exp6file_app	"docker-entrypoint.s..."	21 minutes ago	Up 21 minutes	0.0.0.0:3002->3000/tcp
[::]:3002->3000/tcp					
fff2347b77b2	exp6file_nodeapp	"docker-entrypoint.s..."	34 minutes ago	Up 34 minutes	0.0.0.0:3001->3000/tcp
[::]:3001->3000/tcp					
91c80bc197e1	mysql:5.7	"docker-entrypoint.s..."	10 hours ago	Up 10 hours	3306/tcp, 33060/tcp
ea81f68bde87	portainer/portainer-ce:latest	"/portainer"	13 days ago	Up 11 hours	0.0.0.0:8000->8000/tcp
[::]:8000->8000/tcp, 0.0.0.0:9443->9443/tcp, [::]:9443->9443/tcp, 9000/tcp					
924e1452e147	ghcr.io/k3d-io/k3d-proxy:5.8.3	"/bin/sh -c nginx-pr..."	2 weeks ago	Up 11 hours	80/tcp, 0.0.0.0:39691-
>9443/tcp					
0fa96e9ee63a	rancher/k3s:v1.31.5-k3s1	"/bin/k3d-entrypoint..."	2 weeks ago	Up 11 hours	k3d-mycluster-serverlb
7883abb665c2	containerization-project-1_database	"docker-entrypoint.s..."	2 weeks ago	Up 11 hours	k3d-mycluster-server-0
					postgres db

5. Check volumes

```
vanshika_munj665@SKTOP-107G070: /mnt/c/Users/Vanshika_Munj665/Desktop/C&D Lab/exp6 file/wp-compose-lab$ docker volume ls
```

DRIVER	VOLUME NAME
local	1f8be83fedb2b0464a949686b4e5fa32b9cdc509aee793c30d9300eab7a1e62
local	5d2e578b14128a809f370a10f802e14d67d6d660d7789719064c2af5b96f5d95
local	7ca4b803667911817f228440136bab6419fdc5750f7729c025fb39d4d9b52b57
local	11a9fe249bdf6a609686d75709d056a86bd9fc309463dbc643ba35089b40176
local	14cb18358c4dcfa09657cf0a68447f708a9da4ae8089507f406ca1b1c26983b8
local	25feb-class_mysql_data
local	25feb-class_wp_content
local	26feb-class_mysql_data
local	26feb-class_wp_content
local	49af95d171db743ae8a15940af4e2a4553d97f7fac2e37cf5c94d217ffbf2a2
local	63d456d51a0cb6f49071702398658f92c22f819641ac0da5333ec291a5e28ee7
local	ae41b0e6b4e33ef5938627e214c938a307600f75d1a47bf268e3a41a412bd0d6
local	afc2a5a7de2eaf163fe9e26d5caff53c4ffc2af38a82985b728294c715a5254
local	app-volume
local	b01530fb0ae69c9eb0a1a8f009957d3c3d1be33564abe8ef7776e336975f1cf9
local	containerization-project-1_postgres_data
local	de8ff57a54f4941d6fa842eb2b9a9dbf43fbf73818b10051d12ac75d9d465181
local	e227b701b7b90d3c1f9d8eb905b50249d99fdec65cfb6755d72f9f7c1629e8bf
local	ec5ad73b9dcdbd2565006a3aeb3326e0c2dab77ce4a37f92c565cab363d2d9a97
local	fb058b5137c74e4a6cab010bc33b66d7f5502847ecf0cc48e6732d3bb305f64c
local	k3d-mycluster-images
local	mydata
local	mysql-data
local	myvolume
local	pgdata
local	portainer_data
local	postgres-data
local	redis-data
local	wp-compose-lab_db_data
local	wp-compose-lab_wp_data

6. Stop Application

```
vanshika_munj665@SKTOP-107G070: /mnt/c/Users/Vanshika_Munj665/Desktop/C&D Lab/exp6 file/wp-compose-lab$ docker-compose down
Stopping wordpress_app ... done
Stopping wordpress_db ... done
Removing wordpress_app ... done
Removing wordpress_db ... done
Removing network wp-compose-lab_default
vanshika_munj665@SKTOP-107G070: /mnt/c/Users/Vanshika_Munj665/Desktop/C&D Lab/exp6 file/wp-compose-lab$
```

Scaling in Docker Compose

```

vanshika_munjial@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp6 file/wp-compose-lab$ docker-compose up --scale word
press=3
WARNING: The Docker Engine you're using is running in swarm mode.

Compose does not use swarm mode to deploy services to multiple nodes in a swarm. All containers will be scheduled on the current node
.

To deploy your application across the swarm, use 'docker stack deploy'.

Creating network wp-compose-lab_default with the default driver
Creating wordpress_db ... done
Creating wp-compose-lab_wordpress_1 ... done
Creating wp-compose-lab_wordpress_2 ... done
Creating wp-compose-lab_wordpress_3 ... done
Attaching to wordpress_db, wp-compose-lab_wordpress_2, wp-compose-lab_wordpress_3, wp-compose-lab_wordpress_1
wordpress_db | 2026-04-02 17:32:13+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 5.7.44-1.el7 started.
wordpress_db | 2026-04-02 17:32:13+00:00 [Note] [Entrypoint]: Switching to dedicated user 'mysql'
wordpress_db | 2026-04-02 17:32:13+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 5.7.44-1.el7 started.
wordpress_2 | AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 172.27.0.3. Set the 'Se
rverName' directive globally to suppress this message
wordpress_2 | AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 172.27.0.3. Set the 'Se
rverName' directive globally to suppress this message
wordpress_db | '/var/lib/mysql/mysql.sock' -> '/var/run/mysqld/mysqld.sock'
wordpress_2 | [Thu Apr 02 17:32:13.587926 2026] [mpm_prefork:notice] [pid 1:tid 1] AH00163: Apache/2.4.66 (Debian) PHP/8.3.38 config

```

```

vanshika_munjial@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp6 file$ docker ps
CONTAINER ID   IMAGE                                COMMAND                                CREATED        STATUS        PORTS
25a799c4fd6b   wordpress:latest                    "docker-entrypoint.s..."            About a minute ago Up About a minute 80/tcp
e07671139e7e   wordpress:latest                    "docker-entrypoint.s..."            About a minute ago Up About a minute 80/tcp
e0b64a9cfeae   wordpress:latest                    "docker-entrypoint.s..."            About a minute ago Up About a minute 80/tcp
c4c812ed407a   mysql:5.7                           "docker-entrypoint.s..."            About a minute ago Up About a minute 3306/tcp, 3306
0/tcp
480d688cb751   exp6file_app                        "docker-entrypoint.s..."            29 minutes ago Up 29 minutes 0.0.0.0:3002->
3000/tcp, [::]:3002->3000/tcp
fff2347b77b2   exp6file_nodeapp                    "docker-entrypoint.s..."            42 minutes ago Up 42 minutes 0.0.0.0:3001->
3000/tcp, [::]:3001->3000/tcp
91c80bc197e1   mysql:5.7                           "docker-entrypoint.s..."            11 hours ago Up 11 hours 3306/tcp, 3306
0/tcp
ea81f68bde87   portainer/portainer-ce:latest       "/portainer"                          13 days ago Up 11 hours 0.0.0.0:8000->
8000/tcp, [::]:8000->8000/tcp, 0.0.0.0:9443->9443/tcp, [::]:9443->9443/tcp
924e1452e147   ghcr.io/k3d-io/k3d-proxy:5.8.3      "/bin/sh -c nginx-pr..."            2 weeks ago Up 11 hours 80/tcp, 0.0.0.
0:39691->6443/tcp
0fa96e9ee63a   rancher/k3s:v1.31.5-k3s1            "/bin/k3d-entrypoint..."            2 weeks ago Up 11 hours k3d-mycluster-serverlb
k3d-mycluster-server-0
7883abb665c2   containerization-project-1_database "docker-entrypoint.s..."            2 weeks ago Up 11 hours 5432/tcp
postgres_db
vanshika_munjial@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp6 file$

```

vanshikamunjal665

CONTAINERIZATION AND DEVOPS LAB

Experiment 7: [🔗](#)

CI/CD using Jenkins, GitHub and Docker Hub

Reference Link:

<https://upessocs.github.io/#dir=/Lectures/Containerization%20and%20DevOps/Lab/&file=Experiment%2007%20CI/CD%20using%20Jenkins,%20GitHub%20and%20Docker%20Hub.md>

1. Aim

To design and implement a complete CI/CD pipeline using Jenkins, integrating source code from GitHub, and building & pushing Docker images to Docker Hub.

2. Objectives

- Understand CI/CD workflow using Jenkins (GUI-based tool)
- Create a structured GitHub repository with application + Jenkinsfile
- Build Docker images from source code
- Securely store Docker Hub credentials in Jenkins
- Automate build & push process using webhook triggers
- Use same host (Docker) as Jenkins agent

3. Theory

What is Jenkins?

Jenkins is a web-based GUI automation server used to:

- Build applications
- Test code
- Deploy software

It provides:

- Dashboard (browser-based UI)
- Plugin ecosystem (GitHub, Docker, etc.)
- Pipeline as Code using `Jenkinsfile`

What is CI/CD?

- **Continuous Integration (CI):**
Code is automatically built and tested after each commit
- **Continuous Deployment (CD):**
Built artifacts (Docker images) are automatically delivered/deployed

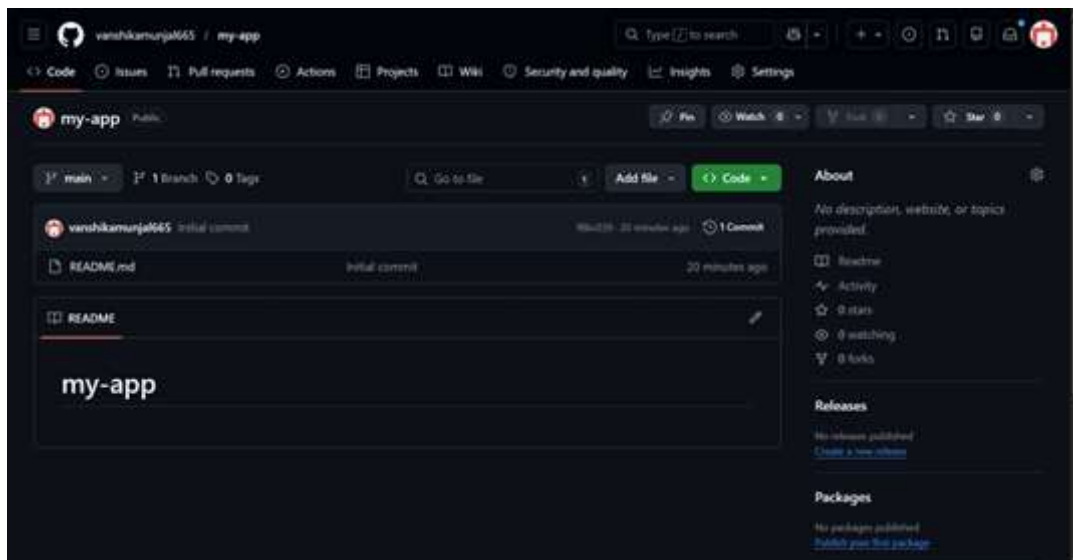
4. Prerequisites

- Docker & Docker Compose installed
- GitHub account
- Docker Hub account
- Basic Linux command knowledge

Part A: GitHub Repository Setup (Source Code + Build Definition)

1. Create Repository

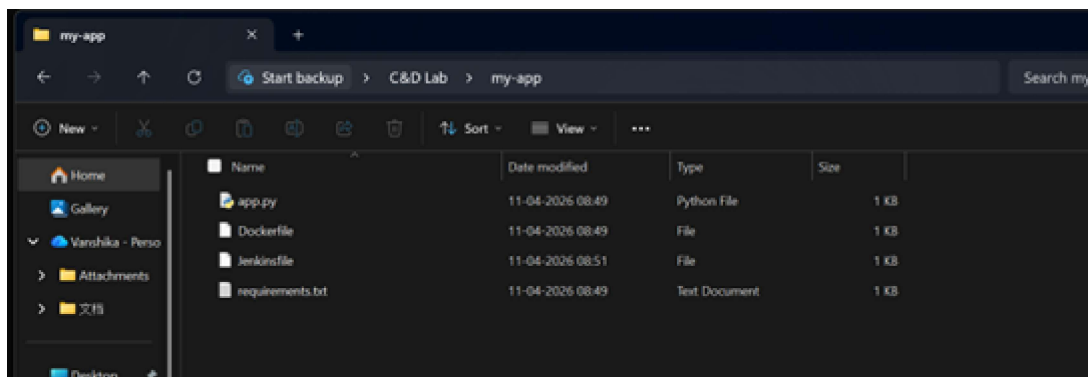
A GitHub repository named **my-app** was created to store the project files.



2. Project Structure

The repository contains the following files:

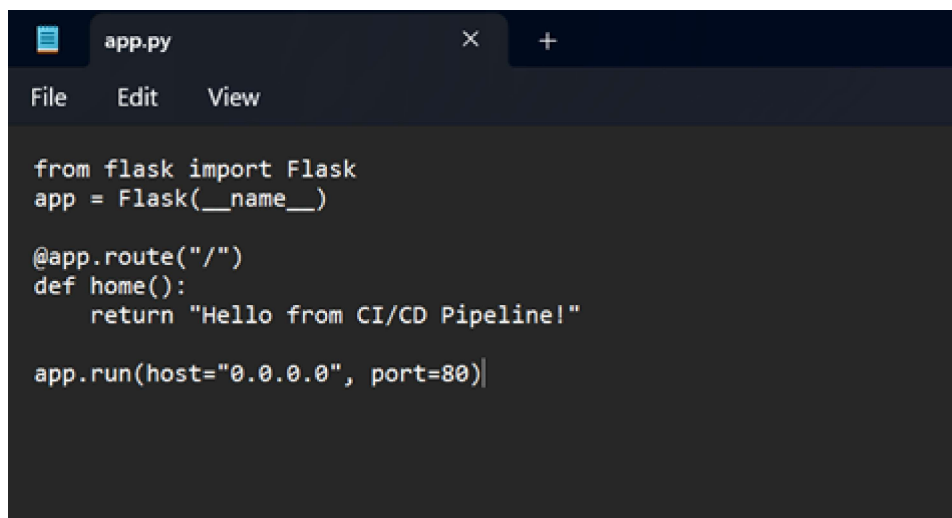
- Dockerfile
- Jenkinsfile
- app.py
- requirements.txt
- README.md



```
PS C:\Users\HP> wsl
vanshika_munjal@DESKTOP-107G070: /mnt/c/Users/HP$ cd "/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab"
vanshika_munjal@DESKTOP-107G070: /mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab$ mkdir my-app
vanshika_munjal@DESKTOP-107G070: /mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab$ cd my-app
vanshika_munjal@DESKTOP-107G070: /mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/my-app$ mv Dockerfile.txt Dockerfile
vanshika_munjal@DESKTOP-107G070: /mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/my-app$ mv Jenkinsfile.txt Jenkinsfile
vanshika_munjal@DESKTOP-107G070: /mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/my-app$
```

3. Application Code

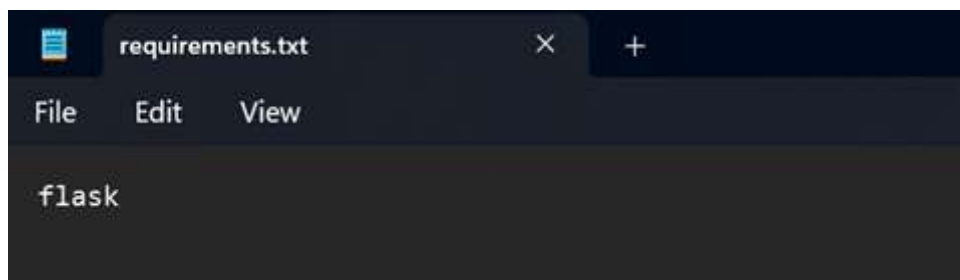
A simple Python application was created using Flask framework.

A screenshot of a code editor window with a dark theme. The title bar shows a file icon, the name 'app.py', and buttons for 'x' and '+'. The menu bar has 'File', 'Edit', and 'View'. The code is a Flask application:

```
from flask import Flask
app = Flask(__name__)

@app.route("/")
def home():
    return "Hello from CI/CD Pipeline!"

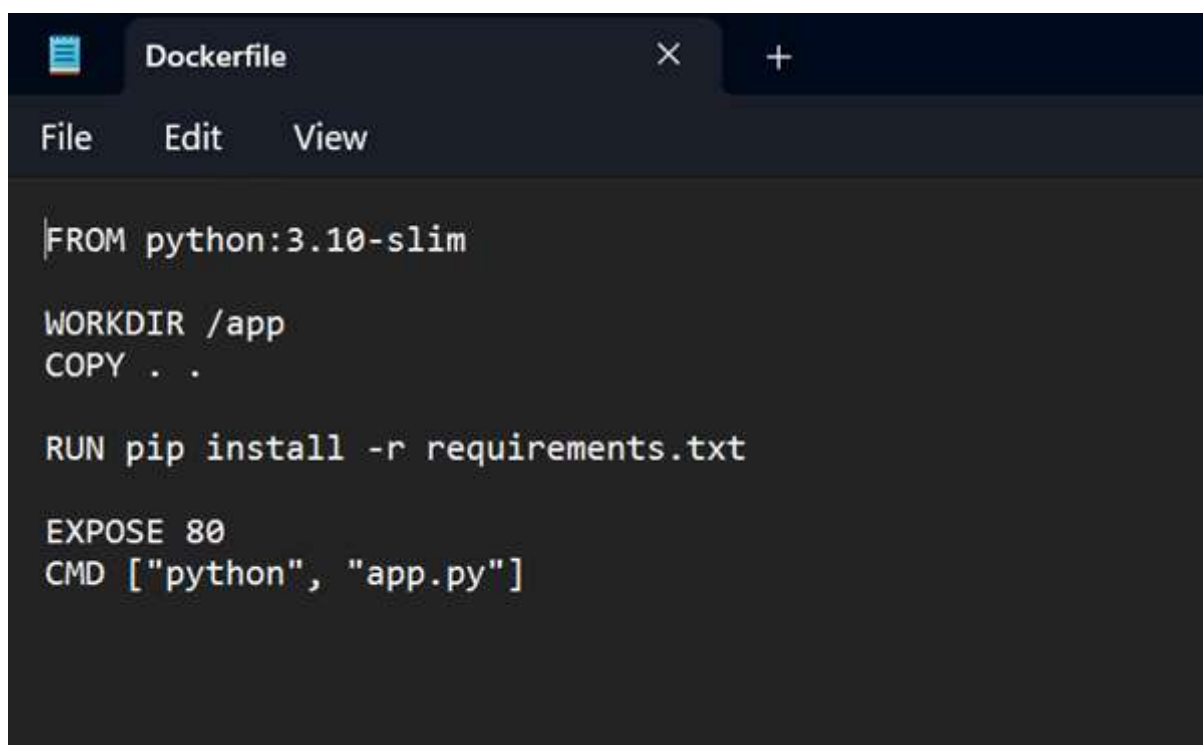
app.run(host="0.0.0.0", port=80)
```

A screenshot of a code editor window with a dark theme. The title bar shows a file icon, the name 'requirements.txt', and buttons for 'x' and '+'. The menu bar has 'File', 'Edit', and 'View'. The code is a single line:

```
flask
```

4. Dockerfile (Build Process)

A Dockerfile was created to containerize the application and define the build steps.

A screenshot of a code editor window with a dark theme. The title bar shows a file icon, the name 'Dockerfile', and buttons for 'x' and '+'. The menu bar has 'File', 'Edit', and 'View'. The code defines the Docker build process:

```
FROM python:3.10-slim

WORKDIR /app
COPY . .

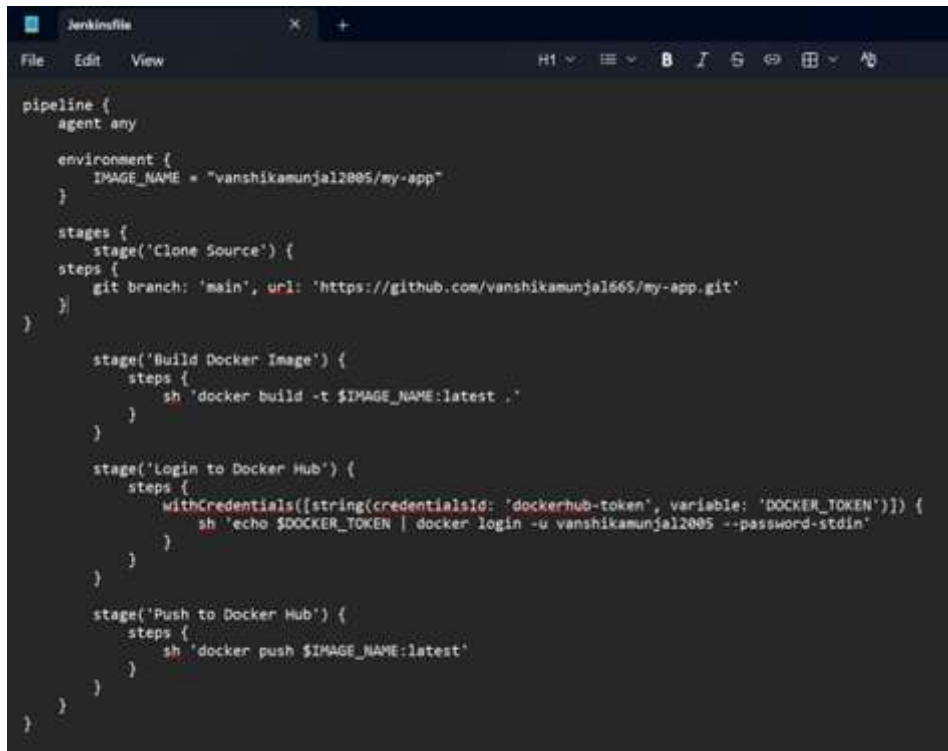
RUN pip install -r requirements.txt

EXPOSE 80
CMD ["python", "app.py"]
```

5. Jenkinsfile (Pipeline Definition in GitHub)

A Jenkinsfile was added to define the CI/CD pipeline including:

- Cloning repository
- Building Docker image
- Logging into Docker Hub
- Pushing Docker image



```
pipeline {
  agent any

  environment {
    IMAGE_NAME = "vanshikamunjal2005/my-app"
  }

  stages {
    stage('Clone Source') {
      steps {
        git branch: 'main', url: 'https://github.com/vanshikamunjal665/my-app.git'
      }
    }

    stage('Build Docker Image') {
      steps {
        sh 'docker build -t $IMAGE_NAME:latest .'
      }
    }

    stage('Login to Docker Hub') {
      steps {
        withCredentials([string(credentialsId: 'dockerhub-token', variable: 'DOCKER_TOKEN')]) {
          sh 'echo $DOCKER_TOKEN | docker login -u vanshikamunjal2005 --password-stdin'
        }
      }
    }

    stage('Push to Docker Hub') {
      steps {
        sh 'docker push $IMAGE_NAME:latest'
      }
    }
  }
}
```

Part B: Jenkins Setup using Docker (Persistent Configuration)

1. Create Docker Compose file

A Docker Compose file was used to run Jenkins container with persistent storage.

```
docker-compose.yml
File Edit View

version: '3.8'

services:
  jenkins:
    image: jenkins/jenkins:lts
    container_name: jenkins
    restart: always
    ports:
      - "8080:8080"
      - "50000:50000"
    volumes:
      - jenkins_home:/var/jenkins_home
      - /var/run/docker.sock:/var/run/docker.sock
    user: root

volumes:
  jenkins_home:
```

2. Run Jenkins

Jenkins was started using Docker and accessed via:
<http://localhost:8080>

```
vanshika_munjal@DESKTOP-1076070:/mnt/c/Users/vanshika_munjal/Desktop/L&D Lab/my-app$ docker-compose up -d
WARNING: The Docker Engine you're using is running in swarm mode.

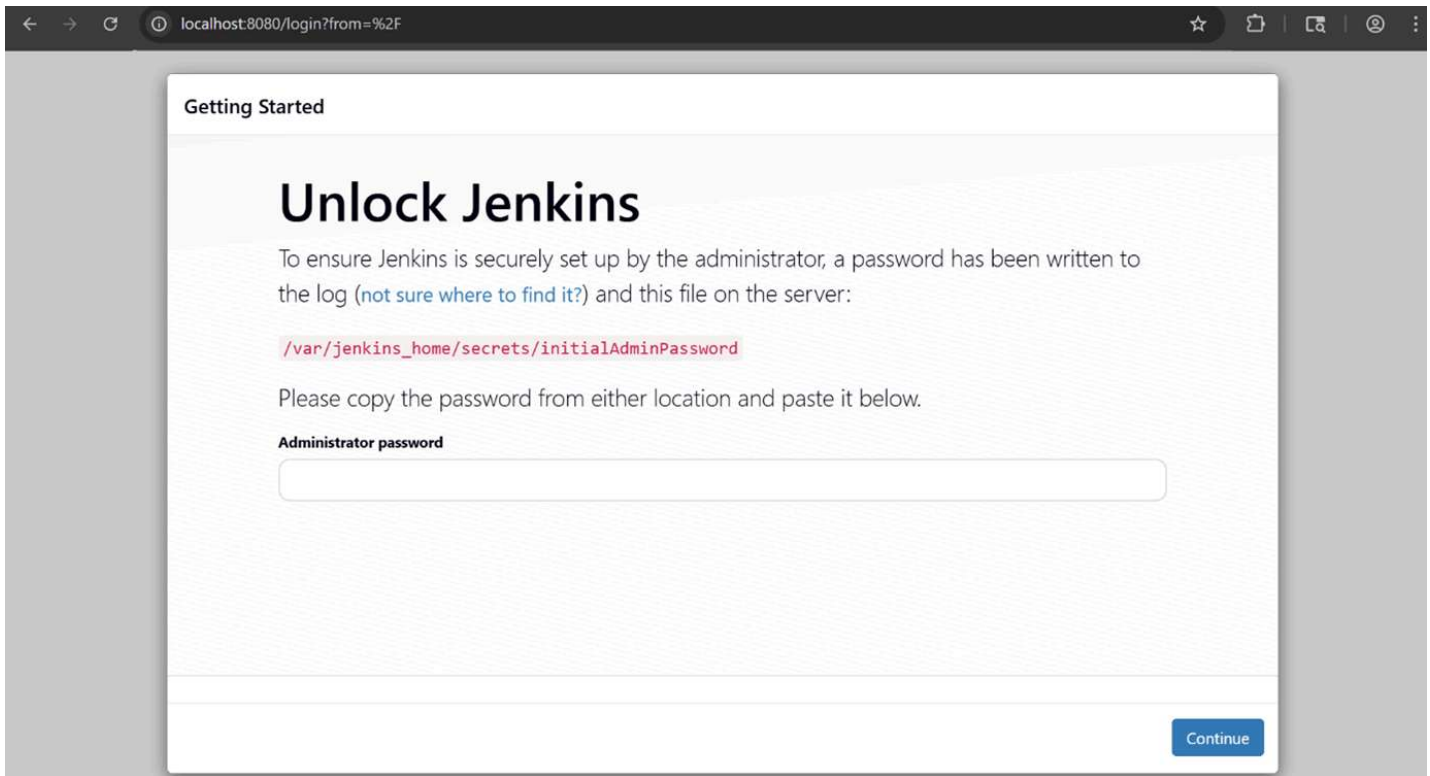
Compose does not use swarm mode to deploy services to multiple nodes in a swarm. All containers will be scheduled on the current node
.

To deploy your application across the swarm, use 'docker stack deploy'.

Creating network "my-app_default" with the default driver
Creating volume "my-app_jenkins_home" with default driver
Pulling jenkins (jenkins/jenkins:lts)...
lts: Pulling from jenkins/jenkins
53c88f1dfb7: Pull complete
4d2125ccf71c: Pull complete
69503609dc01: Pull complete
6da7907f70f4: Pull complete
6327a7366a6f: Pull complete
01938f7c040b: Pull complete
b7b154fd69a1: Pull complete
876217fdee20: Pull complete
fbae626c1b7a: Pull complete
b19825c62ed8: Pull complete
ec7478903548: Pull complete
e682a95ad036: Pull complete
Digest: sha256:c4090806090ca98491d4bf66182f5e3b015a8232f2acf2df209a212a5801aa8e
Status: Downloaded newer image for jenkins/jenkins:lts
Creating jenkins ... done
```

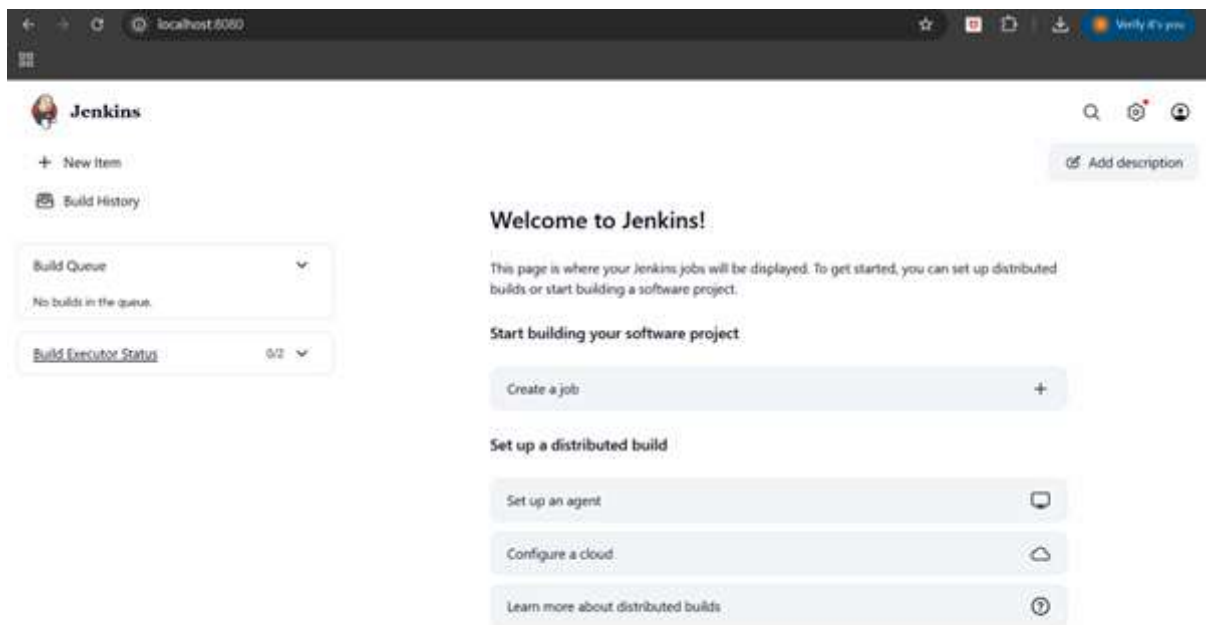
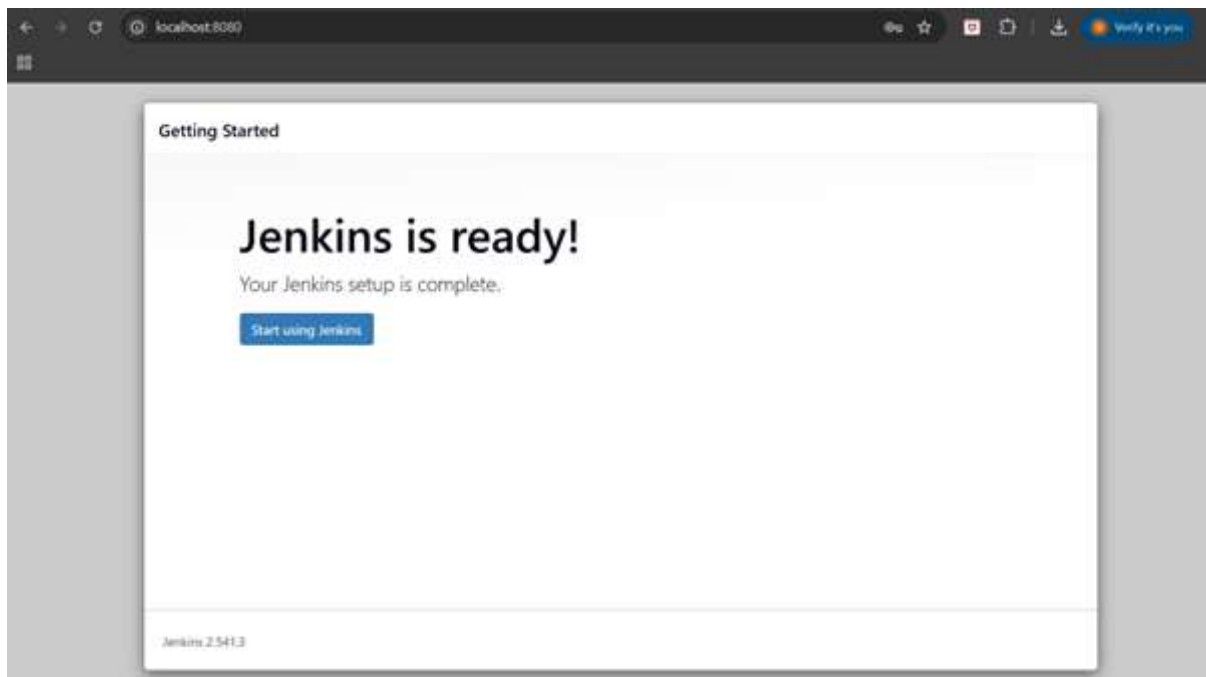
3. Unlock Jenkins

Jenkins was unlocked using the initial admin password from the container.



```
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika_Munjal/Desktop/C&D Lab/my-app$ docker exec -it jenkins cat /var/jenkins_home/secrets/initialAdminPassword
5d6329f4ff3d44098406f5aa14f702e5
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika_Munjal/Desktop/C&D Lab/my-app$ |
```





Part C: Jenkins Configuration

1. Add Docker Hub Credentials

Docker Hub access token was generated and added to Jenkins credentials securely.

< Add Secret text

✕

Scope ?
Global (Jenkins, nodes, items, all child items, etc) ▼

Secret
.....

ID ?
dockerhub-token

Description ?
Docker Hub Token

Create

Global

Credentials that should be available everywhere

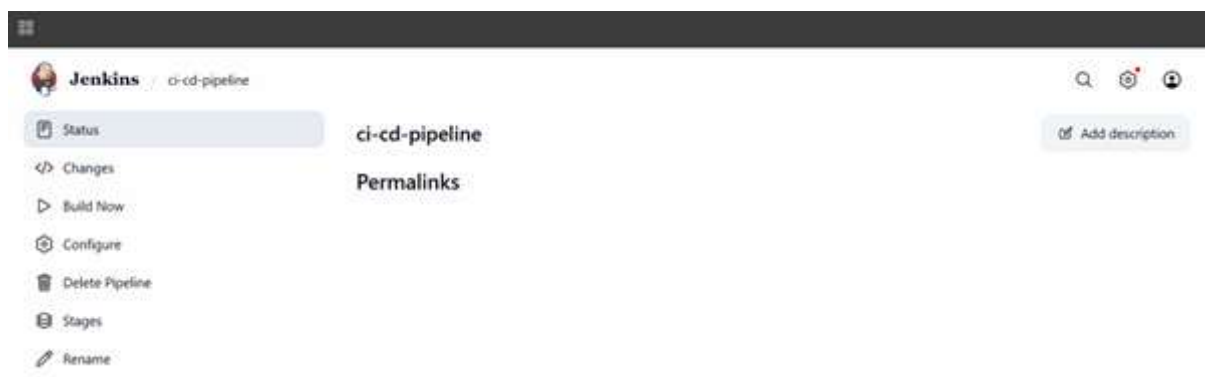
+ Add Credentials

🔑 dockerhub-token
Docker Hub Token

✎ ...

2. Create Pipeline Job

A pipeline job named `ci-cd-pipeline` was created and configured using the Jenkinsfile.

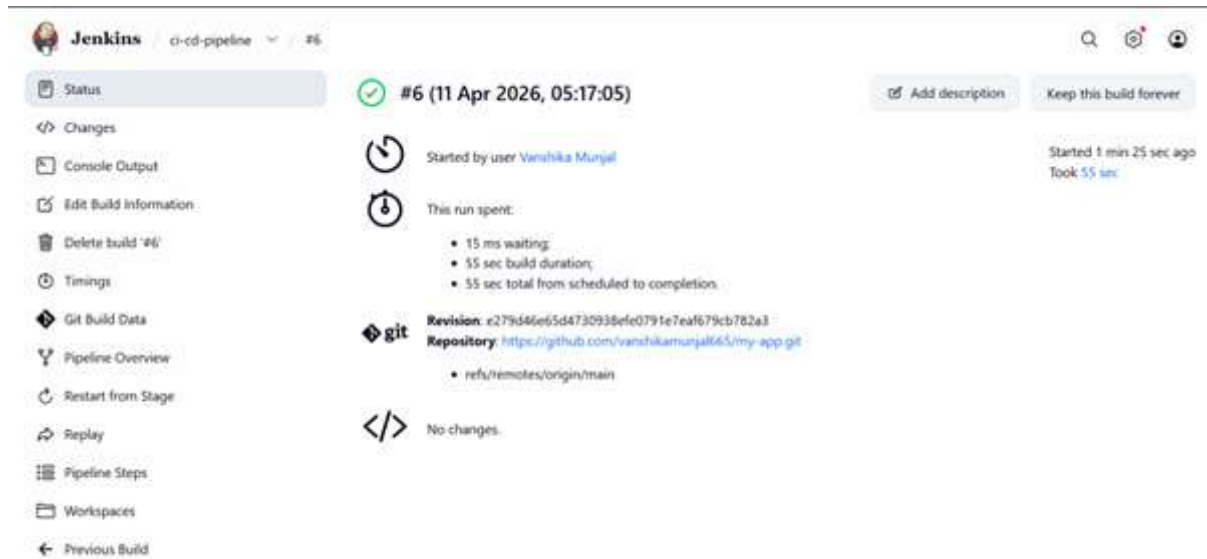


Part D: Build Jenkins

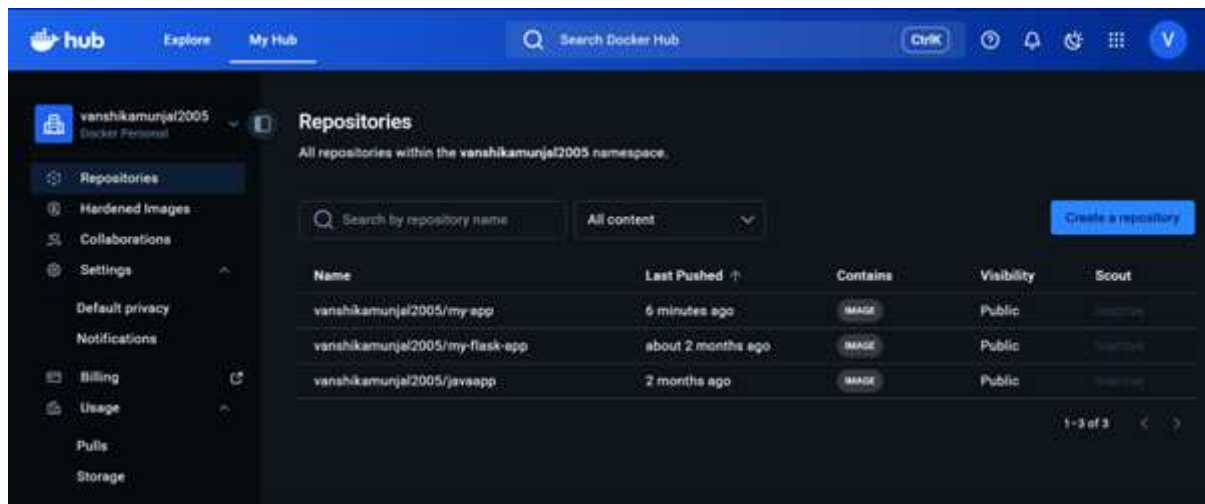
1. Build

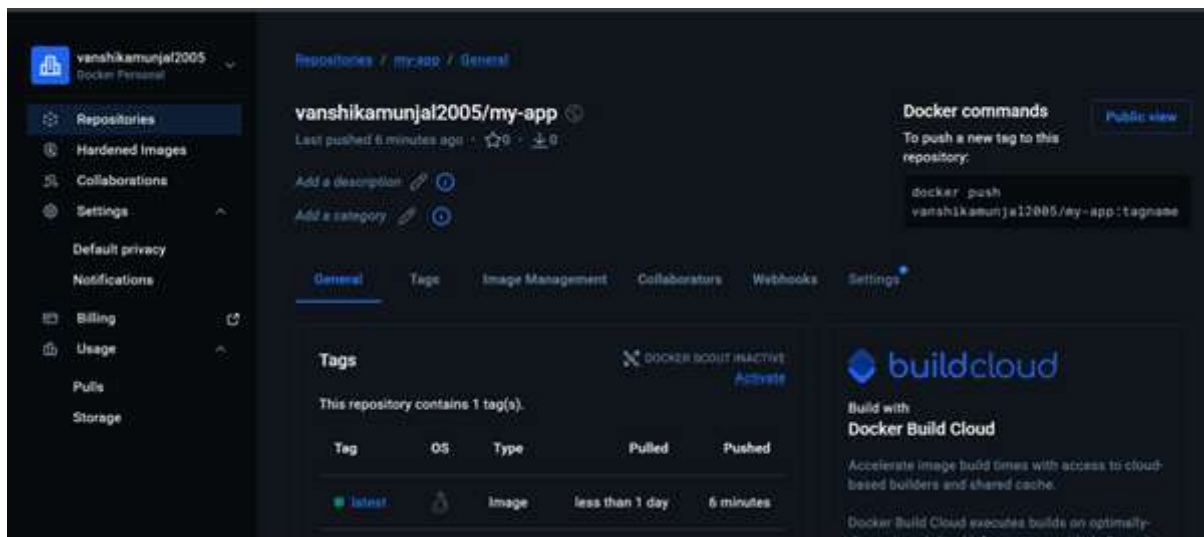
The pipeline was executed successfully, performing the following steps:

- Cloned code from GitHub
- Built Docker image
- Logged into Docker Hub
- Pushed image to Docker Hub



2. Result (DockerHub Screenshot)





Observations

- Jenkins GUI simplifies CI/CD management
- GitHub acts as source + pipeline definition
- Docker ensures consistent builds
- Webhook enables automation

Result

Successfully implemented a complete CI/CD pipeline where:

- Source code and pipeline are maintained in GitHub
- Jenkins automatically detects changes
- Docker image is built on host agent
- Image is securely pushed to Docker Hub

Conclusion

A CI/CD pipeline was successfully implemented using Jenkins, GitHub, and Docker Hub. The application was containerized using Docker and automatically built and deployed through Jenkins pipeline.

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CONTAINERIZATION AND DEVOPS LAB

Experiment 9: [🔗](#)

Ansible

Reference Link:

<https://upessocs.github.io/#dir=/Lectures/Containerization%20and%20DevOps/Lab/&file=Experiment%2009%20Ansible.md>

Theory

Problem Statement

Managing infrastructure manually across multiple servers leads to configuration drift, inconsistent environments, and time-consuming repetitive tasks. Scaling from one server to hundreds becomes nearly impossible with manual SSH-based administration.

What is Ansible?

- Ansible is an open-source **automation tool** for **configuration management, application deployment, and orchestration**.
- It follows an **agentless architecture**, using SSH for Linux and WinRM for Windows.
- Uses **YAML-based playbooks** to define automation tasks.

Ansible is software that enables cross-platform automation and orchestration at scale and has become the standard choice among enterprise automation solutions.

How Ansible Solves the Problem

- **Agentless Architecture:** No software installation required on managed nodes
- **Idempotency:** Running playbooks multiple times yields the same result
- **Declarative Syntax:** Describe desired state, not the steps to achieve it
- **Push-based:** Initiates changes from control node immediately

Key Concepts

Component	Description
Control Node	Machine with Ansible installed
Managed Nodes	Target servers (no Ansible agent needed)
Inventory	Defines the list of managed nodes (e.g., <code>inventory.ini</code>)
Playbooks	YAML files containing a sequence of automation steps
Tasks	Individual actions in playbooks (e.g., installing a package)
Modules	Built-in functionality to perform tasks (e.g., <code>yum</code> , <code>apt</code> , <code>service</code>)
Roles	Pre-defined reusable automation scripts

How does Ansible work?

Ansible uses the concepts of control and managed nodes. It connects from the **control node**, any machine with Ansible installed, to the **managed nodes**, sending commands and instructions to them. The units of code that Ansible executes on the managed nodes are called **modules**. Each module is invoked by a **task**, and an ordered list of tasks together forms a **playbook**. Users write playbooks with tasks and modules to define the desired state of the system. The managed machines are represented in a simple **inventory file** that groups all nodes into different categories. Ansible uses a simple language, **YAML**, to define playbooks in a human-readable format.

Ansible doesn't require installation of any extra agents on managed nodes, making it easy to start using.

Typically, the only requirement is a terminal to execute commands and a text editor to define configuration files.

Benefits of using Ansible

- A free and open-source community project with a huge audience
- Battle-tested over many years as a preferred IT automation tool
- Easy to start and use without needing special coding skills
- Simple deployment workflow without extra agents
- Supports modularity and reusability for advanced automation
- Extensive official documentation with strong community support

Part A: Ansible Installation

1. Install via apt (Debian/Ubuntu)

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\HP> wsl
vanshika_munjai@DESKTOP-107G070:/mnt/c/Users/HP$ pip install ansible
Command 'pip' not found, but can be installed with:
sudo apt install python3-pip
vanshika_munjai@DESKTOP-107G070:/mnt/c/Users/HP$ sudo apt update -y
[sudo] password for vanshika_munjai:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1586 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1896 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [255 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [10.8 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1170 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [345 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [225 kB]
Get:13 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.2 kB]
Get:14 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [22.9 kB]
```

```
vanshika_munjai@DESKTOP-107G070:/mnt/c/Users/HP$ sudo apt install ansible -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libllvml9 libwayland-server0
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  ansible-core ieee-data python3-argcomplete python3-dnspython python3-jmespath python3-kerberos python3-libcloud python3-lockfile
  python3-netaddr python3-ntlm-auth python3-passlib python3-requests-ntlm python3-resolvelib python3-selinux python3-simplejson
  python3-wireshark python3-xmltodict
Suggested packages:
  cowsay sshpass python3-trio python3-aioreg python3-h2 python3-httpx python3-httpcore python-lockfile-doc ipython3
  python-netaddr-docs
The following NEW packages will be installed:
  ansible ansible-core ieee-data python3-argcomplete python3-dnspython python3-jmespath python3-kerberos python3-libcloud
  python3-lockfile python3-netaddr python3-ntlm-auth python3-passlib python3-requests-ntlm python3-resolvelib python3-selinux
  python3-simplejson python3-wireshark python3-xmltodict
0 upgraded, 18 newly installed, 0 to remove and 88 not upgraded.
Need to get 22.0 MB of archives.
After this operation, 330 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/universe amd64 python3-resolvelib all 1.0.1-1 [25.7 kB]
```

```
vanshika_munjai@DESKTOP-107G070:/mnt/c/Users/HP$ ansible --version
ansible [core 2.16.3]
  config file = None
  configured module search path = ['/home/vanshika_munjai/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  ansible collection location = /home/vanshika_munjai/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.12.3 (Main, Jan 22 2026, 20:57:42) [GCC 13.3.0] (/usr/bin/python3)
  jinja2 version = 3.1.2
  libyaml = True
```

Updates the system package list and ensures latest versions are available

```
sudo apt update
```

Installs Ansible on your system and -y auto-confirms installation

```
sudo apt install ansible -y
```

Checks if Ansible is installed properly and displays version and config details

```
ansible --version
```

2. Post Installation Check

```
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/HP$ ansible localhost -m ping
[WARNING]: No inventory was parsed, only implicit localhost is available
localhost | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/HP$ |
```

Tests Ansible on your own system (control node) and Checks if Ansible is working properly before connecting to other servers

```
ansible localhost -m ping
```

Part B: Create Docker Image and Test SSH Login

1. Create SSH key pair in WSL

```
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/HP$ ssh-keygen -t rsa -b 4096
Generating public/private rsa key pair.
Enter file in which to save the key (/home/vanshika_munjal/.ssh/id_rsa):
Created directory '/home/vanshika_munjal/.ssh'.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/vanshika_munjal/.ssh/id_rsa
Your public key has been saved in /home/vanshika_munjal/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:0zqocLJ4L0fhAeZoNevudXS0il/2Dus/+etJTR50bKo vanshika_munjal@DESKTOP-1D7G070
The key's randomart image is:
+---[RSA 4096]---+
|  oo                |
| +..o              |
| ....o            |
| . . . o .S.      |
| .o . +. = .      |
| o o . o *o..E o   |
| .o = *.++ .      |
| o.oBooo+o++=.    |
+---[SHA256]-----+
```

```
ssh-keygen -t rsa -b 4096
```

Explanation: Generates SSH keys.

2. Copy keys to current folder

```
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ cp ~/.ssh/id_rsa.pub .
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ cp ~/.ssh/id_rsa
cp: missing destination file operand after '/home/vanshika_munjal/.ssh/id_rsa'
Try 'cp --help' for more information.
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ cp ~/.ssh/id_rsa .
```

```
cp ~/.ssh/id_rsa.pub .
cp ~/.ssh/id_rsa .
```

Explanation: Copies keys to working directory.

3. Check if copied

```
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ ls
Application Data  NTUSER.DAT{ced7af7f-f208-11ef-9689-a346e8aeb908}.TM.blf
Cookies          NTUSER.DAT{ced7af7f-f208-11ef-9689-a346e8aeb908}.TMContainer000000000000000001.regtrans-ms
Desktop          NTUSER.DAT{ced7af7f-f208-11ef-9689-a346e8aeb908}.TMContainer000000000000000002.regtrans-ms
Documents        NetHood
Downloads        PrintHood
Music            Recent
Pictures         Recent
Public          Recent
Templates       Recent
'Local Settings' SendTo
'My Documents'  SendTo
NTUSER.DAT     Templates
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ chmod 600 id_rsa
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$
```

```
ls
```

Explanation: Verifies files.

4. Copying keys to another directory

```
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/HP$ cd /mnt/c/Users/Vanshika\ Munjal/Desktop/C&D\ Lab/exp9\ file
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp9 file$ cp /mnt/c/Users/HP/id_rsa .
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp9 file$ cp /mnt/c/Users/HP/id_rsa.pub .
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp9 file$ ls
id_rsa id_rsa.pub
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp9 file$ chmod 600 id_rsa
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp9 file$
```

```
cd /mnt/c/Users/Vanshika\ Munjal/Desktop/C&D\ Lab/exp9\ file
cp /mnt/c/Users/HP/id_rsa .
cp /mnt/c/Users/HP/id_rsa.pub .
```

5. Create Dockerfile for Ubuntu SSH Server

```
Dockerfile

FROM ubuntu

RUN apt update -y
RUN apt install -y python3 python3-pip openssh-server
RUN mkdir -p /var/run/sshd

# Configure SSH
RUN mkdir -p /run/sshd && \
    echo 'root:password' | chpasswd && \
    sed -i 's/#PermitRootLogin prohibit-password/PermitRootLogin yes/' /etc/ssh/sshd_config && \
    sed -i 's/#PasswordAuthentication yes/PasswordAuthentication no/' /etc/ssh/sshd_config && \
    sed -i 's/#PubkeyAuthentication yes/PubkeyAuthentication yes/' /etc/ssh/sshd_config

# Create .ssh directory and set proper permissions
RUN mkdir -p /root/.ssh && \
    chmod 700 /root/.ssh

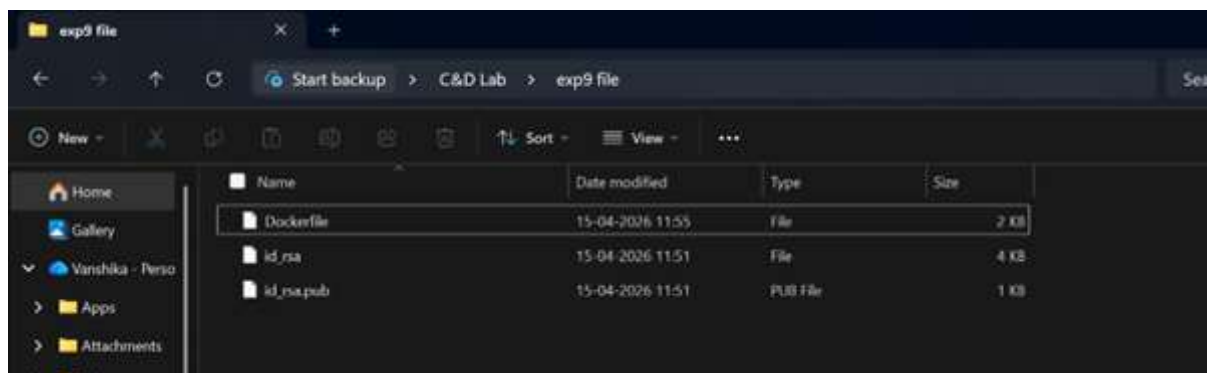
# Copy SSH keys (note: this is not secure for production!)
COPY id_rsa /root/.ssh/id_rsa
COPY id_rsa.pub /root/.ssh/authorized_keys

# Set proper permissions for keys
RUN chmod 600 /root/.ssh/id_rsa && \
    chmod 644 /root/.ssh/authorized_keys

# Fix for SSH login
RUN sed -i 's@session\s*required\s*pam_loginuid.so@session optional pam_loginuid.so@g' /etc/pam.d/sshd

# Expose SSH port
EXPOSE 22

# Start SSH service when container starts
CMD ["/usr/sbin/sshd", "-D"]
```



Explanation: Sets up SSH-enabled Ubuntu container.

6. Build Docker Image


```

vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp9 file$ docker build -t ubuntu-server .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  8.704kB
Step 1/12 : FROM ubuntu
----> 493218ed0f48
Step 2/12 : RUN apt update -y
----> Running in d0be69549a4d

WARNING: apt does not have a stable CLI interface. Use with caution in scripts.

Get:1 http://archive.ubuntu.com/ubuntu noble InRelease [256 kB]
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:3 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1509 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:5 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [34.8 kB]
Get:6 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1998 kB]

Step 11/12 : EXPOSE 22
----> Running in 5d00d43defc2
----> Removed intermediate container 5d00d43defc2
----> 42aa52947460
Step 12/12 : CMD ["/usr/sbin/sshd", "-D"]
----> Running in 7ff2635cc032
----> Removed intermediate container 7ff2635cc032
----> b571ba0909b5
Successfully built b571ba0909b5
Successfully tagged ubuntu-server:latest
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp9 file$ |

```

```
docker build -t ubuntu-server .
```

Explanation: Builds Docker image.

7. Run Docker Container

```

vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp9 file$ docker run -d -p 2222:22 --name ssh-test-server ubuntu-server
b079b6df2ffef41842d3d15a86bfed7777c53e082f7ab3f94a47ed09e709f4c3

```

```
docker run -d -p 2222:22 --name ssh-test-server ubuntu-server
```

Explanation: Runs container with SSH.

8. Find Container IP Address

```

vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp9 file$ docker inspect -f '{{range.NetworkSettings.Networks}}{{.IPAddress}}{{end}}' ssh-test-server
172.18.0.3

```

```
docker inspect -f '{{.NetworkSettings.Networks.IPAddress}}' ssh-test-server
```

Explanation: Gets container IP.

9. Test SSH Connections

Password Authentication

```
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp9 file$ ssh root@localhost -p 2222
The authenticity of host '[localhost]:2222 ([127.0.0.1]:2222)' can't be established.
ED25519 key fingerprint is SHA256:QvHqRuBMLPwbcy2N9ldS+tgN3D3+1LjHL2AMuqDkzyg.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '[localhost]:2222' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.6.87.2-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

root@0079b0df2ffe:~#
```

```
ssh root@localhost -p 2222
```

Key-Based Authentication

```
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp9 file$ ssh -i ~/.ssh/id_rsa root@localhost -p 2222
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.6.87.2-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.
Last login: Wed Apr 15 07:52:00 2026 from 172.18.0.1
root@0079b0df2ffe:~#
```

```
ssh -i ~/.ssh/id_rsa root@localhost -p 2222
```

Explanation: Tests SSH login.

10. Cleanup

```
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp9 file$ docker stop ssh-test-server
ssh-test-server
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp9 file$ docker rm ssh-test-server
ssh-test-server
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp9 file$
```

```
docker stop ss-test-server
```

```
docker rm -f ssh-test-server
```

Part C: Ansible with Docker

1. Start Multiple Containers to act as a server

```

vanshika_munjial@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp9 file$ for i in {1..4}; do
echo -e "\n Creating server${i}\n"
docker run -d --rm -p 220${i}:22 --name server${i} ubuntu-server
echo -e "IP of server${i} is $(docker inspect -f '{{range.NetworkSettings.Networks}}{{.IPAddress}}{{end}}' server${i})"
done

Creating server1
53d67c062e3b52f0e4a4f21731b7d3f080f2b2995b0d423acafbb7d57ac2e4e9
IP of server1 is 172.18.0.3

Creating server2
4833924307ec8f104e0dc6f4a268be6ff46793e9ccb08103d8af9e20eca892d
IP of server2 is 172.18.0.4

Creating server3
01b7f8e3519e4dd762380532caf81d7880ee67ea7b1033f6f4856d856082b6c8
IP of server3 is 172.18.0.5

Creating server4
540b02d7b6955b3b398c46e9f559529d043a2912673f5a975c06c09438e6394b
IP of server4 is 172.18.0.6
vanshika_munjial@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp9 file$ |

```

```

vanshika_munjial@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp9 file$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
540b02d7b695   ubuntu-server  "/usr/sbin/sshd -D"      57 seconds ago Up 57 seconds 0.0.0.0:2204->22/tcp,
[...]:2204->22/tcp
01b7f8e3519e   ubuntu-server  "/usr/sbin/sshd -D"      58 seconds ago Up 57 seconds 0.0.0.0:2203->22/tcp,
[...]:2203->22/tcp
4833924307ec   ubuntu-server  "/usr/sbin/sshd -D"      58 seconds ago Up 58 seconds 0.0.0.0:2202->22/tcp,
[...]:2202->22/tcp
53d67c062e3b   ubuntu-server  "/usr/sbin/sshd -D"      59 seconds ago Up 58 seconds 0.0.0.0:2201->22/tcp,
[...]:2201->22/tcp

```

```

for i in {1..4}; do
    docker run -d -p 220${i}:22 --name server${i} ubuntu-server
done

```

Explanation: Creates 4 server containers.

2. Create Ansible Inventory

```

vanshika_munjial@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp9 file$ echo "[servers]" > inventory.ini
vanshika_munjial@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp9 file$ for i in {1..4}; do
docker inspect -f '{{range.NetworkSettings.Networks}}{{.IPAddress}}{{end}}' server${i} >> inventory.ini
done
vanshika_munjial@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp9 file$ cat << EOF >> inventory.ini

[servers:vars]
ansible_user=root
ansible_ssh_private_key_file=~/.ssh/id_rsa
ansible_python_interpreter=/usr/bin/python3
EOF

```

```
echo "[servers]" > inventory.ini
for i in {1..4}; do
    docker inspect -f '{{range.NetworkSettings.Networks}}{{.IPAddress}}{{end}}' server${i} >> inventory.ini
done
```

Explanation: Adds server IPs.

3. Add Variables

```
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp9 file$ echo "[servers]" > inventory.ini
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp9 file$ for i in {1..4}; do
    docker inspect -f '{{range.NetworkSettings.Networks}}{{.IPAddress}}{{end}}' server${i} >> inventory.ini
done
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp9 file$ cat << EOF >> inventory.ini
[servers:vars]
ansible_user=root
ansible_ssh_private_key_file=~/.ssh/id_rsa
ansible_python_interpreter=/usr/bin/python3
EOF
```

```
cat << EOF >> inventory.ini
```

```
[servers:vars]
ansible_user=root
ansible_ssh_private_key_file=~/.ssh/id_rsa
ansible_python_interpreter=/usr/bin/python3
EOF
```

Explanation: Defines connection settings.

4. Review content of inventory.in

```
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp9 file$ cat inventory.ini
[servers]
172.18.0.3
172.18.0.4
172.18.0.5
172.18.0.6

[servers:vars]
ansible_user=root
ansible_ssh_private_key_file=~/.ssh/id_rsa
ansible_python_interpreter=/usr/bin/python3
```

```
cat inventory.ini
```

5. Manual SSH Test

```

vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp9 file$ ssh -i ~/.ssh/id_rsa root@172.18.0.3
The authenticity of host '172.18.0.3 (172.18.0.3)' can't be established.
ED25519 key fingerprint is SHA256:QvHqRuBMLPwbcy2N9ldS+tgN3D3+1LjHL2AMuqDwzyg.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:1: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '172.18.0.3' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.6.87.2-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

root@53d67c062e3b:~#

```

```
ssh -i ~/.ssh/id_rsa root@<IP>
```

6. Ansible Ping Test

```

vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp9 file$ ansible all -i inventory.ini -m ping
The authenticity of host '172.18.0.4 (172.18.0.4)' can't be established.
ED25519 key fingerprint is SHA256:QvHqRuBMLPwbcy2N9ldS+tgN3D3+1LjHL2AMuqDwzyg.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:1: [hashed name]
  ~/.ssh/known_hosts:4: [hashed name]
  ~/.ssh/known_hosts:5: [hashed name]
  ~/.ssh/known_hosts:6: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? 172.18.0.3 | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
172.18.0.5 | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
172.18.0.6 | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
yes
172.18.0.4 | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp9 file$

```

```
ansible all -i inventory.ini -m ping
```

Explanation: Checks connectivity.

7. Create Playbook (update.yml)


```

GNU nano 7.2                                playbook.yml *
----
- name: Configure servers
  hosts: all
  become: yes

  tasks:
    - name: Update packages
      apt:
        update_cache: yes

    - name: Install packages
      apt:
        name: ["vim", "htop", "wget"]
        state: present

    - name: Create test file
      copy:
        dest: /root/ansible_test.txt
        content: "Configured by Ansible on {{ inventory_hostname }}"

```

8. Run Playbook

```

vanshika_munjal@DESKTOP-107G070: /mnt/c/Users/Vanshika_Munjal/Desktop/C&D Lab/exp9 file$ ansible-playbook -i inventory.ini playbook.yml
PLAY [Configure servers] *****

TASK [Gathering Facts] *****
ok: [172.18.0.6]
ok: [172.18.0.4]
ok: [172.18.0.5]
ok: [172.18.0.3]

TASK [Update packages] *****
ok: [172.18.0.6]
ok: [172.18.0.5]
ok: [172.18.0.4]
ok: [172.18.0.3]

TASK [Install packages] *****
changed: [172.18.0.6]
changed: [172.18.0.5]
changed: [172.18.0.4]
changed: [172.18.0.3]

TASK [Create test file] *****
changed: [172.18.0.4]
changed: [172.18.0.5]
changed: [172.18.0.3]
changed: [172.18.0.6]

PLAY RECAP *****
172.18.0.3      : ok=4    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
172.18.0.4      : ok=4    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
172.18.0.5      : ok=4    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
172.18.0.6      : ok=4    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
vanshika_munjal@DESKTOP-107G070: /mnt/c/Users/Vanshika_Munjal/Desktop/C&D Lab/exp9 file$

```

ansible-playbook -i inventory.ini update.yml

9. Verify Changes using Ansible

```
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp9 file$ ansible all -i inventory.ini -m command -a "cat /root/ansible_test.txt"
172.18.0.3 | CHANGED | rc=0 >>
Configured by Ansible on 172.18.0.3
172.18.0.4 | CHANGED | rc=0 >>
Configured by Ansible on 172.18.0.4
172.18.0.5 | CHANGED | rc=0 >>
Configured by Ansible on 172.18.0.5
172.18.0.6 | CHANGED | rc=0 >>
Configured by Ansible on 172.18.0.6
```

```
ansible all -i inventory.ini -m command -a "cat /root/ansible_test.txt"
```

10. Verify changes manually via Docker

```
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp9 file$ for i in {1..4}; do
docker exec server${i} cat /root/ansible_test.txt
done
Configured by Ansible on 172.18.0.3Configured by Ansible on 172.18.0.4Configured by Ansible on 172.18.0.5Configured by Ansible on 172
.18.0.6vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp9 file$ |
```

```
for i in {1..4}; do
    docker exec server${i} cat /root/ansible_test.txt
done
```

11. Cleanup

```
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp9 file$ for i in {1..4}; do docker rm -f server${i};
done; done
server1
server2
server3
server4
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp9 file$ |
```

```
for i in {1..4}; do docker rm -f server${i}; done
```



Conclusion

In this experiment, Docker containers were used as servers and configured using Ansible. SSH authentication was implemented, an inventory file was created, and automation was achieved using playbooks across multiple nodes.

vanshikamunjal665

CONTAINERIZATION AND DEVOPS LAB

Experiment 10:

SonarQube

Reference Link:

<https://upessocs.github.io/#dir=/Lectures/Containerization%20and%20DevOps/Lab/&file=Experiment%2010%20SonarQube.md>

Theory

Problem Statement

Code quality issues (bugs, vulnerabilities, code smells) are often discovered late in the development cycle, making them expensive to fix. Manual code reviews are inconsistent and do not scale.

What is SonarQube?

SonarQube is an open-source platform for continuous inspection of code quality. It performs automatic reviews with static analysis to detect bugs, code smells, and security vulnerabilities.

How SonarQube Solves the Problem:

- **Continuous Inspection:** Scans code with every commit, providing immediate feedback
- **Quality Gates:** Defines pass/fail criteria for code quality
- **Technical Debt Quantification:** Measures effort needed to fix issues
- **Multi-language Support:** Supports 20+ programming languages

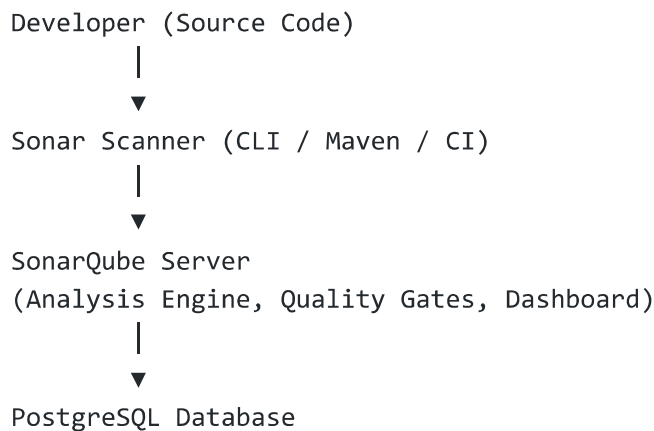
- **Visual Analytics:** Dashboard showing code quality metrics and trends

Key Concepts:

- **Quality Gate:** Set of conditions that code must meet before deployment
- **Technical Debt:** Estimated time to fix all issues
- **Code Smells:** Maintainability issues that do not affect functionality
- **Vulnerabilities:** Security-related issues
- **Bugs:** Code that might break or behave unexpectedly
- **Coverage:** Percentage of code covered by tests
- **Duplications:** Repeated code blocks

Hands-on Lab Setup

Lab Architecture



1. Create Experiment Directory

```
mkdir sonarqube-exp10
cd sonarqube-exp10
```

```
PS C:\Users\HP> wsl
vanshika_munjal@DESKTOP-1D7G07D:/mnt/c/Users/HP$ cd "/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab"
vanshika_munjal@DESKTOP-1D7G07D:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab$ mkdir sonarqube-exp10
vanshika_munjal@DESKTOP-1D7G07D:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab$ cd sonarqube-exp10
```

2. Create docker-compose.yml

```
version: '3.8'

services:
  sonar-db:
    image: postgres:13
    container_name: sonar-db
    restart: unless-stopped
    environment:
      POSTGRES_USER: sonar
      POSTGRES_PASSWORD: sonar
      POSTGRES_DB: sonarqube
    volumes:
      - sonar-db-data:/var/lib/postgresql/data
    networks:
      - sonarqube-lab

  sonarqube:
    image: sonarqube:lts-community
    container_name: sonarqube
    restart: unless-stopped
    depends_on:
      - sonar-db
    ports:
      - "9000:9000"
    environment:
      SONAR_JDBC_URL: jdbc:postgresql://sonar-db:5432/sonarqube
      SONAR_JDBC_USERNAME: sonar
      SONAR_JDBC_PASSWORD: sonar
    volumes:
      - sonar-data:/opt/sonarqube/data
      - sonar-extensions:/opt/sonarqube/extensions
    networks:
      - sonarqube-lab

volumes:
  sonar-db-data:
  sonar-data:
  sonar-extensions:

networks:
  sonarqube-lab:
    driver: bridge
```



```
vanshika_munjal@DESKTOP-1D7G070: /mnt/c/Users/Vanshika_Munjal/Desktop/C&O Lab/sonarqube-expl6$ nano docker-compose.yml
vanshika_munjal@DESKTOP-1D7G070: /mnt/c/Users/Vanshika_Munjal/Desktop/C&O Lab/sonarqube-expl6$ |
```

```
docker-compose.yml
File Edit View

version: '3.8'

services:
  sonar-db:
    image: postgres:13
    container_name: sonar-db
    restart: unless-stopped
    environment:
      POSTGRES_USER: sonar
      POSTGRES_PASSWORD: sonar
      POSTGRES_DB: sonarqube
    volumes:
      - sonar-db-data:/var/lib/postgresql/data
    networks:
      - sonarqube-lab

  sonarqube:
    image: sonarqube:lts-community
    container_name: sonarqube
    restart: unless-stopped
    depends_on:
      - sonar-db
    ports:
      - "9000:9000"
    environment:
      SONAR_JDBC_URL: jdbc:postgresql://sonar-db:5432/sonarqube
      SONAR_JDBC_USERNAME: sonar
      SONAR_JDBC_PASSWORD: sonar
    volumes:
      - sonar-data:/opt/sonarqube/data
      - sonar-extensions:/opt/sonarqube/extensions
    networks:
      - sonarqube-lab

volumes:
  sonar-db-data:
  sonar-data:
  sonar-extensions:
```

3. Start Container

```
docker-compose up -d
```

```

vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika_Munjal/Desktop/C&D Lab/sonarqube-exp10$ docker-compose up -d
WARNING: The Docker Engine you're using is running in swarm mode.

Compose does not use swarm mode to deploy services to multiple nodes in a swarm. All containers will be scheduled on the current node
.

To deploy your application across the swarm, use 'docker stack deploy'.

Creating network "sonarqube-exp10_sonarqube-lab" with driver "bridge"
Creating volume "sonarqube-exp10_sonar-db-data" with default driver
Creating volume "sonarqube-exp10_sonar-data" with default driver
Creating volume "sonarqube-exp10_sonar-extensions" with default driver
Pulling sonar-db (postgres:13)...
13: Pulling from library/postgres
d7ecdd7702a: Pull complete
80e8d95fd53: Pull complete
fcfd49225910: Pull complete
907bdcac0ea3: Pull complete
0977f34b018c0: Pull complete
4d2c7a7b3dba: Pull complete
8cb000c698bc: Pull complete
ba168ff08d42: Pull complete
bad194149a82: Pull complete
5f1cded65589: Pull complete
7d0450bb210f: Pull complete
23a39b1003e9: Pull complete
21f7a31efcb7: Pull complete
ae52c0c0a35: Pull complete
Digest: sha256:4689940c683801b4ab839ab3b0a0a3555a5fe425371422310944e89eca7d8066
Status: Downloaded newer image for postgres:13

```

4. Check Containers

docker ps

```

vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika_Munjal/Desktop/C&D Lab/sonarqube-exp10$ docker network ls
NETWORK ID          NAME                DRIVER              SCOPE
7643b49574c5        app-network         bridge              local
f7ead84b553c        bridge              bridge              local
6c8375a3ca04        containerization-project-1_default bridge              local
6bfcffe93f47        docker_gwbridge     bridge              local
cfcaf012bac0        exp6file_default    bridge              local
21b3242599f7        host                host                local
pmvdlnm6v9xq        ingress             overlay             swarm
531b76a88b56        k3d-mycluster       bridge              local
98086cd8d357        lan_net             macvlan             local
4d4b0137f908        my-app_default      bridge              local
49c767ae24e0        my-network          bridge              local
22cfb0a54a9a        myapp-network       bridge              local
7464a52939aa        none                null                local
bee6452cb7ca        sonarqube-exp10_sonarqube-lab bridge              local
36f38d42404e        wp-compose-lab_default bridge              local
aa03ec1108af        wp-net              bridge              local
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika_Munjal/Desktop/C&D Lab/sonarqube-exp10$ |

```

5. Create Sample Java Application

Create project folder

```

mkdir -p sample-java-app/src/main/java/com/example
cd sample-java-app

```

Create Java File

```
package com.example;

import java.util.ArrayList;
import java.util.List;

public class Calculator {

    public int divide(int a, int b) {
        return a / b;
    }

    public int add(int a, int b) {
        int result = a + b;
        int unused = 100;
        return result;
    }

    public String getUser(String userId) {
        String query = "SELECT * FROM users WHERE id = " + userId;
        return query;
    }

    public int multiply(int a, int b) {
        int result = 0;
        for (int i = 0; i < b; i++) {
            result = result + a;
        }
        return result;
    }

    public int multiplyAlt(int a, int b) {
        int result = 0;
        for (int i = 0; i < b; i++) {
            result = result + a;
        }
        return result;
    }

    public void processUser(String name, String email, String phone,
                           String address, String city, String state,
                           String zip, String country) {
        System.out.println("Processing: " + name);
    }

    public String getName(String name) {
        return name.toUpperCase();
    }
}
```

```

public void riskyOperation() {
    try {
        int x = 10 / 0;
    } catch (Exception e) {
    }
}
}

```

```

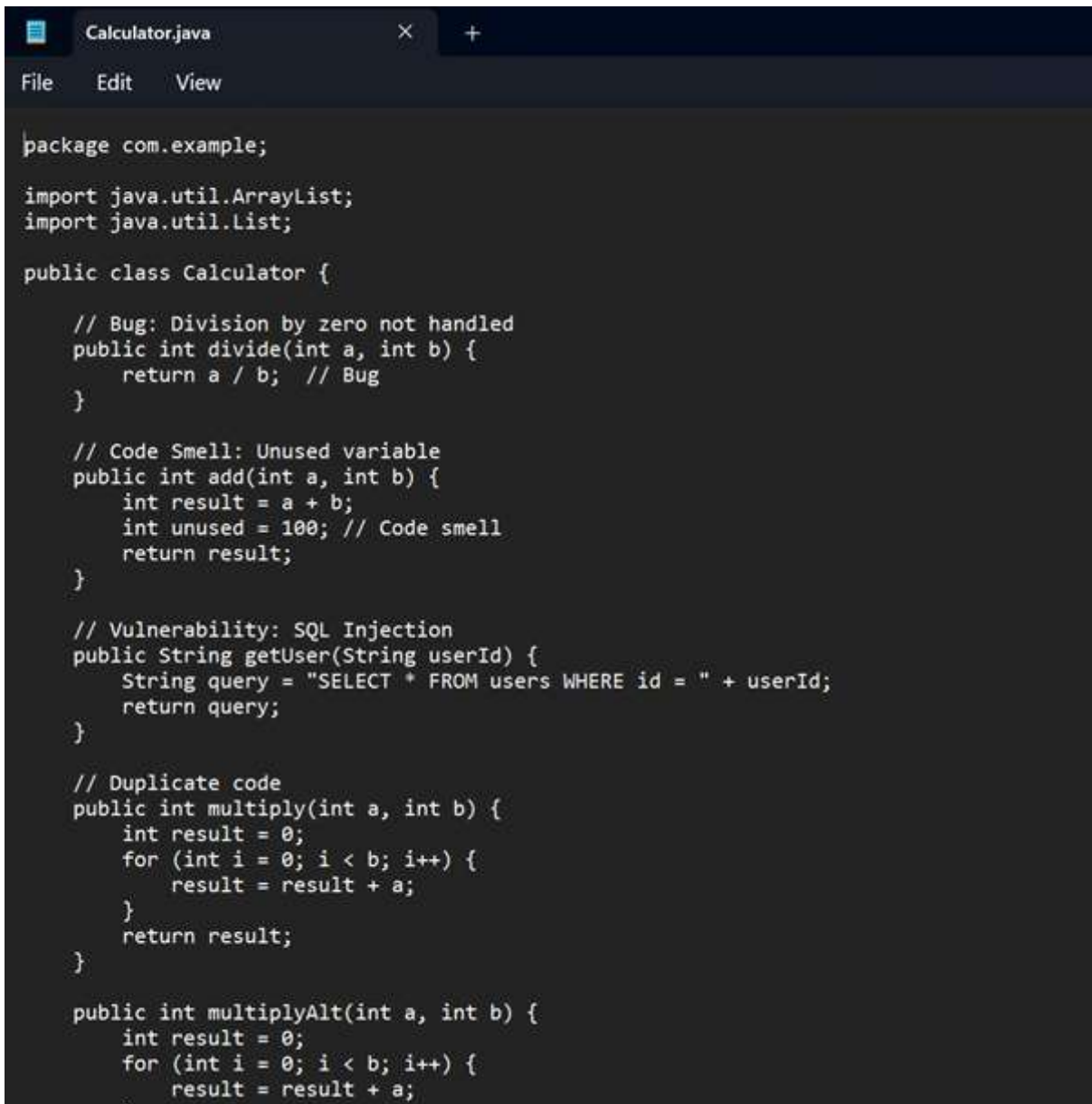
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika_Munjal/Desktop/C&D Lab/sonarqube-expl0$ mkdir -p sample-java-app/src/main/java/
com/example
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika_Munjal/Desktop/C&D Lab/sonarqube-expl0$ cd sample-java-app
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika_Munjal/Desktop/C&D Lab/sonarqube-expl0/sample-java-app$

```

```

vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika_Munjal/Desktop/C&D Lab/sonarqube-expl0/sample-java-app$ nano src/main/java/com/
example/Calculator.java

```



```

Calculator.java
File Edit View

package com.example;

import java.util.ArrayList;
import java.util.List;

public class Calculator {

    // Bug: Division by zero not handled
    public int divide(int a, int b) {
        return a / b; // Bug
    }

    // Code Smell: Unused variable
    public int add(int a, int b) {
        int result = a + b;
        int unused = 100; // Code smell
        return result;
    }

    // Vulnerability: SQL Injection
    public String getUser(String userId) {
        String query = "SELECT * FROM users WHERE id = " + userId;
        return query;
    }

    // Duplicate code
    public int multiply(int a, int b) {
        int result = 0;
        for (int i = 0; i < b; i++) {
            result = result + a;
        }
        return result;
    }

    public int multiplyAlt(int a, int b) {
        int result = 0;
        for (int i = 0; i < b; i++) {
            result = result + a;
        }
    }
}

```

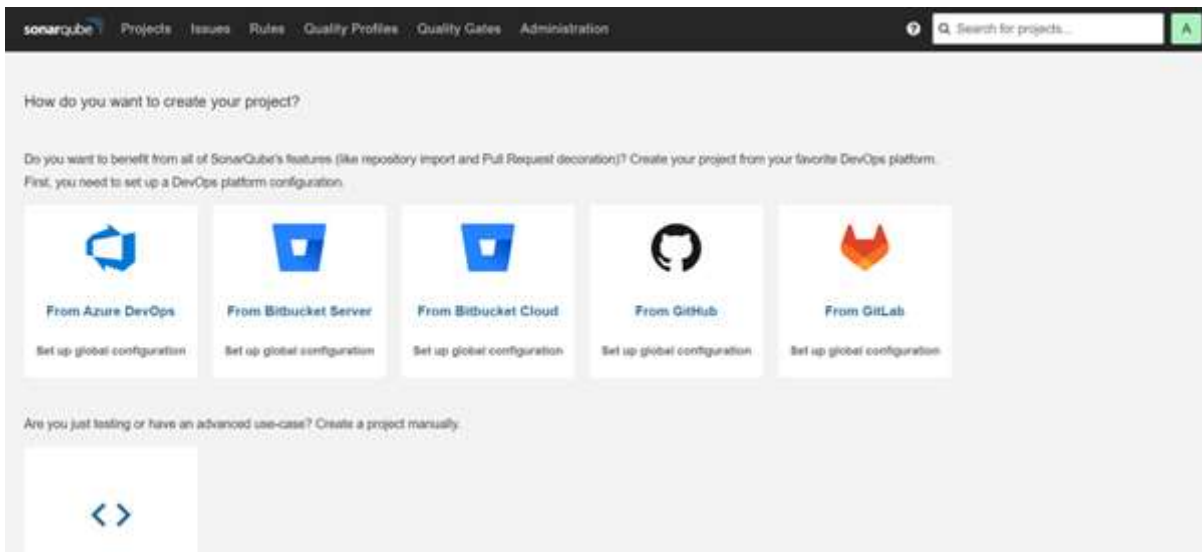
6. Install SonarQube Scanner

```
docker run -d \
--name sonar-scanner \
--network sonarqube-exp10_sonarqube-lab \
-v "$(pwd)/sample-java-app":/usr/src \
sonarsource/sonar-scanner-cli:latest \
sleep infinity
```

```
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/Vanshika_Munjal/Desktop/C60_Lab/sonarqube-exp10/sample-java-app$ docker run -d \
--name sonar-scanner \
--network sonarqube-exp10_sonarqube-lab \
-v "$(pwd)/sample-java-app":/usr/src \
sonarsource/sonar-scanner-cli:latest \
sleep infinity
e3f0f7ce20c6cb21a97134558797e7691d9ef42b7f3ba53084b3af2818a95d75
```

7. Login to SonarQube and Generate Token

- Open: <http://localhost:9000>
- Login: admin/admin
- Go to My Account → Security
- Generate Token



Name	Type	Project	Last use	Created	Expiration	
token1	User		Never	April 18, 2026	May 18, 2026	Revoke

8. Create sonar-project.properties


```
sonar.projectKey=sample-java-app
sonar.projectName=Sample Java Application
sonar.sources=src
sonar.host.url=http://sonarqube:9000
sonar.login=YOUR_TOKEN
```

```
vanshika_munjai@DESKTOP-107G070:/mnt/c/Users/Vanshika_Munjai/Desktop/C&D Lab/sonarqube-expl0/sample-java-app$ nano sonar-project.properties
```

```
GNU nano 7.2 sonar-project.properties *
sonar.projectKey=sample-java-app
sonar.projectName=Sample Java Application
sonar.sources=src
sonar.host.url=http://sonarqube:9000
sonar.login=YOUR_TOKEN
```

9. Run SonarQube Scanner

```
docker exec sonar-scanner sonar-scanner \
-Dsonar.host.url=http://sonarqube:9000 \
-Dsonar.login=YOUR_TOKEN \
-Dsonar.projectKey=sample-java-app \
-Dsonar.sources=src
```

```
vanshika_munjai@DESKTOP-107G070:/mnt/c/Users/Vanshika_Munjai/Desktop/C&D Lab/sonarqube-expl0/sample-java-app$ docker exec sonar-scanner sonar-scanner \
-Dsonar.host.url=http://sonarqube:9000 \
-Dsonar.login=squ_eb5a67a701243509ec81eaf7da4aceb5ef632c67 \
-Dsonar.projectKey=sample-java-app \
-Dsonar.projectName="Sample Java App" \
-Dsonar.sources=src
04:20:33.620 INFO Scanner configuration file: /opt/sonar-scanner/conf/sonar-scanner.properties
04:20:33.634 INFO Project root configuration file: /usr/src/sonar-project.properties
04:20:33.662 INFO SonarScanner CLI 8.0.1.6346
04:20:33.671 INFO Linux 6.6.87.2-microsoft-standard-WSL2 amd64
04:20:43.510 INFO Communicating with SonarQube Server 9.9.8.100196
04:20:36.466 INFO Load global settings
04:20:36.579 INFO Load global settings (done) | time=115ms
04:20:36.582 INFO Server id: 74C15348-AZ2etQ-Krsqwni13MG91
04:20:36.593 INFO User cache: /opt/sonar-scanner/.sonar/cache
04:20:36.600 INFO Load/download plugins
04:20:36.601 INFO Load plugins index
04:20:36.664 INFO Load plugins index (done) | time=62ms
04:20:38.275 INFO Load/download plugins (done) | time=1675ms
04:20:38.930 INFO Process project properties
04:20:38.943 INFO Process project properties (done) | time=13ms
04:20:38.953 INFO Execute project builders
04:20:38.955 INFO Execute project builders (done) | time=2ms
04:20:38.962 INFO Project key: sample-java-app
04:20:38.963 INFO Base dir: /usr/src
04:20:38.963 INFO Working dir: /usr/src/.scannerwork
04:20:38.980 INFO Load project settings for component key: 'sample-java-app'
04:20:39.166 WARN SCM provider autodetection failed. Please use "sonar.scm.provider" to define SCM of your project, or disable the SCM Sensor in the project settings.
04:20:39.184 INFO Load quality profiles
04:20:39.314 INFO Load quality profiles (done) | time=130ms
04:20:39.345 INFO Load active rules
04:20:42.374 INFO Load active rules (done) | time=3029ms
```

10. Analyze Results

- Open SonarQube Dashboard
- View Bugs, Vulnerabilities, Code Smells
- Check Quality Gate

The SonarQube dashboard for 'Sample Java App' shows a 'Passed' status with 'All conditions passed'. The 'Overall Code' section displays the following metrics:

- 1 Bug** (Reliability: D)
- 0 Vulnerabilities** (Security: A)
- 0 Security Hotspots** (Security Review: A)
- 1h 36min Debt**
- 11 Code Smells** (Maintainability: B)

The 'Issues' view shows 12 issues. The left sidebar filters show 11 issues in new code. The main list displays the following issues:

- Remove this unused import java.util.ArrayList.** (Code Smell, Minor, Open, Not assigned, 2min effort, Comment, 8 minutes ago, L3, unused)
- Remove this unused import java.util.List.** (Code Smell, Minor, Open, Not assigned, 2min effort, Comment, 8 minutes ago, L4, unused)
- Remove this useless assignment to local variable "unused".** (Code Smell, Major, Open, Not assigned, 15min effort, Comment, 8 minutes ago, L18, cert, owt, unused)
- Remove this unused "unused" local variable.** (Code Smell, Minor, Open, Not assigned, 5min effort, Comment, 8 minutes ago, L18, unused)
- Immediately return this expression instead of assigning it to the temporary variable "query".** (Code Smell, Minor, Open, Not assigned, 2min effort, Comment, 8 minutes ago, L22, query)
- Update this method so that its implementation is not identical to "multiply" on line 27.** (8 minutes ago, L35)

11. Generate Report using SonarQube API

```
curl -u YOUR_TOKEN: \
"http://localhost:9000/api/issues/search?projectKeys=sample-java-app&types=BUG"
```

```
vanshika_munj665@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/sonarqube-expl0/sample-java-app$ curl -u squ_eb5a67a701243509ec81eaf7da4aceb5ef632c67: "http://localhost:9000/api/issues/search?projectKeys=sample-java-app&types=BUG"
{"total":1,"p":1,"ps":100,"paging":{"pageIndex":1,"pageSize":100,"total":1},"effortTotal":5,"issues":[{"key":"A22e0fhEy5J5cHj05PFe","rule":"java:S3518","severity":"CRITICAL","component":"sample-java-app:src/main/java/com/example/Calculator.java","project":"sample-java-app","line":58,"hash":"0551c559a67ca914d3740295088af2b7","textRange":{"startLine":58,"endLine":58,"startOffset":25,"endOffset":26},"flows":[],"status":"OPEN","message":"Make sure this expression can\u0027t be zero before doing this division.","effort":"5min","debt":"5min","author":"","tags":["cert","cwe","denial-of-service"],"creationDate":"2026-04-18T04:20:39+0000","updateDate":"2026-04-18T04:20:39+0000","type":"BUG","scope":"MAIN","quickFixAvailable":false,"messageFormattings":[]},"components":[{"key":"sample-java-app:src/main/java/com/example/Calculator.java","enabled":true,"qualifier":"FIL","name":"Calculator.java","longName":"src/main/java/com/example/Calculator.java","path":"src/main/java/com/example/Calculator.java"},{"key":"sample-java-app","enabled":true,"qualifier":"TRM","name":"Sample Java App","longName":"Sample Java App"}],"facets":[]}]vanshika_munj665@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/sonarqube-expl0/sample-java-app$
```

Conclusion

SonarQube was successfully deployed using Docker and used to analyze a Java application. The tool detected bugs and code smells, and results were visualized through the dashboard and API.

vanshikamunjal665

CONTAINERIZATION AND DEVOPS LAB

Experiment 11: [🔗](#)

Orchestration using Docker Compose & Docker Swarm (Continuation of Experiment 6)

Reference Link:

<https://upessocs.github.io/#dir=/Lectures/Containerization%20and%20DevOps/Lab/&file=Experiment%2011%20Container%20Orchestration%20with%20Docker%20Stack.md> —

Theory

PART A – CONCEPT CONTINUATION (Simple Explanation)

From Experiment 6, we already know:

Tool	What it does	Limitation
<code>docker run</code>	Runs a single container	Manual, no coordination
Docker Compose	Runs multiple containers together	Single machine, no auto-healing

New Concept: Orchestration

Orchestration = Automatic management of containers

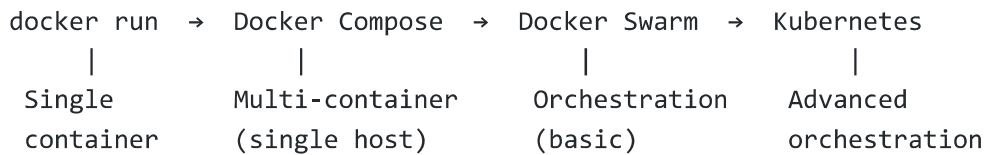
Think of it like a **restaurant manager**:

- Decides how many waiters are needed (**scaling**)
- Replaces a sick waiter immediately (**self-healing**)
- Distributes customers evenly (**load balancing**)

What Orchestration Adds:

Feature	What it means
Scaling	Increase/decrease number of containers
Self-healing	Restart failed containers automatically
Load balancing	Distribute traffic across containers
Multi-host	Run containers across multiple machines

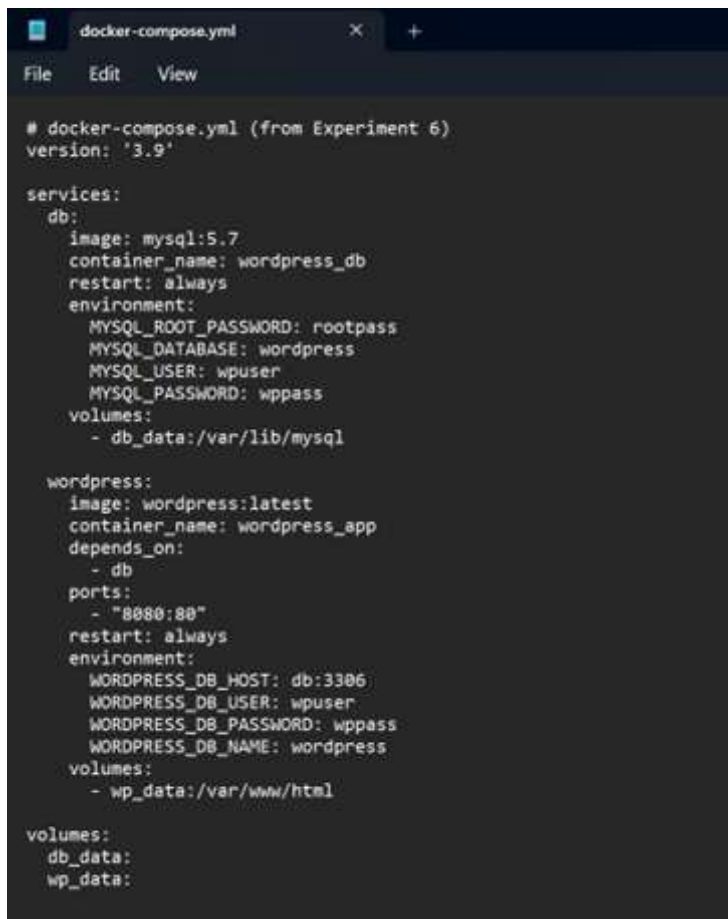
Progression Path



Part B: Practical (Extension of Experiment 6)

1. Create docker-compose.yml (from experiment 6)

We use the same docker-compose file created in Experiment 6 for WordPress and MySQL setup.



```
# docker-compose.yml (from Experiment 6)
version: '3.9'

services:
  db:
    image: mysql:5.7
    container_name: wordpress_db
    restart: always
    environment:
      MYSQL_ROOT_PASSWORD: rootpass
      MYSQL_DATABASE: wordpress
      MYSQL_USER: wpuser
      MYSQL_PASSWORD: wppass
    volumes:
      - db_data:/var/lib/mysql

  wordpress:
    image: wordpress:latest
    container_name: wordpress_app
    depends_on:
      - db
    ports:
      - "8080:80"
    restart: always
    environment:
      WORDPRESS_DB_HOST: db:3306
      WORDPRESS_DB_USER: wpuser
      WORDPRESS_DB_PASSWORD: wppass
      WORDPRESS_DB_NAME: wordpress
    volumes:
      - wp_data:/var/www/html

volumes:
  db_data:
  wp_data:
```

2. Stop any existing compose setup

This ensures no previous containers interfere with Swarm deployment.

```
docker compose down -v
```



```
PS C:\Users\HP> wsl
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/HP$ docker compose down -v
Docker version 28.2.2, build 28.2.2-0ubuntu1~24.04.1
```

3. Verify no containers are running

Check that all containers are stopped.

```
docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED NAMES	STATUS	PORTS
bef98000fb4c	containerization-project-1_database	"docker-entrypoint.s..."	7 days ago postgres_db	Up 20 seconds	5432/tcp
554cbb7049b8	containerization-project-1_backend	"docker-entrypoint.s..."	7 days ago backend_api	Up 18 seconds	0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp
26ee230615b8	sonarqube:lts-community	"/opt/sonarqube/dock..."	7 days ago sonarqube	Up 20 seconds	0.0.0.0:9000->9000/tcp, [::]:9000->9000/tcp
dda77d469ab9	postgres:13	"docker-entrypoint.s..."	7 days ago sonar-db	Up 20 seconds	5432/tcp
d5f70fbel443	jenkins/jenkins:lts	"/usr/bin/tini -- /u..."	2 weeks ago jenkins	Up 20 seconds	0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp, 0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp
25a799c4fd6b	wordpress:latest	"docker-entrypoint.s..."	3 weeks ago wp-compose-lab_wordpress_2	Up 20 seconds	80/tcp
e07671139e7e	wordpress:latest	"docker-entrypoint.s..."	3 weeks ago wp-compose-lab_wordpress_3	Up 20 seconds	80/tcp
e0b64a9cfeae	wordpress:latest	"docker-entrypoint.s..."	3 weeks ago wp-compose-lab_wordpress_1	Up 20 seconds	80/tcp
c4c812ed407a	mysql:5.7	"docker-entrypoint.s..."	3 weeks ago wordpress_db	Up 20 seconds	3306/tcp, 33060/tcp
ea81f68bde87	portainer/portainer-ce:lts	"/portainer"	5 weeks ago portainer	Up 21 seconds	0.0.0.0:8000->8000/tcp, [::]:8000->8000/tcp, 0.0.0.0:9443->9443/tcp, [::]:9443->9443/tcp, 9000/tcp
924e1452e147	ghcr.io/k3d-io/k3d-proxy:5.8.3	"/bin/sh -c nginx-pr..."	5 weeks ago k3d-mycluster-serverlb	Up 20 seconds	80/tcp, 0.0.0.0:39691->6443/tcp
0fa96e9ee63a	rancher/k3s:v1.31.5-k3s1	"/bin/k3d-entrypoint..."	5 weeks ago k3d-mycluster-server-0	Up 21 seconds	

4. Initialize Docker Swarm

This command enables swarm mode and makes the system a manager node.

```
docker swarm init --advertise-addr $(hostname -I | awk '{print $1}')
```

```
vanshika_munjial@DESKTOP-107G070:/mnt/c/Users/HP$ docker swarm init
Error response from daemon: This node is already part of a swarm. Use "docker swarm leave" to leave this swarm and join another one.
vanshika_munjial@DESKTOP-107G070:/mnt/c/Users/HP$ docker swarm init
Error response from daemon: This node is already part of a swarm. Use "docker swarm leave" to leave this swarm and join another one.
vanshika_munjial@DESKTOP-107G070:/mnt/c/Users/HP$ docker swarm leave --force
Node left the swarm.
vanshika_munjial@DESKTOP-107G070:/mnt/c/Users/HP$ docker swarm init
Error response from daemon: could not choose an IP address to advertise since this system has multiple addresses on different interfaces (10.255.255.254 on lo and 172.17.39.232 on eth0) - specify one with --advertise-addr
vanshika_munjial@DESKTOP-107G070:/mnt/c/Users/HP$ docker swarm init --advertise-addr $(hostname -I | awk '{print $1}')
Swarm initialized: current node (cx3ddtch0vmvt9t5tlaym9wy) is now a manager.

To add a worker to this swarm, run the following command:

    docker swarm join --token SWMTKN-1-48dp5fmljxz4j9ex3ke9zup2ygev3s9xo2reed3ojw9yegndab-cdblhfoh3a767b9d9fzk13hjr 172.17.39.232:2377

To add a manager to this swarm, run 'docker swarm join-token manager' and follow the instructions.
vanshika_munjial@DESKTOP-107G070:/mnt/c/Users/HP$
```

5. Verify Swarm is active

This shows the node status in the swarm.

```
docker node ls
```

```
vanshika_munjial@DESKTOP-107G070:/mnt/c/Users/HP$ docker node ls
ID                HOSTNAME          STATUS    AVAILABILITY    MANAGER STATUS    ENGINE VERSION
cx3ddtch0vmvt9t5tlaym9wy *  DESKTOP-107G070  Ready     Active           Leader             28.2.2
vanshika_munjial@DESKTOP-107G070:/mnt/c/Users/HP$
```

6. Deploy as a Stack (not just Compose)

Deploy the application using swarm stack instead of docker compose.

```
docker stack deploy -c docker-compose.yml wpstack
```

```
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp11 file$ docker stack deploy -c docker-compose.yml wpstack
Ignoring unsupported options: restart
Ignoring deprecated options:
container_name: Setting the container name is not supported.
Since --detach=false was not specified, tasks will be created in the background.
In a future release, --detach=false will become the default.
Creating network wpstack_default
Creating service wpstack_db
Creating service wpstack_wordpress
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp11 file$ |
```

7. Verify the deployment

Check all services created in the stack.

```
docker service ls
```

```
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp11 file$ docker service ls
ID            NAME          MODE          REPLICAS  IMAGE          PORTS
yz62orme0zs9 wpstack_db    replicated    1/1        mysql:5.7
cx5cmzqed9dt wpstack_wordpress replicated    1/1        wordpress:latest *:8888->80/tcp
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp11 file$ |
```

8. See detailed tasks (containers) for a service

Displays container-level details for a specific service.

```
docker service ps wpstack_wordpress
```

```
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp11 file$ docker service ps wpstack_wordpress
ID            NAME          IMAGE          NODE          DESIRED STATE  CURRENT STATE  ERROR  PORTS
wkbhfswpf9a9 wpstack_wordpress.1 wordpress:latest DESKTOP-107G070 Running        Running 2 hours ago
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp11 file$ |
```

9. See all running containers

List all running containers.

```
docker ps
```

```

vanshika_munjal@DESKTOP-107G070: /mnt/c/Users/Vanshika_Munjal/Desktop/C&O Lab/exp11 file$ docker ps

```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
33360cbec2f7	wordpress:latest	"docker-entrypoint.s..."	2 hours ago	Up 2 hours	80/tcp
563cec3cff94	mysql:5.7	"docker-entrypoint.s..."	2 hours ago	Up 2 hours	3306/tcp, 33060/tcp
bef98000fb4c	containerization-project-1_database	"docker-entrypoint.s..."	7 days ago	Up 3 hours	
554cbb7049b8	containerization-project-1_backend	"docker-entrypoint.s..."	7 days ago	Up 3 hours	
26ee230615b8	sonarqube:lts-community	"/opt/sonarqube/dock..."	7 days ago	Up 3 hours	
dda77d469ab9	postgres:13	"docker-entrypoint.s..."	7 days ago	Up 3 hours	
d5f70fbe1443	jenkins/jenkins:lts	"/usr/bin/tini -- /u..."	2 weeks ago	Up 3 hours	
25a799c4fd6b	wordpress:latest	"docker-entrypoint.s..."	3 weeks ago	Up 3 hours	
e07671139e7e	wordpress:latest	"docker-entrypoint.s..."	3 weeks ago	Up 3 hours	
e0b64a9cfeae	wordpress:latest	"docker-entrypoint.s..."	3 weeks ago	Up 3 hours	
c4c812ed407a	mysql:5.7	"docker-entrypoint.s..."	3 weeks ago	Up 3 hours	
ea81f65bde87	portainer/portainer-ce:lts	"/portainer"	5 weeks ago	Up 3 hours	0.0.0.0:8000->8000/tcp, [::]:8000->8000/tcp, 0.0.0.0:9443->9443/tcp, [::]:9443->9443/tcp
924e1452e147	ghcr.io/k3d-io/k3d-proxy:5.8.3	"/bin/sh -c nginx-pr..."	5 weeks ago	Up 3 hours	80/tcp, 0.0.0.0:39691->6443/tcp
0fa96e9ee63a	rancher/k3s:v1.31.5-k3s1	"/bin/k3d-entrypoint..."	5 weeks ago	Up 3 hours	

```

vanshika_munjal@DESKTOP-107G070: /mnt/c/Users/Vanshika_Munjal/Desktop/C&O Lab/exp11 file$

```

10. Access WordPress

Open the application in browser:

<http://localhost:8080>

Welcome

Welcome to the famous five-minute WordPress installation process! Just fill in the information below and you'll be on your way to using the most extendable and powerful personal publishing platform in the world.

Information needed

Please provide the following information. Do not worry, you can always change these settings later.

Site Title:

Username:
 Usernames can have only alphanumeric characters, spaces, underscores, hyphens, periods, and the @ symbol.

Password:
 Strong

Important: You will need this password to log in. Please store it in a secure location.

Your Email:
 Double-check your email address before continuing.

Search engine: ☐ Discourage search engines from indexing this site

11. Scale the Application

Increase the number of WordPress containers.

```
docker service scale wpstack_wordpress=3
```

```

vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp11 file$ docker service scale wpstack_wordpress=3
wpstack_wordpress scaled to 3
overall progress: 3 out of 3 tasks
1/3: running [=====]
2/3: running [=====]
3/3: running [=====]
verify: Service wpstack_wordpress converged
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp11 file$

```

12. Verify scaling

Check if replicas increased successfully.

```
docker service ls
```

```

vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp11 file$ docker service ls
ID                NAME                MODE                REPLICAS            IMAGE                PORTS
yz62ormeezs9     wpstack_db           replicated          1/1                 mysql:5.7
ca5cwzqed9dt     wpstack_wordpress    replicated          3/3                 wordpress:latest    *:8888->88/tcp
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp11 file$ docker ps | grep wordpress
ac4d929396d9      wordpress:latest     "docker-entrypoint.s" About a minute ago Up About a minute 80/tcp
                  wpstack_wordpress.3.xd8ok7ld82jffj8ophmt87t9u8
bd2a3e70bbfe      wordpress:latest     "docker-entrypoint.s" About a minute ago Up About a minute 80/tcp
                  wpstack_wordpress.2.n9rhiie0bhlbytuluhnuk74la
33360cbec2f7      6fe8e9457b10        "docker-entrypoint.s" 3 hours ago        Up 3 hours          80/tcp
                  wpstack_wordpress.1.kkbhfsmpf9a97sskv2lftqwp
25a799c4fd6b      6fe8e9457b10        "docker-entrypoint.s" 3 weeks ago        Up 3 hours
                  wp-compose-lab_wordpress_2
e07671139e7e      6fe8e9457b10        "docker-entrypoint.s" 3 weeks ago        Up 3 hours
                  wp-compose-lab_wordpress_3
e0b64a9cfeae      6fe8e9457b10        "docker-entrypoint.s" 3 weeks ago        Up 3 hours
                  wp-compose-lab_wordpress_1
c4c812ed407a      mysql:5.7            "docker-entrypoint.s" 3 weeks ago        Up 3 hours
                  wordpress_db
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp11 file$

```

13. Test self-healing

a. Kill the container (simulate crash)

Manually stop a container to test recovery.

```
docker kill <container_id>
```

```

vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp11 file$ docker kill ac4d929396d9
ac4d929396d9

```

b. Watch swarm recreate it

Swarm automatically creates a new container.

```
docker service ps wpstack_wordpress
```



```

vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika_Munjal/Desktop/C&D_Lab/exp11_file$ docker service ps wpstack_wordpress
ID                NAME                IMAGE                NODE                DESIRED STATE       CURRENT STATE        ERROR
wkbhfsapf9a9     wpstack_wordpress.1  wordpress:latest     DESKTOP-107G070    Running             Running 3 hours ago
n9rhie0bhlb      wpstack_wordpress.2  wordpress:latest     DESKTOP-107G070    Running             Running 4 minutes ago
j2p3wckh4lgu     wpstack_wordpress.3  wordpress:latest     DESKTOP-107G070    Running             Running 3 seconds ago
xd0ok7ld82jf     \_ wpstack_wordpress.3  wordpress:latest     DESKTOP-107G070    Shutdown            Failed 8 seconds ago  "task: non-zero
exit (137)"

```

c. Verify new container is running

docker ps

```

vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika_Munjal/Desktop/C&D_Lab/exp11_file$ docker ps | grep wordpress
a7177b9d6aaf     wordpress:latest     "docker-entrypoint.s..." 32 seconds ago Up 30 seconds 80/tcp
bd2a3e70bbfe     wordpress:latest     "docker-entrypoint.s..." 5 minutes ago Up 5 minutes 80/tcp
33360cbec2f7     6fe8e9457b10        "docker-entrypoint.s..." 3 hours ago Up 3 hours 80/tcp
25a799c4fd6b     6fe8e9457b10        "docker-entrypoint.s..." 3 weeks ago Up 3 hours
e07671139e7e     6fe8e9457b10        "docker-entrypoint.s..." 3 weeks ago Up 3 hours
e0b64a9cfeae     6fe8e9457b10        "docker-entrypoint.s..." 3 weeks ago Up 3 hours
c4c812ed407a     mysql:5.7            "docker-entrypoint.s..." 3 weeks ago Up 3 hours

```

14. Remove the stack

Deletes all services in the stack.

docker stack rm wpstack

```

vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika_Munjal/Desktop/C&D_Lab/exp11_file$ docker stack rm wpstack
Removing service wpstack_db
Removing service wpstack_wordpress
Removing network wpstack_default

```

15. Verify removal

Ensure all services are removed.

docker service ls


```

vanshika_munjal@DESKTOP-107GQ70:/mnt/c/Users/Vanshika_Munjal/Desktop/C&D Lab/exp11 file$ docker service ls

```

ID	NAME	MODE	REPLICAS	IMAGE	PORTS
vanshika_munjal@DESKTOP-107GQ70:/mnt/c/Users/Vanshika_Munjal/Desktop/C&D Lab/exp11 file\$	docker ps				

```

vanshika_munjal@DESKTOP-107GQ70:/mnt/c/Users/Vanshika_Munjal/Desktop/C&D Lab/exp11 file$ docker ps

```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
bef98880fb4c	containerization-project-1_database	"docker-entrypoint.s..."	7 days ago	Up 3 hours	
554cbb7049b8	containerization-project-1_backend	"docker-entrypoint.s..."	7 days ago	Up 3 hours	
26ee238615b8	sonarqube:lts-community	"/opt/sonarqube/dock..."	7 days ago	Up 3 hours	
dda77d469ab9	postgres:13	"docker-entrypoint.s..."	7 days ago	Up 3 hours	
d5f70fbel443	jenkins/jenkins:lts	"/usr/bin/tini -- /u..."	2 weeks ago	Up 3 hours	
25a790c4fd6b	6fe8e9457b10	"docker-entrypoint.s..."	3 weeks ago	Up 3 hours	
e07671139e7e	6fe8e9457b10	"docker-entrypoint.s..."	3 weeks ago	Up 3 hours	
e0b64a9cfeae	6fe8e9457b10	"docker-entrypoint.s..."	3 weeks ago	Up 3 hours	
c4c812ed407a	mysql:5.7	"docker-entrypoint.s..."	3 weeks ago	Up 3 hours	
ea81f68bde87	portainer/portainer-ce:lts	"/portainer"	5 weeks ago	Up 3 hours	0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp, 0.0.0.0:9443->9443/tcp, [::]:9443->9443/tcp, 9888/tcp
924e1452e147	ghcr.io/k3d-io/k3d-proxy:5.8.3	"/bin/sh -c nginx-pr..."	5 weeks ago	Up 3 hours	88/tcp, 0.0.0.0:39691->6443/tcp
0fa96e9ee63a	rancher/k3s:v1.31.5-k3s1	"/bin/k3d-entrypoint..."	5 weeks ago	Up 3 hours	k3d-mycluster-server-0

```

vanshika_munjal@DESKTOP-107GQ70:/mnt/c/Users/Vanshika_Munjal/Desktop/C&D Lab/exp11 file$

```

Conclusion

In this experiment, we extended Docker Compose by introducing Docker Swarm. We successfully deployed a multi-container application as a stack, scaled services, and observed self-healing behavior. This demonstrates how container orchestration automates deployment, scaling, and management of applications.

vanshikamunjal665

CONTAINERIZATION AND DEVOPS LAB

Experiment 12:

Study and analyse container orchestration using Kubernetes

Reference Link:

<https://upessocs.github.io/#dir=/Lectures/Containerization%20and%20DevOps/Lab/&file=Experiment%2012%20Kubernetes%20Draft.md> —

Theory

Objective

Learn why Kubernetes is used, its basic concepts, and how to deploy, scale, and fix apps using Kubernetes commands.

Why Kubernetes over Docker Swarm?

Reason	Explanation
Industry standard	Most companies use Kubernetes
Powerful scheduling	Automatically decides where to run your app
Large ecosystem	Many tools and plugins available
Cloud-native support	Works on AWS, Google Cloud, Azure, etc.

Core Kubernetes Concepts (Simple Explanation)

Docker Concept	Kubernetes Equivalent	What it means
Container	Pod	A pod is a group of one or more containers. Smallest unit in K8s.
Compose service	Deployment	Describes how your app should run (e.g., number of copies, image used)
Load balancing	Service	Exposes your app to the outside world or other pods
Scaling	ReplicaSet	Ensures a certain number of pod copies are always running

Task 1: Create a Deployment

A Deployment tells Kubernetes:

- Which container image to use (e.g., WordPress)
- How many copies (replicas) to run
- How to identify the pods (labels)

Step 1: Create a file (wordpress-deployment.yaml)

```

wordpress-deployment.yaml
File Edit View

# wordpress-deployment.yaml
apiVersion: apps/v1          # Which Kubernetes API to use
kind: Deployment             # Type of resource
metadata:
  name: wordpress            # Name of this deployment
spec:
  replicas: 2                 # Run 2 identical pods
  selector:
    matchLabels:
      app: wordpress          # Pods with this label belong to this deployment
  template:
    metadata:
      labels:
        app: wordpress        # Label applied to each pod
    spec:
      containers:
      - name: wordpress
        image: wordpress:latest # Docker image
        ports:
        - containerPort: 80     # Port inside the container

```

Step 2: Apply the deployment

```
kubectl apply -f wordpress-deployment.yaml
```

```

PS C:\Users\HP> wsl
vanshika_munjal@DESKTOP-107G070: /mnt/c/Users/HP$ cd "/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp12 file"
vanshika_munjal@DESKTOP-107G070: /mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp12 file$ kubectl apply -f wordpress-deployment.yaml
deployment.apps/wordpress created
vanshika_munjal@DESKTOP-107G070: /mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp12 file$ |

```

What happens? Kubernetes creates 2 pods running WordPress.

Step 3: Check if pods are running

```
kubectl get pods
```

```

vanshika_munjal@DESKTOP-107G070: /mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp12 file$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
apache-7d8d5c5d5-pszfz              1/1     Running   8 (3m12s ago)    36d
apache-7d8d5c5d5-xwptg              1/1     Running   8 (3m12s ago)    36d
apache-7d8d5c5d5-z668q              1/1     Running   8 (3m12s ago)    36d
apache-pod                           1/1     Running   8 (3m12s ago)    36d
wordpress-7d6f6db8d8-fa8lv          1/1     Running   0            117s
wordpress-7d6f6db8d8-mvf9w          1/1     Running   0            117s
vanshika_munjal@DESKTOP-107G070: /mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp12 file$

```

Step 4: Check deployment status

```
kubectl get deployments
```

```
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp12 file$ kubectl get deployments
NAME         READY   UP-TO-DATE   AVAILABLE   AGE
apache       3/3     3             3           36d
wordpress    2/2     2             2           3m57s
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp12 file$
```

Task 2: Expose the deployment as a service

Pods are temporary (they can be deleted or recreated). A service gives them a fixed IP and exposes them to the outside.

Step 1: Create a file (wordpress-service.yaml)

```
wordpress-service.yaml
File Edit View

# wordpress-service.yaml
apiVersion: v1
kind: Service
metadata:
  name: wordpress-service
spec:
  type: NodePort          # Exposes service on a port of each node (VM)
  selector:
    app: wordpress        # Send traffic to pods with this label
  ports:
    - port: 80             # Service port
      targetPort: 80       # Pod port
      nodePort: 30007       # External port (range: 30000-32767)
```

Step 2: Apply the service

```
kubectl apply -f wordpress-service.yaml
```

```
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp12 file$ kubectl apply -f wordpress-service.yaml
service/wordpress-service created
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp12 file$
```

Step 3: Services running in cluster

```
kubectl get services
```

```
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp12 file$ kubectl get services
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
apache	NodePort	10.43.245.92	<none>	80:32468/TCP	36d
kubernetes	ClusterIP	10.43.0.1	<none>	443/TCP	43d
web	NodePort	10.43.24.191	<none>	80:30109/TCP	42d
wordpress-service	NodePort	10.43.63.127	<none>	80:30007/TCP	26s

```
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp12 file$
```

Task 3: Verify everything

Step 1: Check pods

```
kubectl get pods
```

```
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp12 file$ kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
apache-7d8d5c5d5-pszfz	1/1	Running	8 (13m ago)	36d
apache-7d8d5c5d5-xwptg	1/1	Running	8 (13m ago)	36d
apache-7d8d5c5d5-z668q	1/1	Running	8 (13m ago)	36d
apache-pod	1/1	Running	8 (13m ago)	36d
wordpress-7d6f6db8d8-fw8lv	1/1	Running	0	12m
wordpress-7d6f6db8d8-mvf9w	1/1	Running	0	12m

```
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp12 file$
```

Step 2: Check service

```
kubectl get svc
```

```
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp12 file$ kubectl get svc
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
apache	NodePort	10.43.245.92	<none>	80:32468/TCP	36d
kubernetes	ClusterIP	10.43.0.1	<none>	443/TCP	43d
web	NodePort	10.43.24.191	<none>	80:30109/TCP	43d
wordpress-service	NodePort	10.43.63.127	<none>	80:30007/TCP	8m48s

```
vanshika_munjal@DESKTOP-1D7G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp12 file$
```

Step 3: Access WordPress in browser

```
kubectl port-forward service/wordpress-service 8080:80
```

Open in browser:

<http://localhost:8080>

Welcome

Welcome to the famous five-minute WordPress installation process! Just fill in the information below and you'll be on your way to using the most extensible and powerful personal publishing platform in the world.

Information needed

Please provide the following information. Do not worry, you can always change these settings later.

Site Title

Username

Username can have only alphanumeric characters, spaces, underscores, hyphens, periods, and the @ symbol.

Password [Hide](#)

Strong

Important: You will need this password to log in. Please store it in a secure location.

Your Email

Double-check your email address before continuing.

Search engine ☐ Discourage search engines from indexing this site

Task 4: Scale the deployment

Step 1: Increase replicas

```
kubectl scale deployment wordpress --replicas=4
```

```
vanshika_munj665@DESKTOP-107G070:/mnt/c/Users/Vanshika_Munj665/Desktop/C&D Lab/exp12 file$ kubectl scale deployment wordpress --replicas=4
deployment.apps/wordpress scaled
vanshika_munj665@DESKTOP-107G070:/mnt/c/Users/Vanshika_Munj665/Desktop/C&D Lab/exp12 file$
```

Step 2: Verify scaling

```
kubectl get pods
```

```
vanshika_munj665@DESKTOP-107G070:/mnt/c/Users/Vanshika_Munj665/Desktop/C&D Lab/exp12 file$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
apache-7d8d5c5d5-pszfz             1/1     Running   8 (25m ago) 36d
apache-7d8d5c5d5-xmptg             1/1     Running   8 (25m ago) 36d
apache-7d8d5c5d5-z668q             1/1     Running   8 (25m ago) 36d
apache-pod                          1/1     Running   8 (25m ago) 36d
wordpress-7d6f6db8d8-9wnzw         1/1     Running   0           2m15s
wordpress-7d6f6db8d8-cln2q         1/1     Running   0           2m15s
wordpress-7d6f6db8d8-fm8lv         1/1     Running   0           24m
wordpress-7d6f6db8d8-mvf9w         1/1     Running   0           24m
vanshika_munj665@DESKTOP-107G070:/mnt/c/Users/Vanshika_Munj665/Desktop/C&D Lab/exp12 file$
```

Task 5: Self-healing demonstration

Kubernetes automatically replaces failed pods.

Step 1: Delete one pod manually

```
kubectl delete pod <pod-name>
```

```
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp12 file$ kubectl delete pod wordpress-7d6f6db8d8-9wnzw  
pod "wordpress-7d6f6db8d8-9wnzw" deleted from default namespace  
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp12 file$ |
```

Step 2: Check pods again

```
kubectl get pods
```

```
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp12 file$ kubectl get pods  
NAME                                READY   STATUS    RESTARTS   AGE  
apache-7d8d5c5d5-pszfz             1/1     Running   8 (28m ago) 36d  
apache-7d8d5c5d5-xwptg             1/1     Running   8 (28m ago) 36d  
apache-7d8d5c5d5-z668q             1/1     Running   8 (28m ago) 36d  
apache-pod                          1/1     Running   8 (28m ago) 36d  
wordpress-7d6f6db8d8-7j9f98        1/1     Running   0           41s  
wordpress-7d6f6db8d8-cln2q         1/1     Running   0           5m25s  
wordpress-7d6f6db8d8-fm8lv         1/1     Running   0           27m  
wordpress-7d6f6db8d8-mvf9w         1/1     Running   0           27m  
vanshika_munjal@DESKTOP-107G070:/mnt/c/Users/Vanshika Munjal/Desktop/C&D Lab/exp12 file$ |
```

Result: The deleted pod is automatically recreated.

Result

- Successfully deployed WordPress using Kubernetes
- Exposed application using Service
- Verified deployment and service
- Scaled application from 2 to 4 pods
- Demonstrated self-healing

Conclusion

Kubernetes simplifies container orchestration by providing automated deployment, scaling, load balancing, and self-healing, ensuring high availability of applications.